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MIL-M-28787/349(SH)
31 JANUARY 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1553 RTU PROTOCOL MODULE, KEY CODE UKU

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Remote Terminal Unit (RTU) protocol module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.19 failures per million operating hours for class I or 2.87 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be UKU.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 115 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation, used to unused	II	X	10	INF	10	INF	10	INF	MOhm
A3	Isolation, unused to unused	II	X	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D23	Ibd	V	S	-20	20	-20	20	-20	20	uA
D24	Ibd	V	S	-11	111	-11	111	-11	111	uA
D25	Ibd	V	S	0	200	0	200	0	200	uA
D26	Ibd	V	S	-11	111	-11	111	-11	111	uA
D27-D42	Ibd	V	S	0	100	0	100	0	100	uA
E1-E18	Vic	VI	S	-1.5	-0.3	-1.5	-0.3	-1.5	-0.3	V
F1-F16	Ios	VII	S	-225	-40	-225	-40	-225	-40	mA
F17	Ios	VII	S	-100	-20	-100	-20	-100	-20	mA
G1-G23	Iih	VIII	X	-20	20	-20	20	-20	20	uA
G24	Iih	VIII	X	-260	-70	-260	-70	-260	-70	uA
G25	Iih	VIII	X	0	40	0	40	0	40	uA
G26	Iih	VIII	X	-20	50	-20	50	-20	50	uA
G27-G42	Iih	VIII	X	0	20	0	20	0	20	uA
H1-H23	Iil	IX	X	-20	20	-20	20	-20	20	uA
H24	Iil	IX	X	-910	-200	-910	-200	-910	-200	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H25	Iil	IX	X	-800	0	-800	0	-800	0	uA
H26	Iil	IX	X	-810	0	-810	0	-810	0	uA
H27-H42	Iil	IX	X	-400	0	-400	0	-400	0	uA
J1-J60	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J61	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K61	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L57	Iozh	XII	X	-20	20	-20	20	-20	20	uA
M1-M16	Iozl	XIII	X	-400	20	-400	20	-400	20	uA
M17-M57	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
N1-N5	Oscillator	XIV	S	----------See table XIV----------						
P1	Ioo	XV	X	30	323	30	323	30	323	mA
P2	Icc	XV	X	2	25	2	25	2	25	mA
Q1,Q2	Functional	XVI	X	----------See table XVI----------						
R1,R2	Dynamic	XVII	S	----------See table XVII----------						
R3,R4	Dynamic	XVII	Q	----------See table XVII----------						
S	PSFC	XVIII	S	4.8	8.8	4.8	8.8	4.8	8.8	uF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
U1-U60	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	mV
V1-V60	Output crosstalk, low level	XXI	Q	0	400	0	400	0	400	mV
W1-W26	Cin	XXII	Q	2	40	2	40	2	40	pF
X1-X62	Cout	XXIII	Q	2	40	2	40	2	40	pF
Y	Power-on reset test	XXIV	Q	240	1790	240	1790	240	1790	uS
Z1	Vt(+)	XXV	S	---	2.0	---	2.0	---	2.0	V
Z2	Vt(-)	XXV	S	0.5	---	0.5	---	0.5	---	V
Z3	Hysteresis	XXV	S	0.4	---	0.4	---	0.4	---	V

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test tolerance table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/349-1 for class I and M28787/349-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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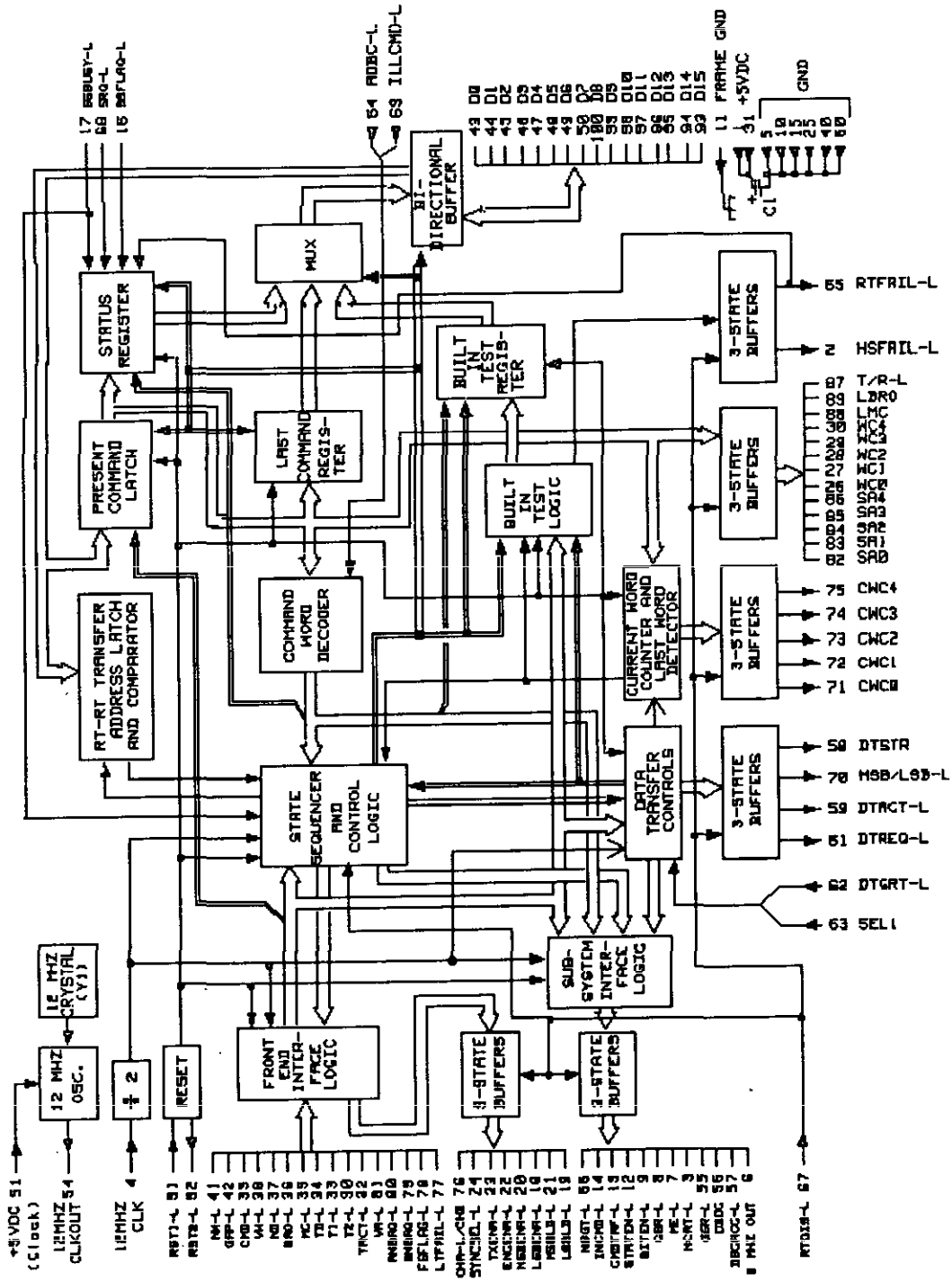


FIGURE 1. UKU functional block diagram.

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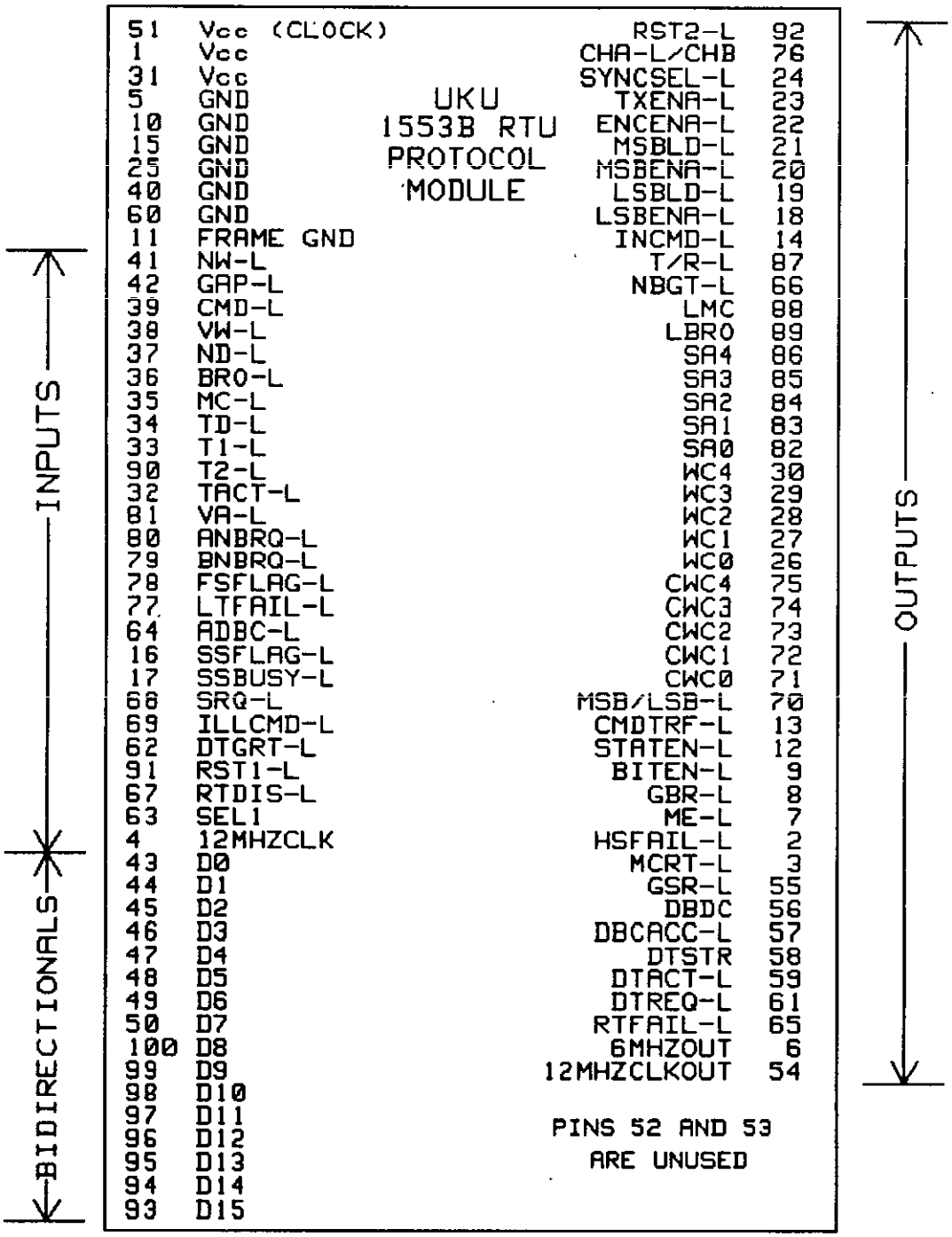


FIGURE 2. UKU pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

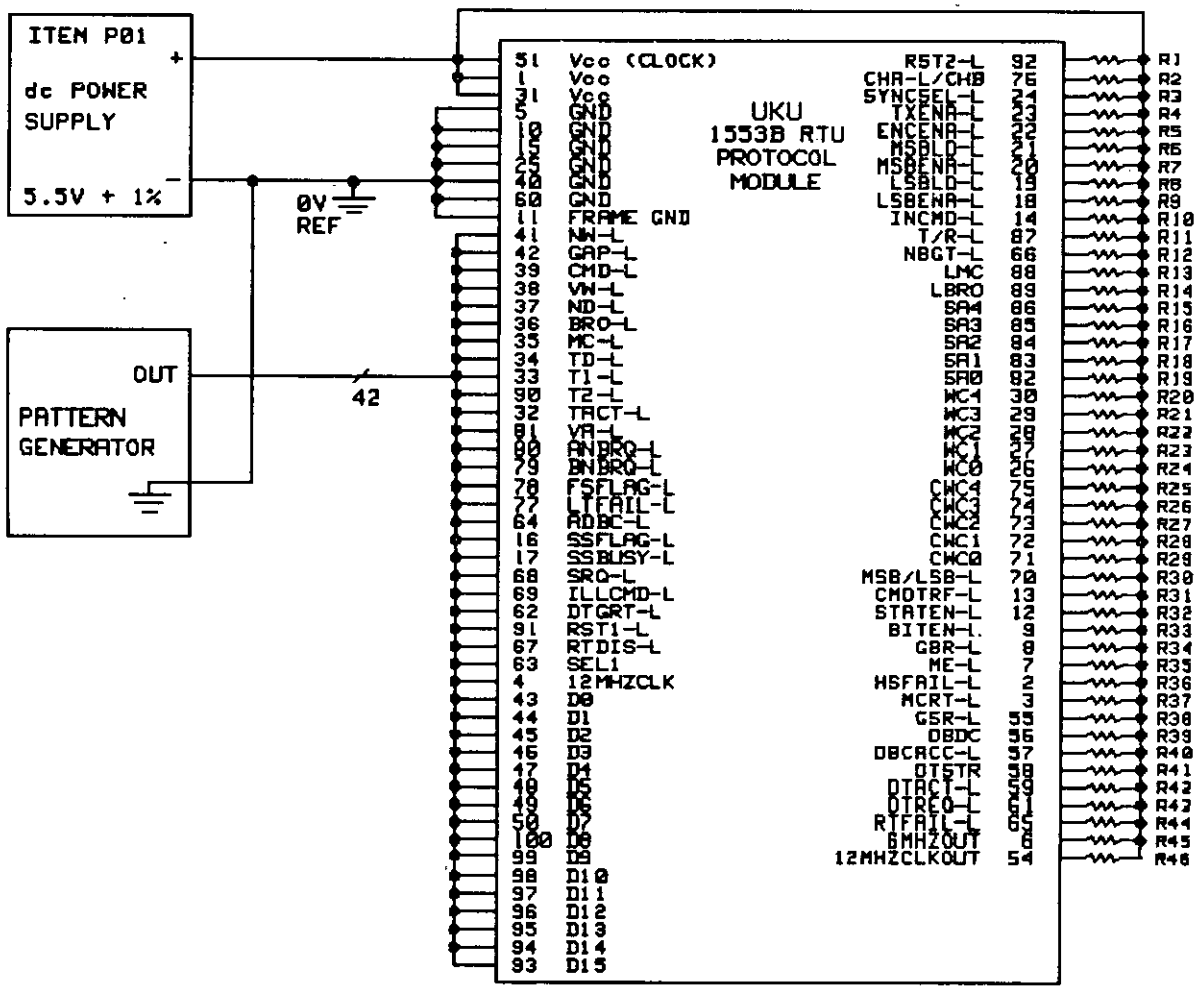
4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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NOTES:

1. R1 = 1.3 KOhm ± 5 percent, 0.25 W; R2-R29, R31-R44 = 3.24 KOhm ± 5 percent, 0.25 W; R30 = R45 = 4.32 KOhm ± 5 percent, 0.25 W; R46 = 511 Ohm ± 5 percent, 0.25 W.

2. Signals D0-D15 are bidirectional. Each will be loaded with a 432 Ohm ± 5 percent, 0.25 W resistor tied to Vcc when programmed to be an output.

3. Set the pattern generator to a pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle, using appropriate patterns to exercise the module.

FIGURE 3. UKU operating temperature, component temperature and life test circuit.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25 ± 5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the 25°± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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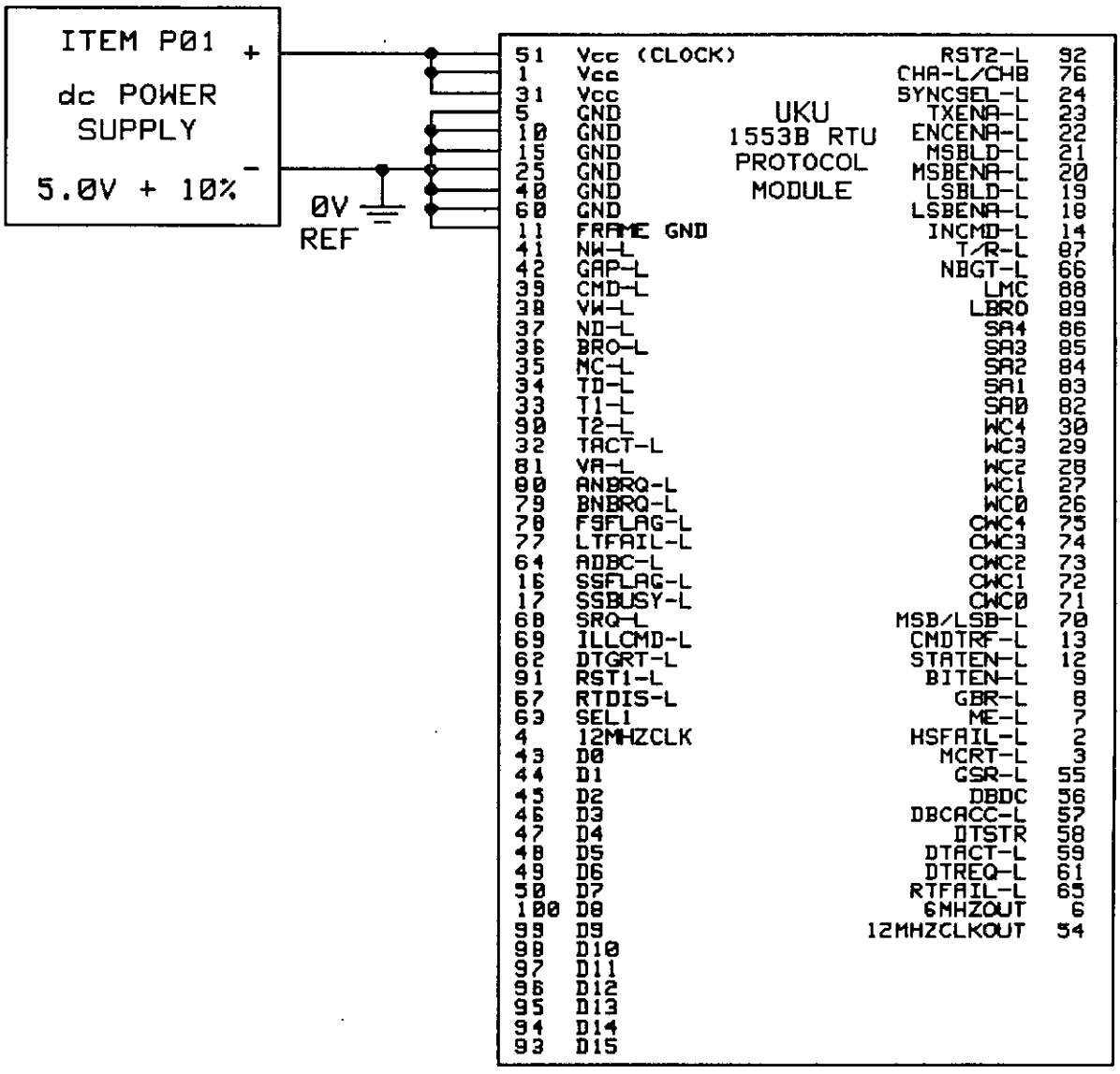


FIGURE 4. UKU vibration test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Voh = output high level voltage.
(s) Vol = output low level voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the class I and class II end-of-life parameters in table I. Operating power supply voltage range: 4.5 V to 5.5 V dc.

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
SSFLAG-L	NA	0.7	2.4	-20	20	20	40
SSBUSY-L	NA	0.7	2.4	-20	20	20	40
TACT-L	NA	0.7	2.4	-20	20	20	40
T1-L	NA	0.7	2.4	-20	20	20	40
TD-L	NA	0.7	2.4	-20	20	20	40
MC-L	NA	0.7	2.4	-20	20	20	40
BRO-L	NA	0.7	2.4	-20	20	20	40
ND-L	NA	0.7	2.4	-20	20	20	40
VW-L	NA	0.7	2.4	-20	20	20	40
CMD-L	NA	0.7	2.4	-20	20	20	40
NW-L	NA	0.7	2.4	-20	20	20	40
GAP-L	NA	0.7	2.4	-20	20	20	40
DTGRT-L	NA	0.7	2.4	-20	20	20	40
ADBC-L	NA	0.7	2.4	-20	20	20	40

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
RTDIS-L	NA	0.7	2.4	-20	20	20	40
SRQ-L	NA	0.7	2.4	-20	20	20	40
ILLCMD-L	NA	0.7	2.4	-20	20	20	40
LTFAIL-L	NA	0.7	2.4	-20	20	20	40
FSFLAG-L	NA	0.7	2.4	-20	20	20	40
BNBRQ-L	NA	0.7	2.4	-20	20	20	40
ANBRQ-L	NA	0.7	2.4	-20	20	20	40
VA-L	NA	0.7	2.4	-20	20	20	40
12MHZCLK	NA	0.7	2.4	-20	20	20	40
RST-L 1/	-1.5	0.5	2.0	-910	-260	100	40
T2-L	-1.5	0.7	2.0	-800	40	200	40
SEL1	NA	0.5	2.4	-810	50	100	40
D0-D15 2/	-1.5	0.7	2.0	-400	20	100	40

1/ Schmitt trigger input.
2/ These signals are bidirectional.

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
HSFAIL-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
MCRT-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
ME-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
GBR-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
BITEN-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
STATEN-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
CMDTRF-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
INCMD-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
LSBENA-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
LSBLD-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
MSBENA-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
MSBLD-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
ENCENA-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
TXENA-L	2.4	0.4	-0.4	1.6	20	-20	NA	40

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
SYNCSEL-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
WC0-WC4	2.4	0.4	-0.4	1.6	20	-20	NA	40
GSR-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
DBDC	2.4	0.4	-0.4	1.6	20	-20	NA	40
DBCACC-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
DTSTR	2.4	0.4	-0.4	1.6	20	-20	NA	40
DTACT-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
DTREQ-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
NBGT-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
CWC0-CWC4	2.4	0.4	-0.4	1.6	20	-20	NA	40
CHA-L/CHB	2.4	0.4	-0.4	1.6	20	-20	NA	40
SA0-SA4	2.4	0.4	-0.4	1.6	20	-20	NA	40
T/R-L	2.4	0.4	-0.4	1.6	20	-20	NA	40
LMC	2.4	0.4	-0.4	1.6	20	-20	NA	40

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)	Cout max (pF)
LBRO	2.4	0.4	-0.4	1.6	20	-20	NA	40
RTFAIL-L	2.4	0.4	-0.39	1.6	20	-20	NA	40
6MHZOUT	2.4	0.4	-0.38	1.2	20	-20	NA	40
MSB/LSB-L	2.4	0.4	-0.38	1.2	20	-20	NA	40
RST2-L	2.4	0.4	-0.39	4.0	NA	NA	-100	40
12MHZCLKOUT	Vcc-0.4	0.4	-10.0	10.0	NA	NA	NA	40
D0-D15 1/	2.4	0.4	-3.0	12.0	20	-400	-225	40

1/ These signals are bidirectional.

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
1	tplh	12MHZCLK(-) to 6MHZOUT (+)	-	-	-	155	155	155	nS	5
2	tphl	12MHZCLK(-) to 6MHZOUT (-)	-	-	-	145	145	145	nS	5
3	tphl	12MHZCLK to NBGT-L (-)	-	-	-	245	245	245	nS	7b
4	tphl	12MHZCLK to CHA-L/ CHB(-)	-	-	-	245	245	245	nS	7b
5	tplh	12MHZCLK to CHA-L/ CHB(+)	-	-	-	245	245	245	nS	
6	tphl	12MHZCLK to DTACT-L(-) (SEL1 low)	-	-	-	200	200	200	nS	9b
7	tphl	12MHZCLK to DTACT-L(-) (SEL1 high)	-	-	-	200	200	200	nS	7b
8	tphl	12MHZCLK to MSBENA-L(-)	-	-	-	240	240	240	nS	7b
9	tphl	12MHZCLK to LSBENA-L(-) (SEL1 low)	-	-	-	240	240	240	nS	9b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
10	tphl	12MHZCLK to LSBENA-L(-) (SEL1 high)	-	-	-	260	260	260	nS	7b
11	tplh	12MHZCLK to DTSTR(+)	-	-	-	245	245	245	nS	7b
12	tplh	12MHZCLK to DTSTR(+) (SEL1 high, 2nd pulse)	-	-	-	245	245	245	nS	7b
13	tphl	12MHZCLK to MSB/LSB-L (-) (SEL1 high)	-	-	-	245	245	245	nS	7b
14	tphl	12MHZCLK to MSBLD-L(-)	-	-	-	245	245	245	nS	7d
15	tphl	12MHZCLK to LSBLD-L(-) (SEL 1 low)	-	-	-	245	245	245	nS	9d
16	tphl	12MHZCLK to LSBLD-L(-) (SEL1 high)	-	-	-	245	245	245	nS	7d
17	tphl	12MHZCLK to STATEN-L(-) (SEL1 low)	-	-	-	250	250	250	nS	9d
18	tphl	12MHZCLK to STATEN-L(-) (SEL1 high)	-	-	-	250	250	250	nS	7d

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
19	tphl	12MHZCLK to BITEN-L(-) (SEL1 low)	-	-	-	250	250	250	nS	14b
20	tphl	12MHZCLK to BITEN-L(-) (SEL1 high)	-	-	-	250	250	250	nS	14b
21	tphl	12MHZCLK to GBR-L(-)	-	-	-	255	255	255	nS	7d
22	tphl	12MHZCLK to GSR-L(-)	-	-	-	255	255	255	nS	
23	tphl	12MHZCLK to ME-L(-)	-	-	-	215	215	215	nS	8
24	tplh	12MHZCLK to ME-L(+)	-	-	-	305	305	305	nS	
25	tphl	12MHZCLK to HSFAIL-L(-)	-	-	-	215	215	215	nS	
26	tplh	12MHZCLK to HSFAIL-L(+)	-	-	-	305	305	305	nS	
27	tphl	12MHZCLK to INCMD-L(-)	-	-	-	200	200	200	nS	7b
28	tplh	12MHZCLK to INCMD-L(+)	-	-	-	235	235	235	nS	
29	tphl	12MHZCLK to CMDTRF-L(-)	-	-	-	215	215	215	nS	7b
30	tplh	12MHZCLK to CMDTRF-L(+)	-	-	-	265	265	265	nS	

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
31	tphl	12MHZCLK to DTREQ-L(-)	-	-	-	215	215	215	nS	7b
32	tplh	12MHZCLK to DTREQ-L(+)	-	-	-	230	230	230	nS	
33	tphl	12MHZCLK to SYNCSEL-L(-)	-	-	-	220	220	220	nS	7d
34	tplh	12MHZCLK to SYNCSEL-L(+)	-	-	-	270	270	270	nS	
35	tphl	12MHZCLK to TXENA-L(-)	-	-	-	255	255	255	nS	7d
36	tplh	12MHZCLK to TXENA-L(+)	-	-	-	220	220	220	nS	
37	tphl	12MHZCLK to ENCENA-L(-)	-	-	-	265	265	265	nS	7d
38	tplh	12MHZCLK to ENCENA-L(+)	-	-	-	290	290	290	nS	
39	tplh	12MHZCLK to DBDC(+)	-	-	-	255	255	255	nS	
40	tphl	12MHZCLK to DBDC(-)	-	-	-	300	300	300	nS	12e
41	tphl	12MHZCLK to DBCACC-L(-)	-	-	-	225	225	225	nS	17
42	tplh	12MHZCLK to DBCACC-L(+)	-	-	-	305	305	305	nS	
43	tphl	12MHZCLK to MCRT-L(-)	-	-	-	215	215	215	nS	15

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
44	tpd	12MHZCLK to CWC0(+) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
45	tpd	12MHZCLK to CWC1(+) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
46	tpd	12MHZCLK to CWC2(+) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
47	tpd	12MHZCLK to CWC3(+) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
48	tpd	12MHZCLK to CWC4(+) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
49	tpd	12MHZCLK to CWC0(-) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
50	tpd	12MHZCLK to CWC1(-) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
51	tpd	12MHZCLK to CWC2(-) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
52	tpd	12MHZCLK to CWC3(-) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
53	tpd	12MHZCLK to CWC4(-) (Increment cleared NBGT-L)	-	-	-	245	245	245	nS	7c
54	tplh	12MHZCLK to RTFAIL-L(+)	-	-	-	215	215	215	nS	18
55	tphl	LTFAIL-L(-) to RTFAIL-L(-) (TXENA low)	-	-	-	110	110	110	nS	18
56	tphl	12MHZCLK to RTFAIL-L(-) (FSFLAG low TXENA low)	-	-	-	215	215	215	nS	18
57	tpd	12MHZCLK to LMC(+)	-	-	-	295	295	295	nS	7b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
58	tpd	12MHZCLK to LMC(-)	-	-	-	295	295	295	nS	7b
59	tpd	12MHZCLK to LBRO(+)	-	-	-	295	295	295	nS	7b
60	tpd	12MHZCLK to LBRO(-)	-	-	-	295	295	295	nS	7b
61	tpd	12MHZCLK to T/R-L(+)	-	-	-	295	295	295	nS	7b
62	tpd	12MHZCLK to T/R-L(-)	-	-	-	295	295	295	nS	7b
63	tpd	12MHZCLK to SA0(+)	-	-	-	295	295	295	nS	7b
64	tpd	12MHZCLK to SA1(+)	-	-	-	295	295	295	nS	7b
65	tpd	12MHZCLK to SA2(+)	-	-	-	295	295	295	nS	7b
66	tpd	12MHZCLK to SA3(+)	-	-	-	295	295	295	nS	7b
67	tpd	12MHZCLK to SA4(+)	-	-	-	295	295	295	nS	7b
68	tpd	12MHZCLK to SA0(-)	-	-	-	295	295	295	nS	7b
69	tpd	12MHZCLK to SA1(-)	-	-	-	295	295	295	nS	7b
70	tpd	12MHZCLK to SA2(-)	-	-	-	295	295	295	nS	7b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
71	tpd	12MHZCLK to SA3(-)	-	-	-	295	295	295	nS	7b
72	tpd	12MHZCLK to SA4(-)	-	-	-	295	295	295	nS	7b
73	tpd	12MHZCLK to WC0(+)	-	-	-	295	295	295	nS	7b
74	tpd	12MHZCLK to WC1(+)	-	-	-	295	295	295	nS	7b
75	tpd	12MHZCLK to WC2(+)	-	-	-	295	295	295	nS	7b
76	tpd	12MHZCLK to WC3(+)	-	-	-	295	295	295	nS	7b
77	tpd	12MHZCLK to WC4(+)	-	-	-	295	295	295	nS	7b
78	tpd	12MHZCLK to WC0(-)	-	-	-	295	295	295	nS	7b
79	tpd	12MHZCLK to WC1(-)	-	-	-	295	295	295	nS	7b
80	tpd	12MHZCLK to WC2(-)	-	-	-	295	295	295	nS	7b
81	tpd	12MHZCLK to WC3(-)	-	-	-	295	295	295	nS	7b
82	tpd	12MHZCLK to WC4(-)	-	-	-	295	295	295	nS	7b

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
83	tphz tplz 4/	RTDIS-L(-) to OUTPUTS (Tri-state)	-	-	-	130	130	130	nS	
84	tpzh tpzl	RTDIS-L(+) to OUTPUTS (Active)	-	-	-	160	160	160	nS	
85	tphl	RST1-L(-) to RST2-L(-)	-	-	-	50	50	50	nS	
86	tpd	RST1-L(-) to MODULE RESET	-	-	-	310	310	310	nS	
87	tplh	RST1-L(+) to RST2-L(+)	-	-	-	50	50	50	nS	
88	tpv	12MHZCLK to D15-D0(+/-) (SEL1-L low)	-	-	-	255	255	255	nS	5
89	tpz 4/	12MHZCLK to D15-D0 Disable (SEL1-L low DTACT-L high)	-	-	-	235	235	235	nS	5
90	tpv	12MHZCLK to D15-D8(+/-) (SEL1-L high)	-	-	-	255	255	255	nS	6
91	tpz 4/	12MHZCLK to D15-D8 Disable (SEL1-L high MSB/ LSB-L high)	-	-	-	235	235	235	nS	

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
92	tpv	12MHZCLK to D7-D0(+/-) (SEL1-L high)	-	-	-	255	255	255	nS	6
93	tpz 4/	12MHZCLK to D7-D0 Disable (SEL1-L high DTACT-L high)	-	-	-	235	235	235	nS	
94	PWL	NBGT-L	77	77	77	256	256	256	nS	7b
95	PWL	DTACT-L (SEL1 low)	450	450	450	570	570	570	nS	9b
96	PWL	DTACT-L (SEL1 high)	900	900	900	1070	1070	1070	nS	7b
97	PWL	MSBENA-L	450	450	450	570	570	570	nS	7b
98	PWL	LSBENA-L	450	450	450	570	570	570	nS	7b
99	PWH	DTSTR-H	122	122	122	210	210	210	nS	7b
100	PWL	MSB/LSB-L	450	450	450	570	570	570	nS	7b
101	PWL	MSBLD-L	122	122	122	210	210	210	nS	7d
102	PWL	LSBLD-L	122	122	122	210	210	210	nS	7d
103	PWL	STATEN-L (SEL1 low)	405	405	405	570	570	570	nS	9d

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
104	PWL	STATEN-L (SEL1 high)	905	905	905	1070	1070	1070	nS	7d
105	PWL	BITEN-L (SEL1 low)	405	405	405	570	570	570	nS	14b
106	PWL	BITEN-L (SEL1 high)	905	905	905	1070	1070	1070	nS	14b
107	PWL	GBR-L	400	400	400	600	600	600	nS	7d
108	PWL	GSR-L	400	400	400	600	600	600	nS	
109	PWL	MCRT-L	106	106	106	360	360	360	nS	15
110	PWL	NW-L	200	200	200	-	-	-	nS	19b
111	PWL	T1-L	200	200	200	-	-	-	nS	19c
112	PWL	ANBRQ-L	200	200	200	-	-	-	nS	19a
113	PWL	BNBRQ-L	200	200	200	-	-	-	nS	19a
114	PWL	BRO-L	1000	1000	1000	-	-	-	nS	
115	PWL	MC-L	1000	1000	1000	-	-	-	nS	
116	PWL	GAP-L	1000	1000	1000	-	-	-	nS	
117	tsu	ANBRQ-L, BNBRQ-L to BRO-L,MC-L, GAP-L	0	0	0	-	-	-	nS	19a
118	th	ANBRQ-L, BNBRQ-L to BRO-L,MC-L, GAP-L	1000	1000	1000	-	-	-	nS	19a

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
119	tsu	CMD-L,VW-L, VA-L,BRO-L, MC-L, GAP-L to NW-L(-)	166	166	166	-	-	-	nS	19b
120	th	CMD-L,VW-L, VA-L,BRO-L, MC-L,GAP-L to NW-L(-)	500	500	500	-	-	-	nS	19b
121	tsu	DTREQ-L to DTGRT-L	-	-	-	1500	1500	1500	nS	19c
122	tsu	T1-L to ADBC-L, SRQ-L, SSFLAG-L, SSBUSY-L	166	166	166	-	-	-	nS	19c
123	th	T1-L to ADBC-L, SRQ-L, SSFLAG-L, SSBUSY-L	333	333	333	-	-	-	nS	19c
124	tsu	ILLCMD-L to 12MHZCLK(-)	80	80	80	-	-	-	nS	19d
125	th	ILLCMD-L to 12MHZCLK(-)	235	235	235	-	-	-	nS	19d
126	tpv	MSBENA-L(-) to D15-D8 (Valid)	-	-	-	80	80	80	nS	20

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/ 3/	Minimum limit 1/			Maximum limit 1/			Unit	Figs
			-55°	25°	85°	-55°	25°	85°		
127	tpz 4/	MSBENA-L(+) to D15-D8 (Invalid)	0	0	0	-	-	-	nS	20
128	tpv	LSBENA-L(-) to D7-D0 (Valid)	-	-	-	80	80	80	nS	20
129	tpz 4/	LSBENA-L(+) to D7-D0 (Invalid)	0	0	0	-	-	-	nS	20

- See footnotes on next page.

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1/ The limits for class I at 0, 25 and 60°C are the same as class II limits at -55, 25 and 85°C.

2/ For pulses and delays from other signals besides 6MHZOUT, it is assumed that 12MHZCLK is exactly 12 MHz. That is, for each 6MHZOUT clock cycle, 166 nS is used for the period.

3/ (-) indicates a negative going pulse. (+) indicates a positive going pulse.

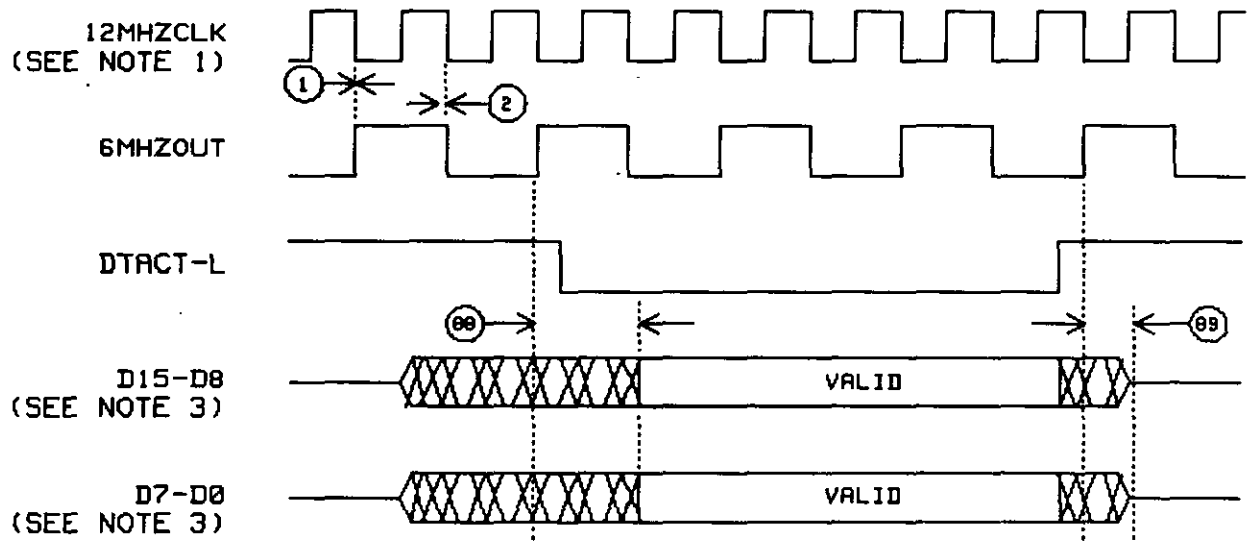
4/ Qualification (Q) test.

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6.4 Module function. The UKU module is a class II, 2A format B module. When used in conjunction with two UKT modules, it provides all the protocol and timing requirements necessary to make the UKU a complete dual-redundant RTU per MIL-STD-1553B. It operates on a +5 V dc power supply and can be used with either an 8 bit or 16 bit data bus structure which is pin programmable. The UKU module consists of a hybrid, a 12 MHz crystal, and a power supply filter capacitor. Inside the hybrid there is an oscillator circuit, a large scale semi-custom Complementary Metal-Oxide Semiconductor (CMOS) logic gate array, two data bus buffers, miscellaneous Low-power Schottky Transistor-Transistor Logic (LS TTL) logic, bypass capacitors, and a Resistor-Capacitor (R-C) circuit for power on clear.

6.4.1 Module timing. The UKU module timing is illustrated on figures 5 through 18.

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NOTES:

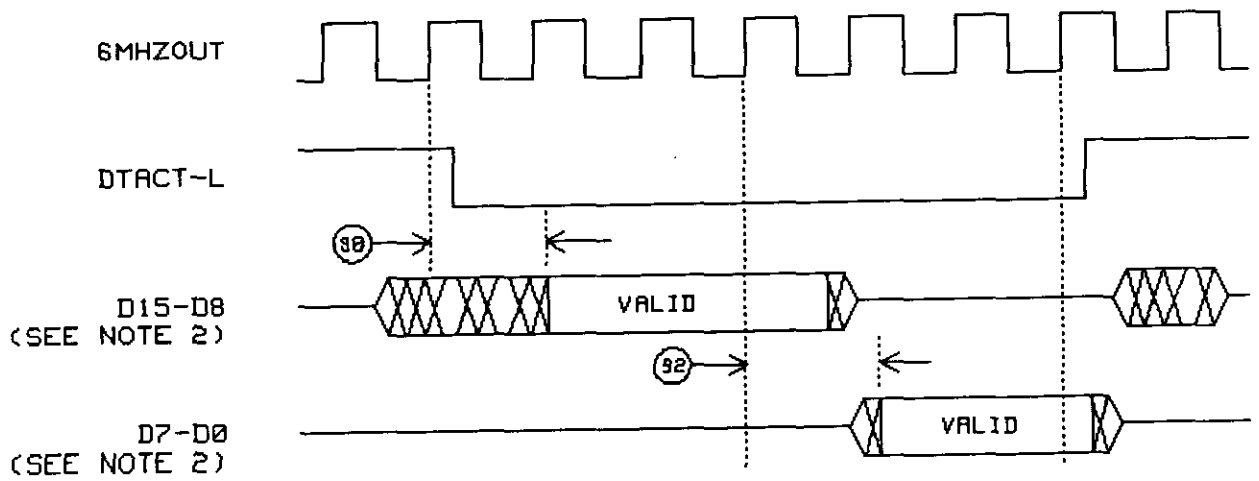
1. Delays from 12MHZCLK to 6MHZOUT must be added to obtain correct limits as in 6.3.3 for remaining timing diagrams.

2. SEL1 is low.

3. Output is present only on the transfer of the status word, built-in-test word or last command word. At all other times it is an input.

FIGURE 5. UKU driving data bus (16 bit operation).

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NOTES:

1. SEL1 is high.

2. Output is present only on the transfer of the status word, built-in-test word or last command word. At all other times it is an input.

FIGURE 6. UKU driving data bus (8 bit operation).

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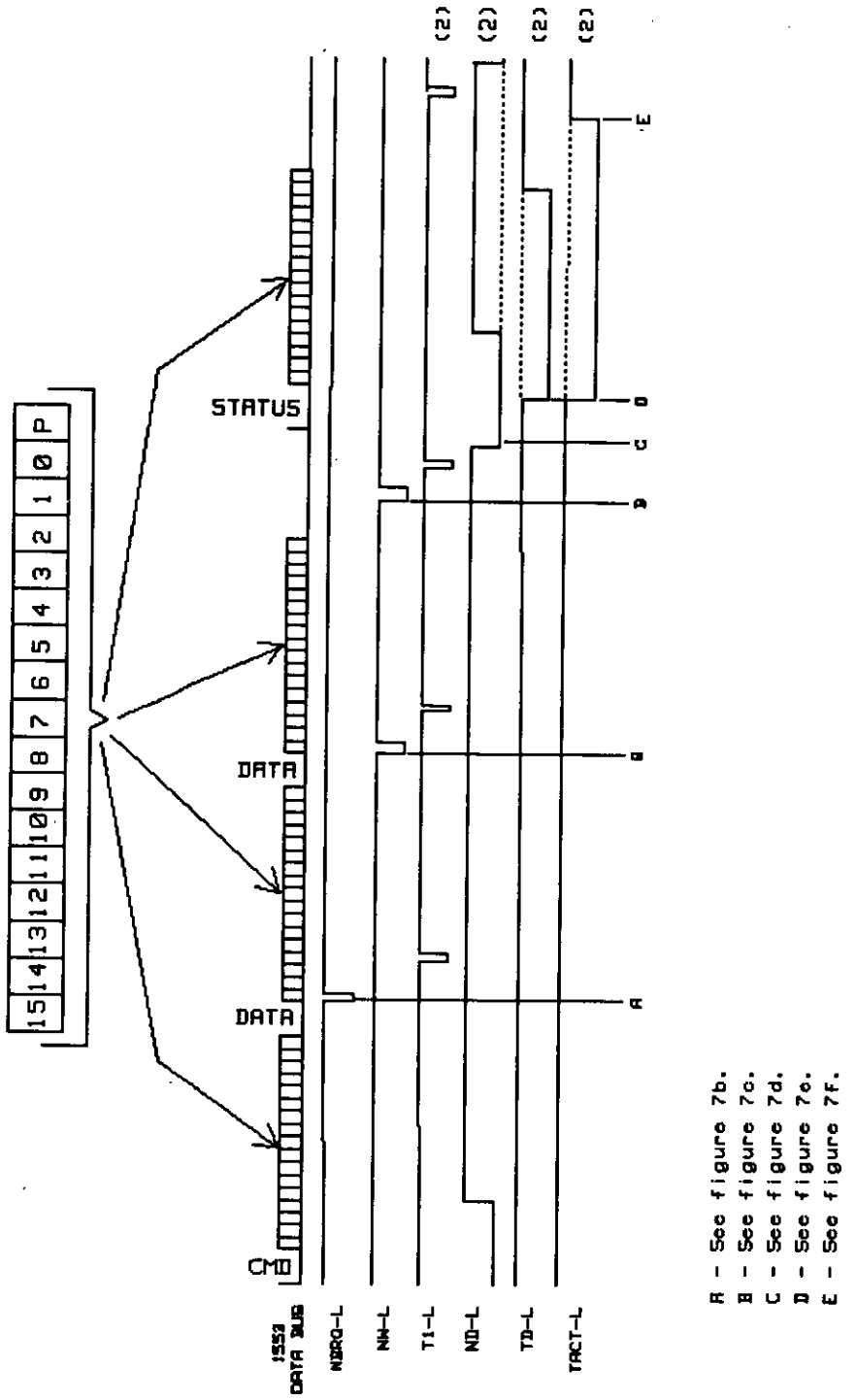


FIGURE 7a. Receive two words (8 bit mode).

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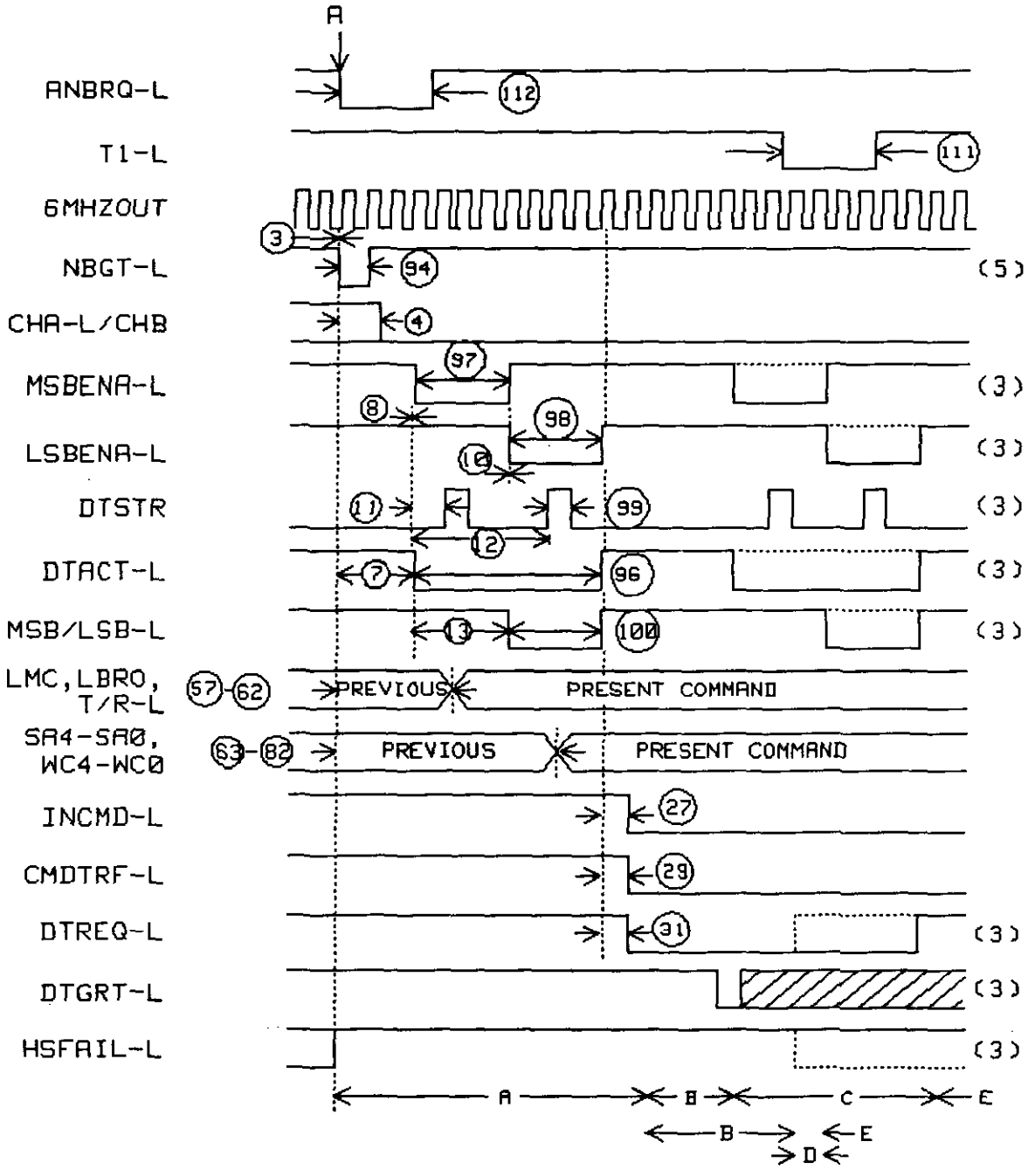


FIGURE 7b. Receive two words (8 bit mode).

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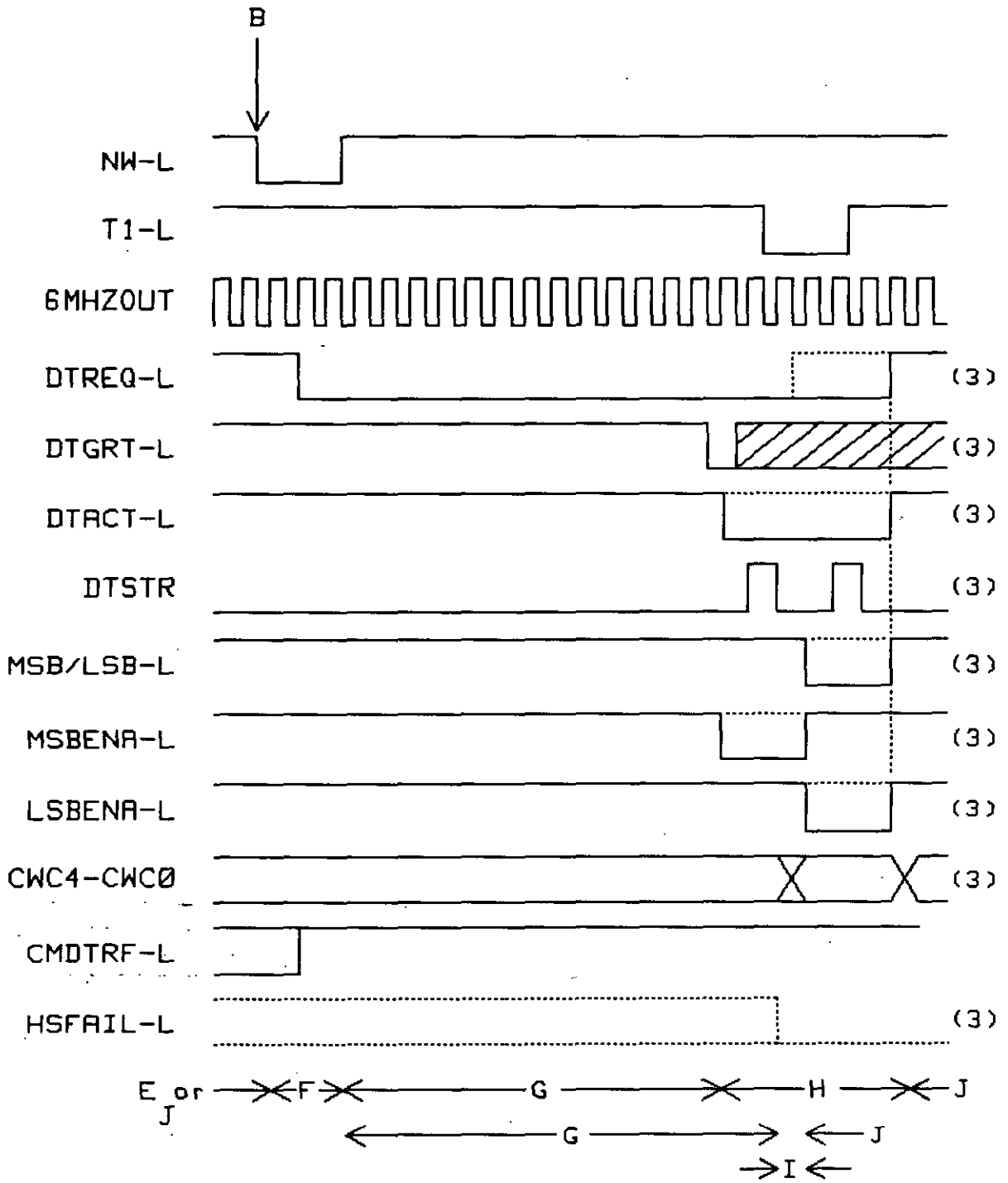


FIGURE 7c. Receive two words (8 bit mode).

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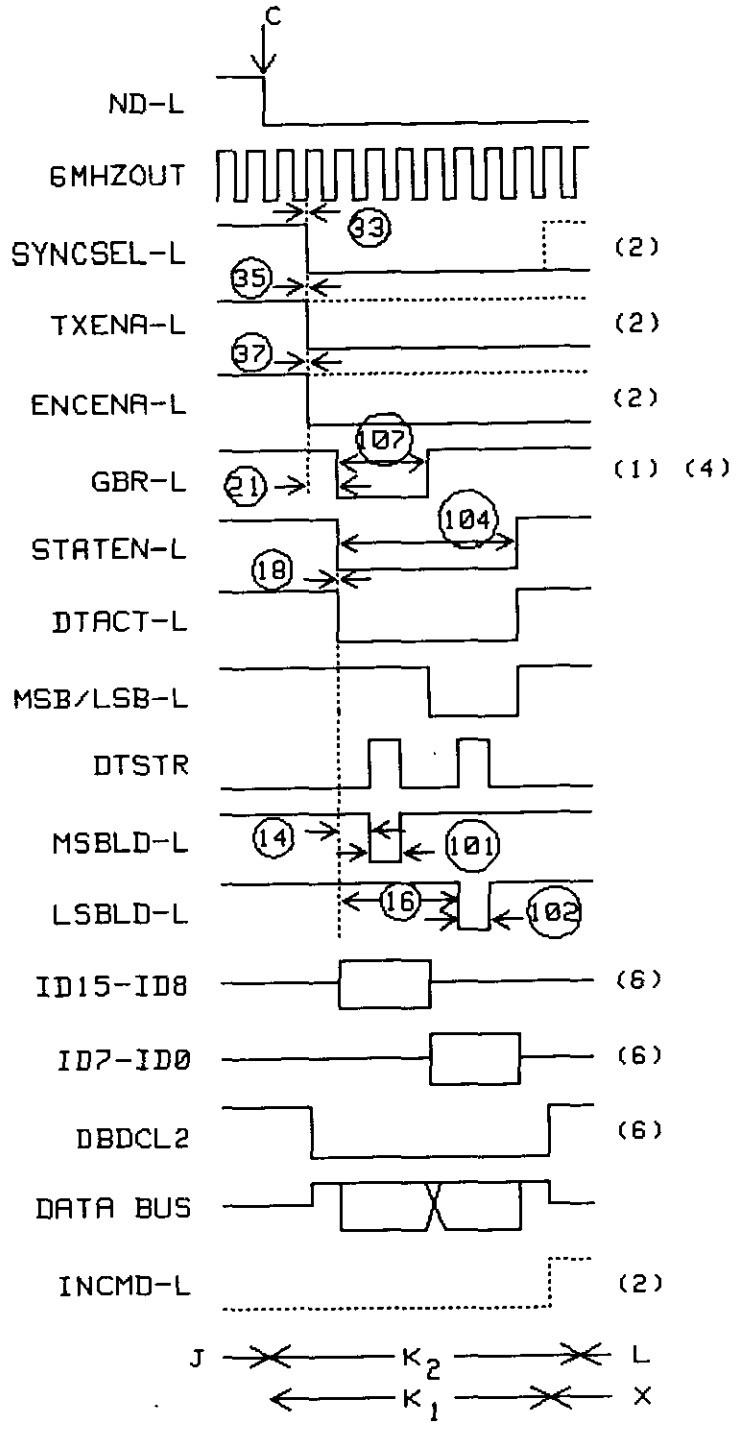


FIGURE 7d. Receive two words (8 bit mode).

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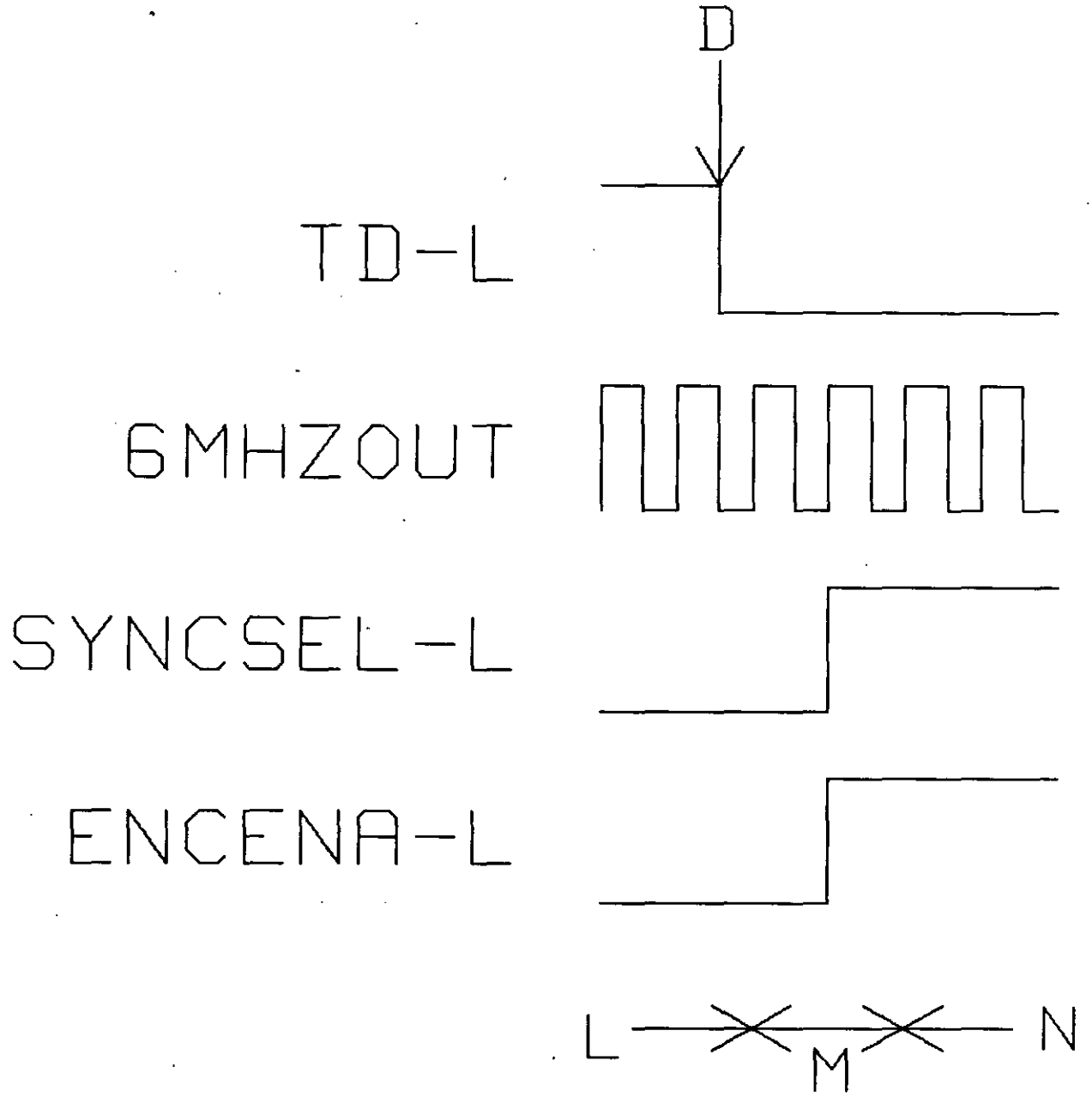


FIGURE 7e. Receive two words (8 bit mode).

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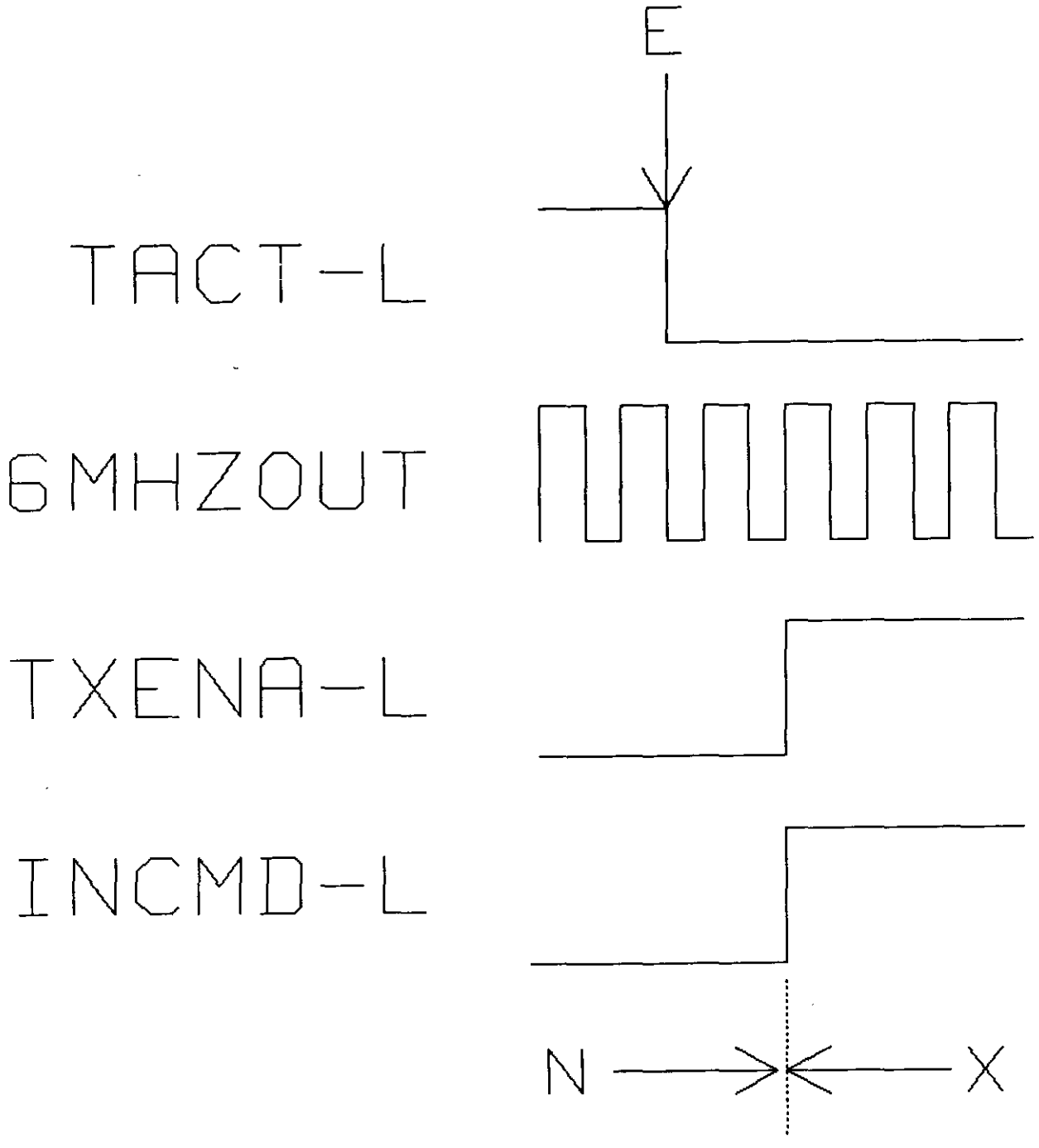


FIGURE 7f. Receive two words (8 bit mode).

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NOTES:

1. These timing diagrams represent the sequence of events that occur on a typical receive two word command. The number of data words could vary from 1 to 32 and all that will be repeated is sequence J-F-G-H-J or sequence J-F-G-I-J. If error conditions exist, the message servicing is aborted as shown on figure 8. If the command word is a mode command to "synchronize with data", the sequence is the same as to receive one word except that GBR-L signal is masked and a new signal, GSR-L, is generated identical to GBR-L. These diagrams reflect an 8 bit data bus structure.

2. Dashed lines (referenced by (2)) refer to the use of a broadcast receive command. Note that the status word is not output on the 1553 data bus.

3. Dashed lines (referenced by (3)) represent the sequence of events that occur if DTREQ-L is asserted and T1-L comes in prior to or at the same time as DTGRT-L. Once set, the HSFAIL-L signal remains set until cleared by the next NBGT-L or RST2-L. The maximum recommended response time of DTGRT-L to DTREQ-L for a transfer to the subsystem is:

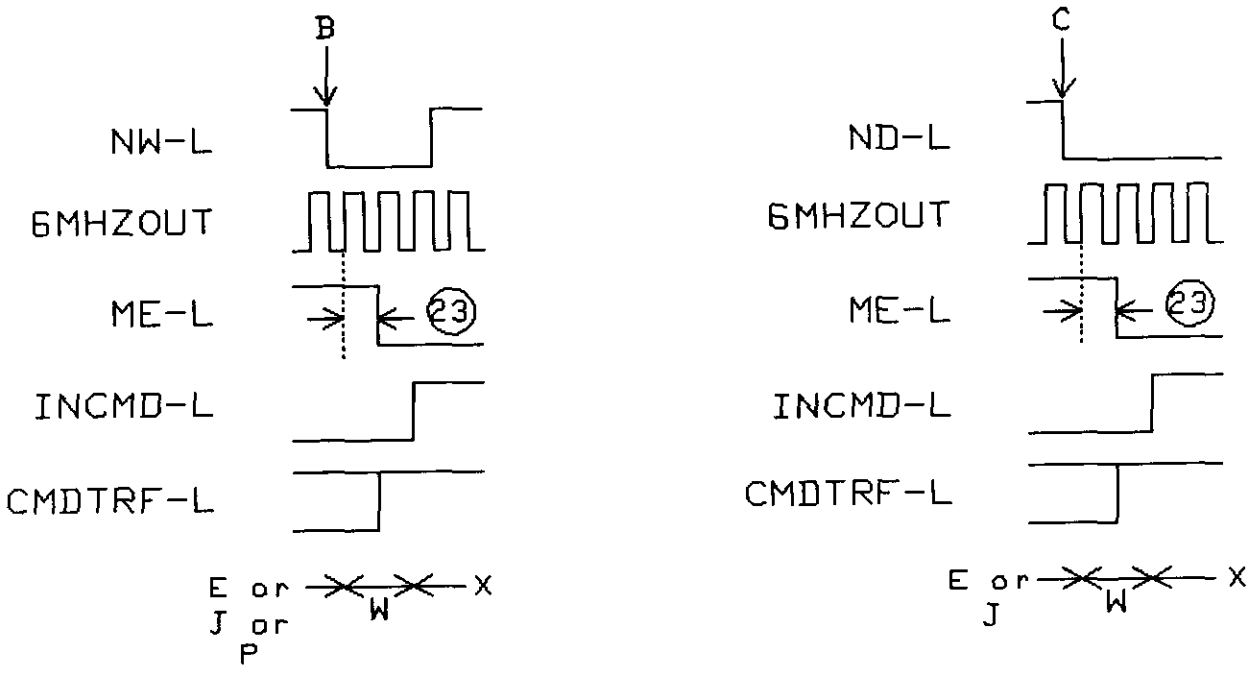
$\leq$ 1 uS for COMMAND (B)
$\leq$ 2.33 uS for DATA (G)

4. The good block received signal (GBR-L), which is present only on non-mode receive command, is suppressed if HSFAIL-L is asserted or if the command word is defined as illegal through the use of the ILLCMD-L input. The same is true for GSR-L (good sync received).

5. NBGT-L pulse clears CWC4-CWC0, ME-L and HSFAIL-L signals.

6. Internal signals are shown for reference only.

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NOTE:

1. The timing diagrams on figures 7 and 9 represent the sequence of events that occur on a typical receive two word command. If error conditions exist, the message servicing is aborted as shown above.

FIGURE 8. Cancellation of message servicing.

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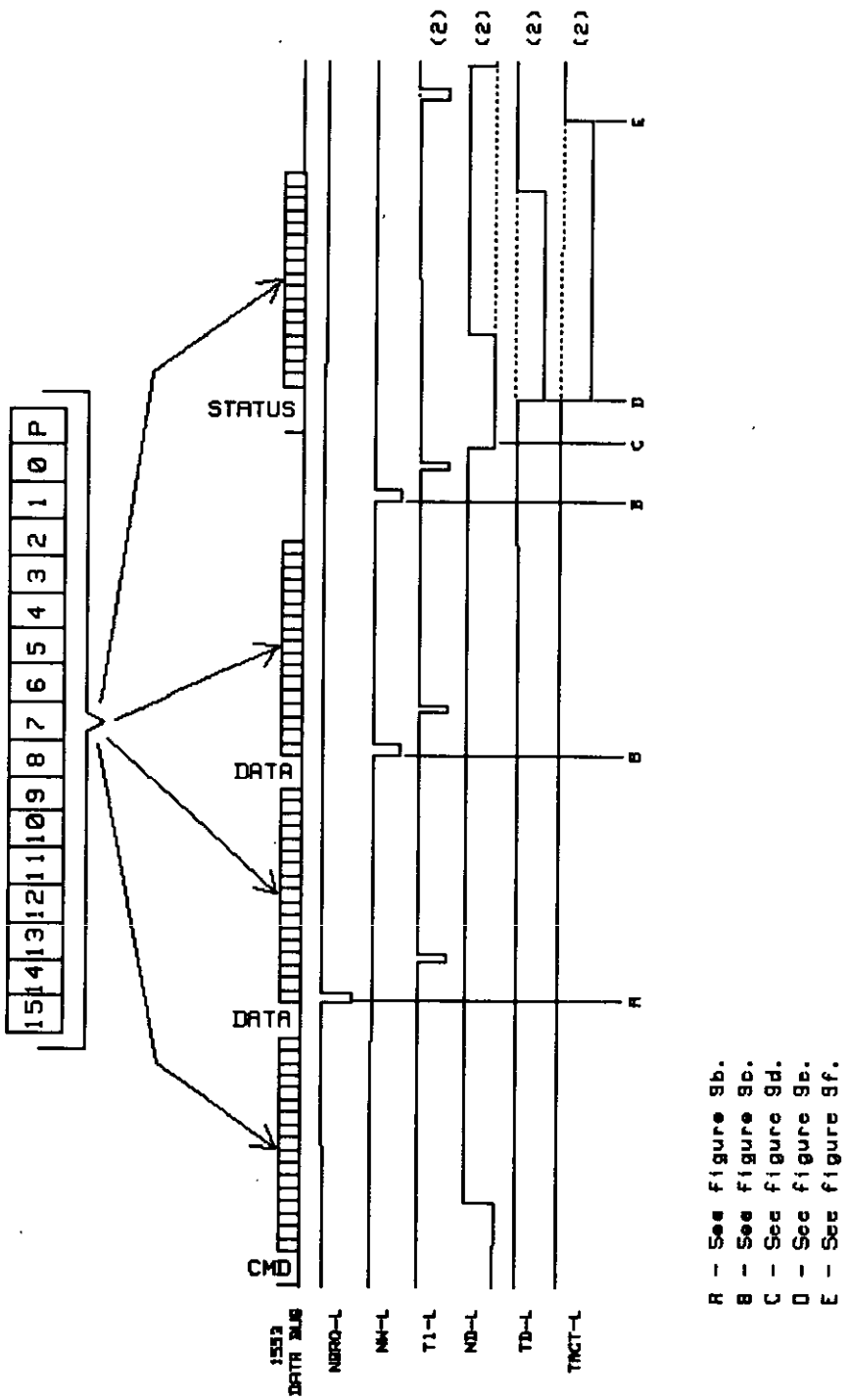


FIGURE 9a. Receive two words (16 bit mode).

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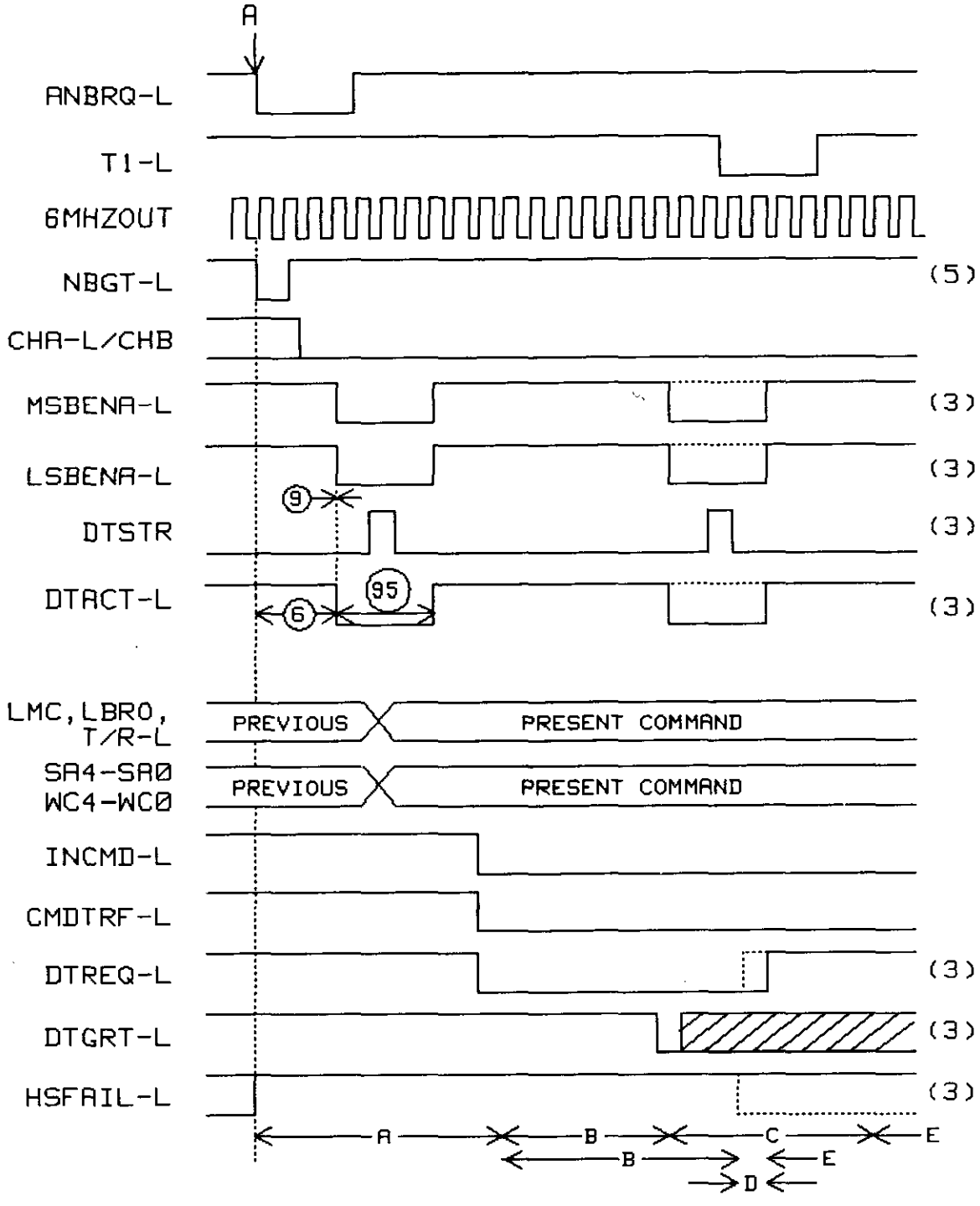


FIGURE 9b. Receive two words (16 bit mode).

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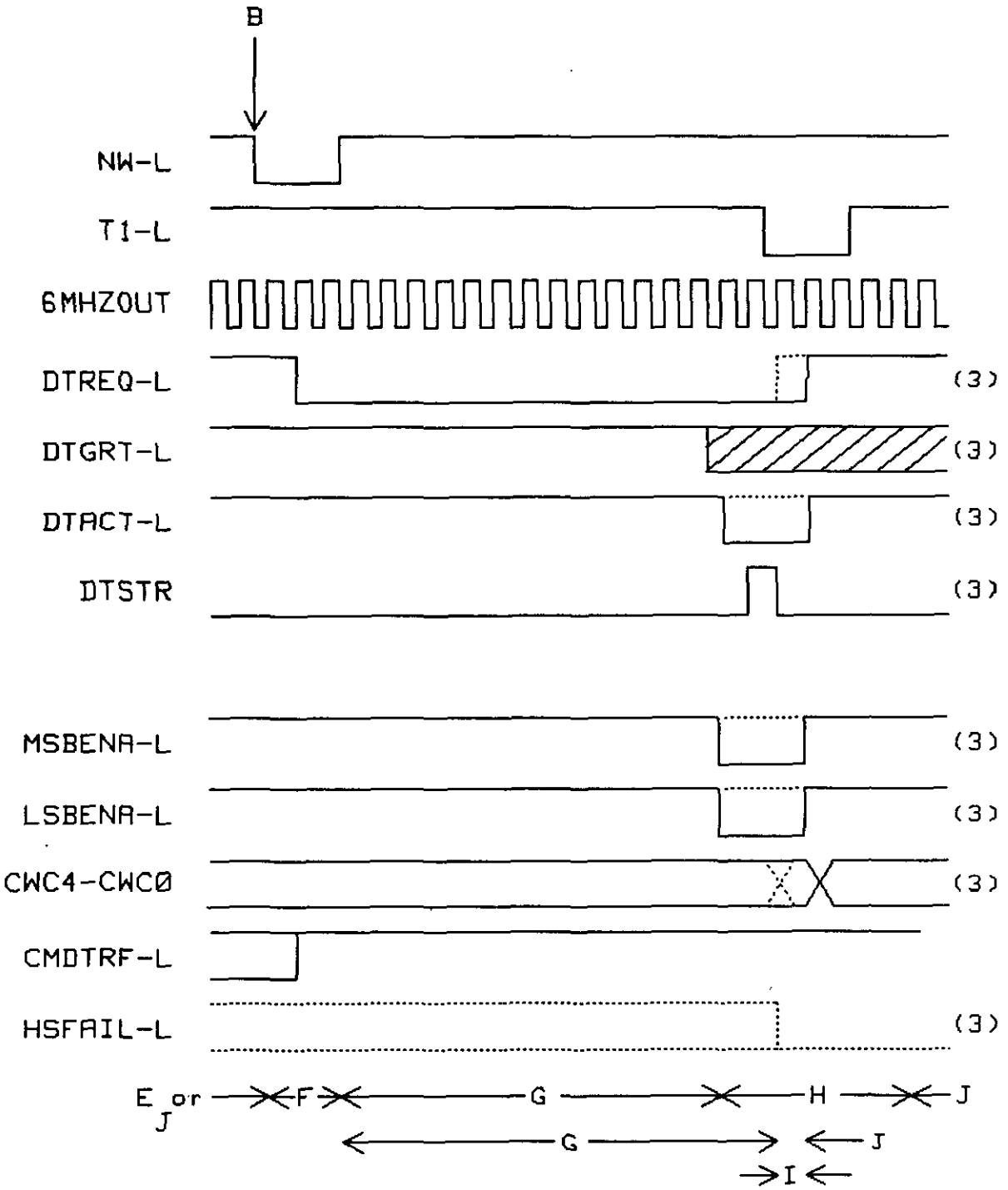


FIGURE 9c. Receive two words (16 bit mode).

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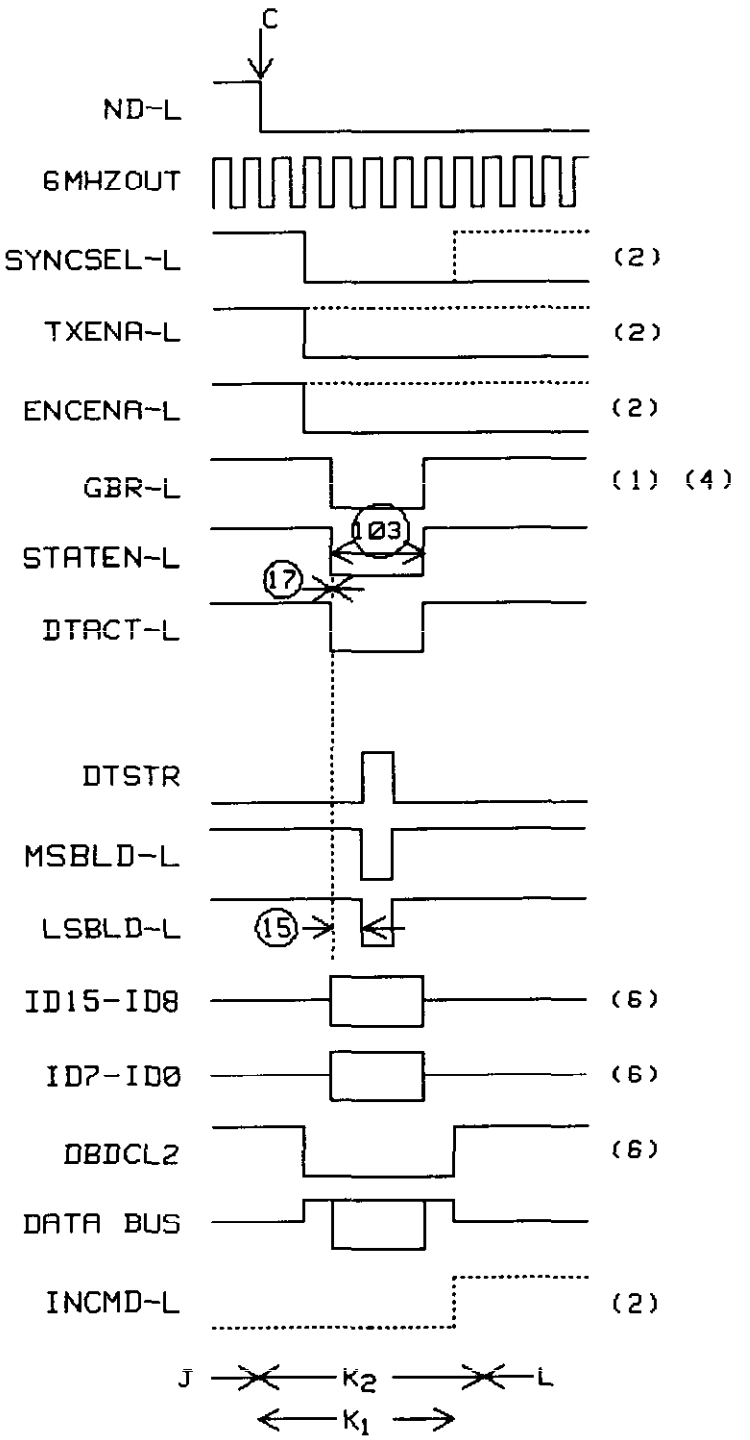


FIGURE 9d. Receive two words (16 bit mode).

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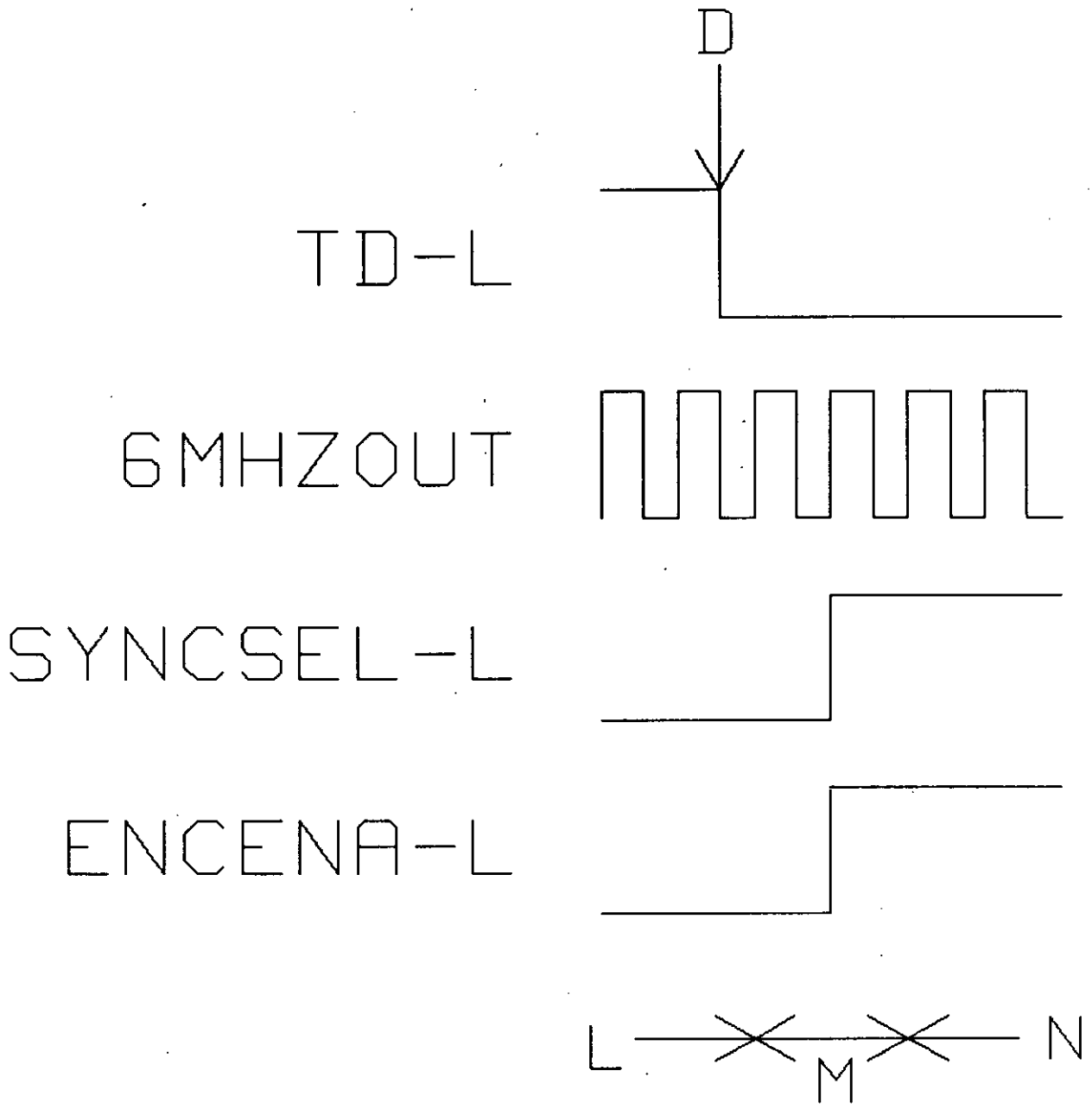


FIGURE 9e. Receive two words (16 bit mode).

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Check the source to verify that this is the current version before use.

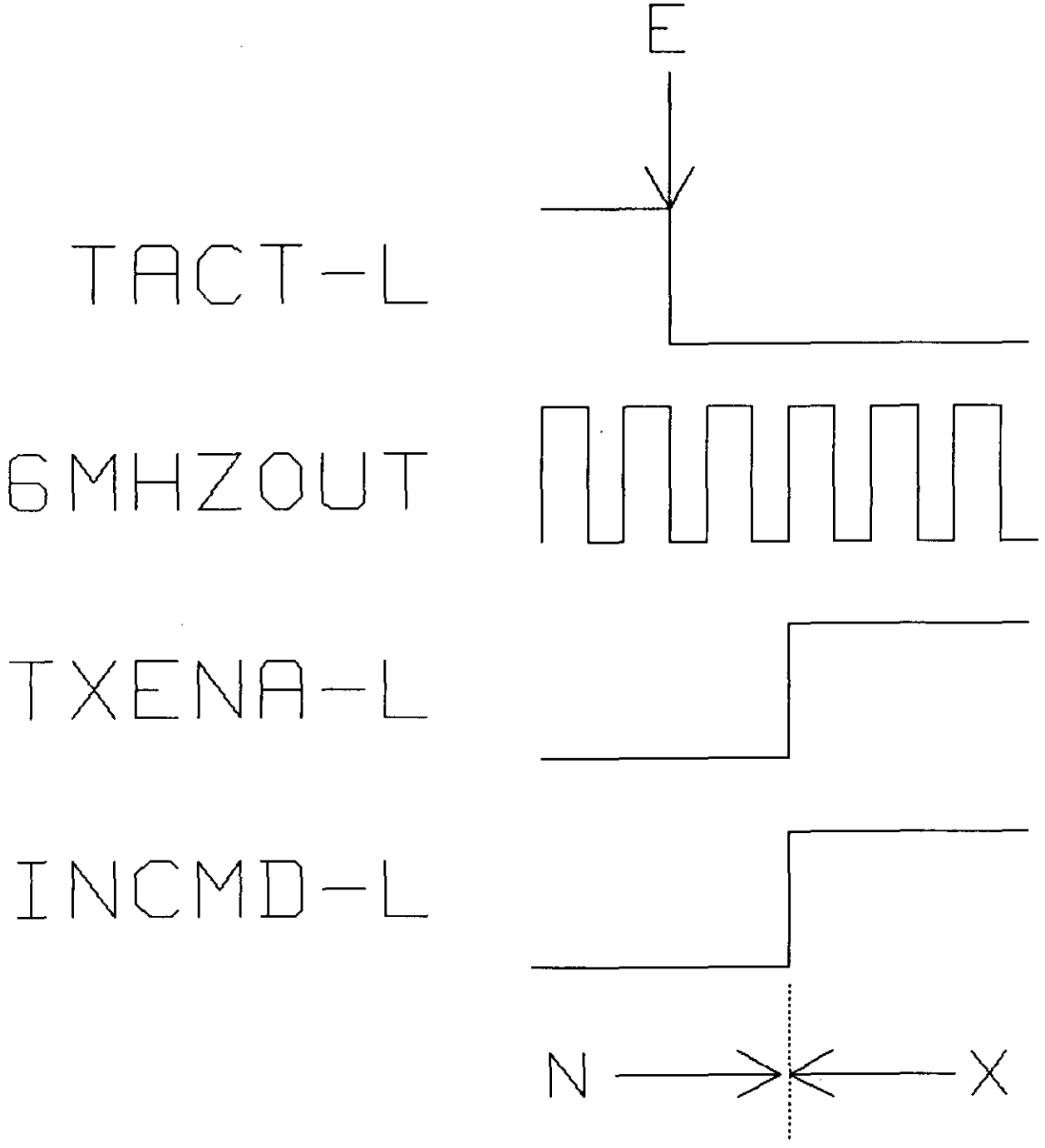


FIGURE 9f. Receive 2 words (16 bit mode).

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NOTES:

1. These timing diagrams represent the sequence of events that occur on a typical receive two word command. The number of data words could vary from 1 to 32 and all that will be repeated is sequence J-F-G-H-J or sequence J-F-G-I-J. If error conditions exist, the message servicing is aborted as shown on figure 8. If the command word is a mode command to "synchronize with data", then the sequence is the same as to receive one word with the exception that GBR-L signal is masked and a new signal, GSR-L, will be generated identical to GBR-L. These diagrams reflect a 16 bit data bus structure.

2. Dashed lines (referenced by (2)) refer to the use of a broadcast receive command. Note that the status word is not output on the 1553 data bus.

3. Dashed lines (referenced by (3)) represent the sequence of events that occur if DTREQ-L is asserted and T1-L comes in prior to or at the same time as DTGRT-L. Once set, the HSFAIL-L signal remains set until cleared by the next NBGT-L or RST2-L. The maximum recommended response time of DTGRT-L to DTREQ-L for a transfer to the subsystem is:

$\leq$ 1.5 uS for COMMAND (B)
$\leq$ 2.33 uS for DATA (G)

4. The good block received signal (GBR-L), which is present only on non-mode receive command, is suppressed if HSFAIL-L is asserted or if the command word is defined as illegal through the use of the ILLCMD-L input. The same is true for GSR-L (good sync received).

5. NBGT-L pulse clears CWC4-CWC0, ME-L and HSFAIL-L signals.

6. Internal signals are shown for reference only.

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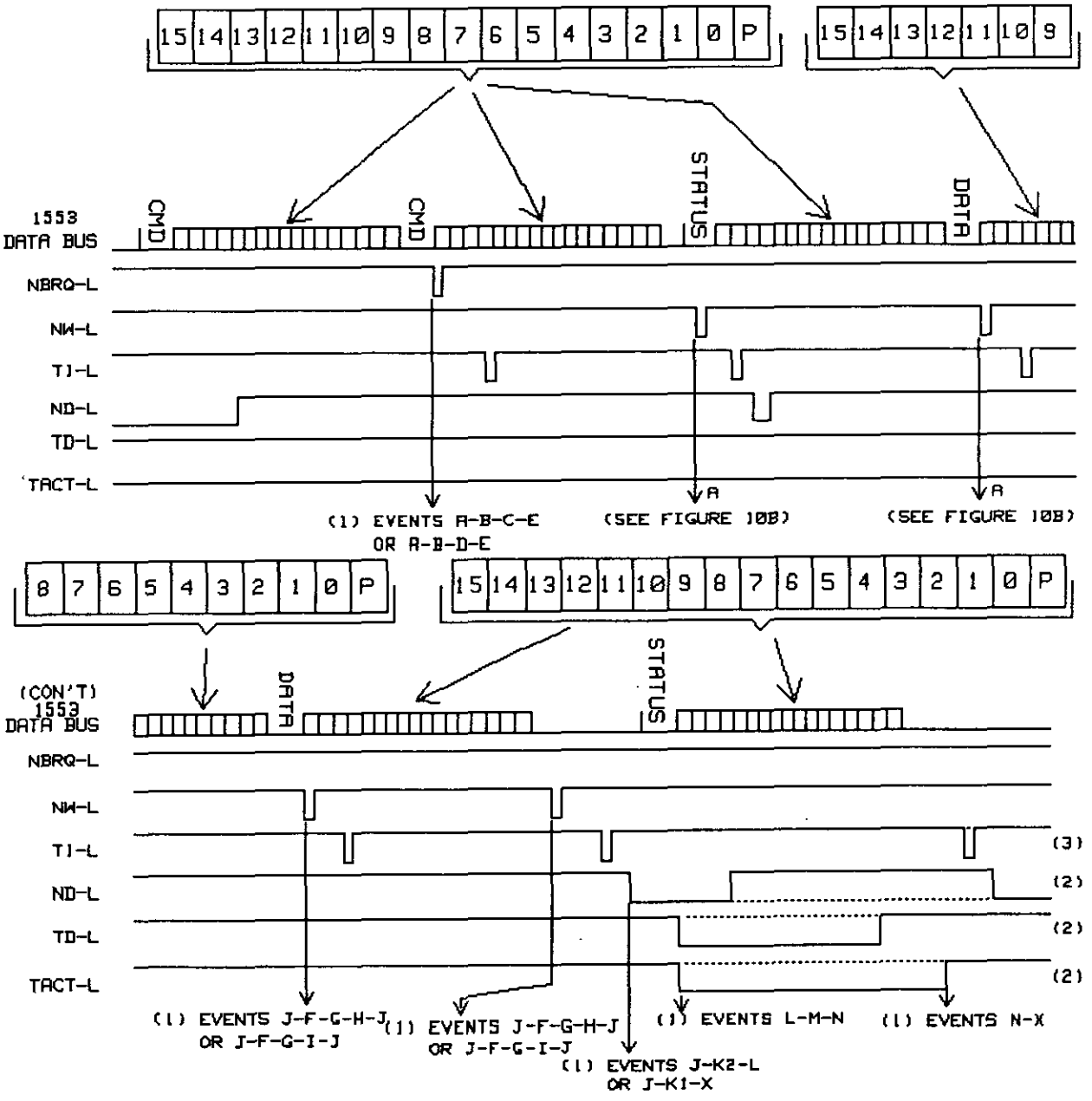


FIGURE 10a. RT-RT transfer to receive two words.

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Check the source to verify that this is the current version before use.

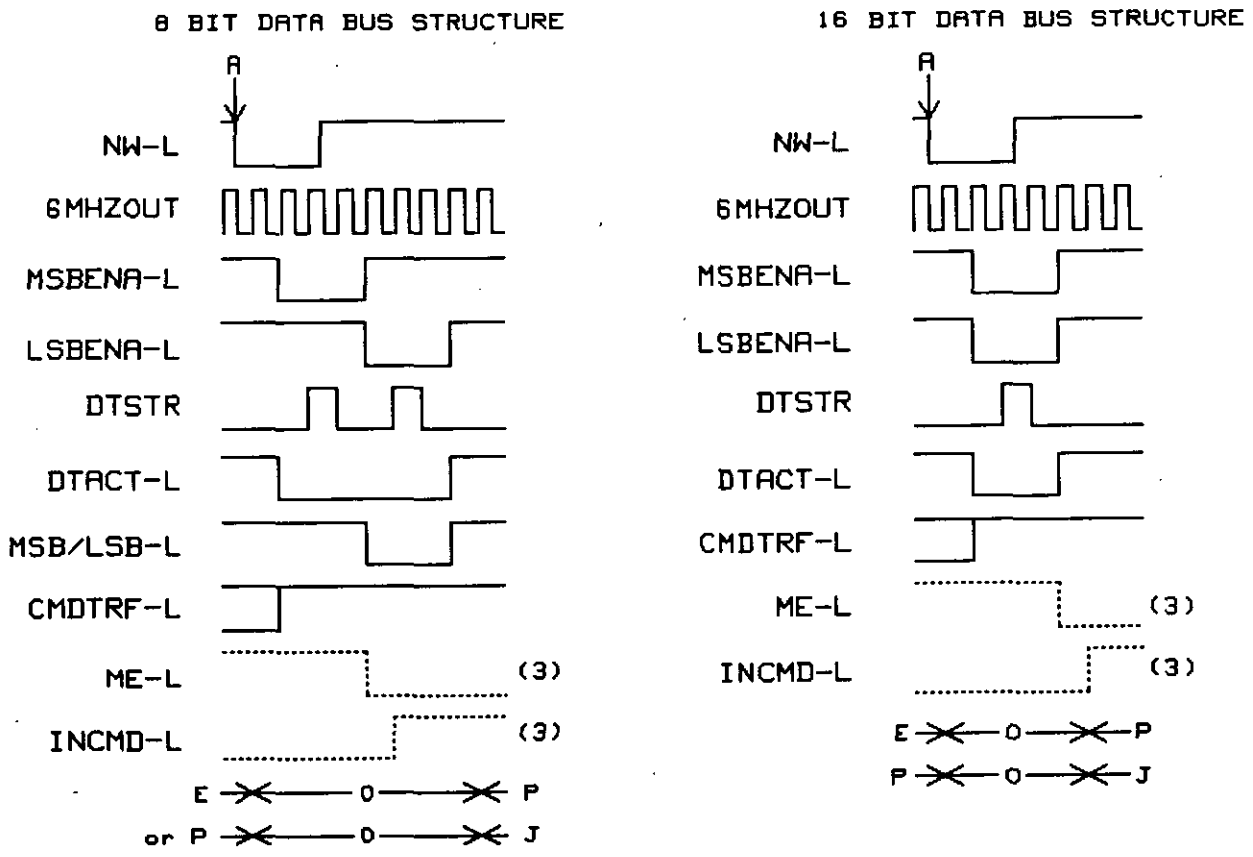


FIGURE 10b. RT-RT transfer to receive two words.

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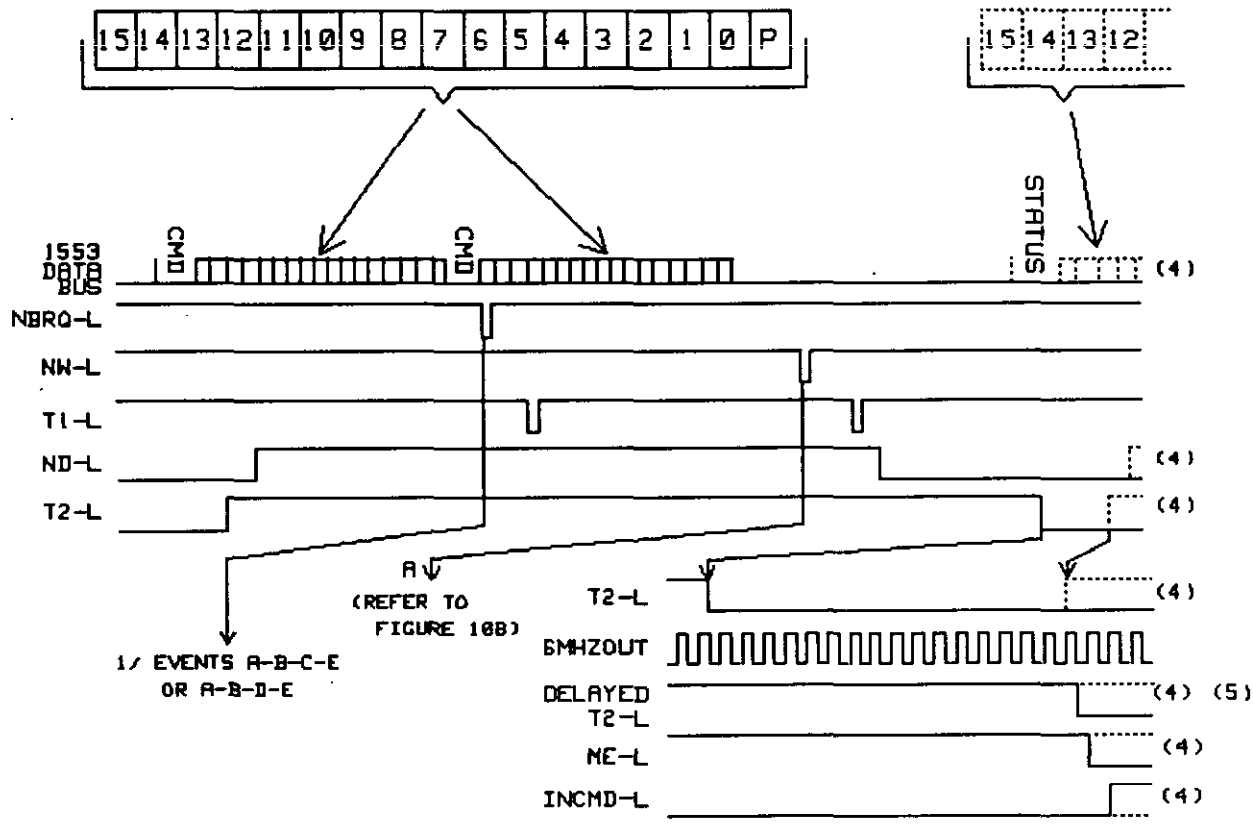
NOTES:

1. All referenced events refer to identical sequences on the "receive two words" timing diagram (figures 7 and 9).

2. Dashed lines (referenced by (2)) refer to the use of a broadcast receive command. Note that the status word is not output on the 1553 data bus.

3. Dashed lines (referenced by (3)) represent the sequence of events that occur if the T/R-L bit is low on the first transfer or if the responding RT status word address does not match the address of the RT told to transmit on the second transfer. Note: if this occurs, all future sequences are aborted.

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NOTES:

1. All referenced events refer to identical sequences on the "receive two words" timing diagram (figures 7 and 9).

2. Dashed lines (referenced by (4)) represent the latest time the transmitting RT could respond without causing a timeout. That time is 19.5 uS measured from the zero-crossing of the parity bit of the last command word to the zero-crossing of the status sync.

3. Internal signal is shown for reference only.

FIGURE 11. RT-RT transfer to receive without 2nd RT response.

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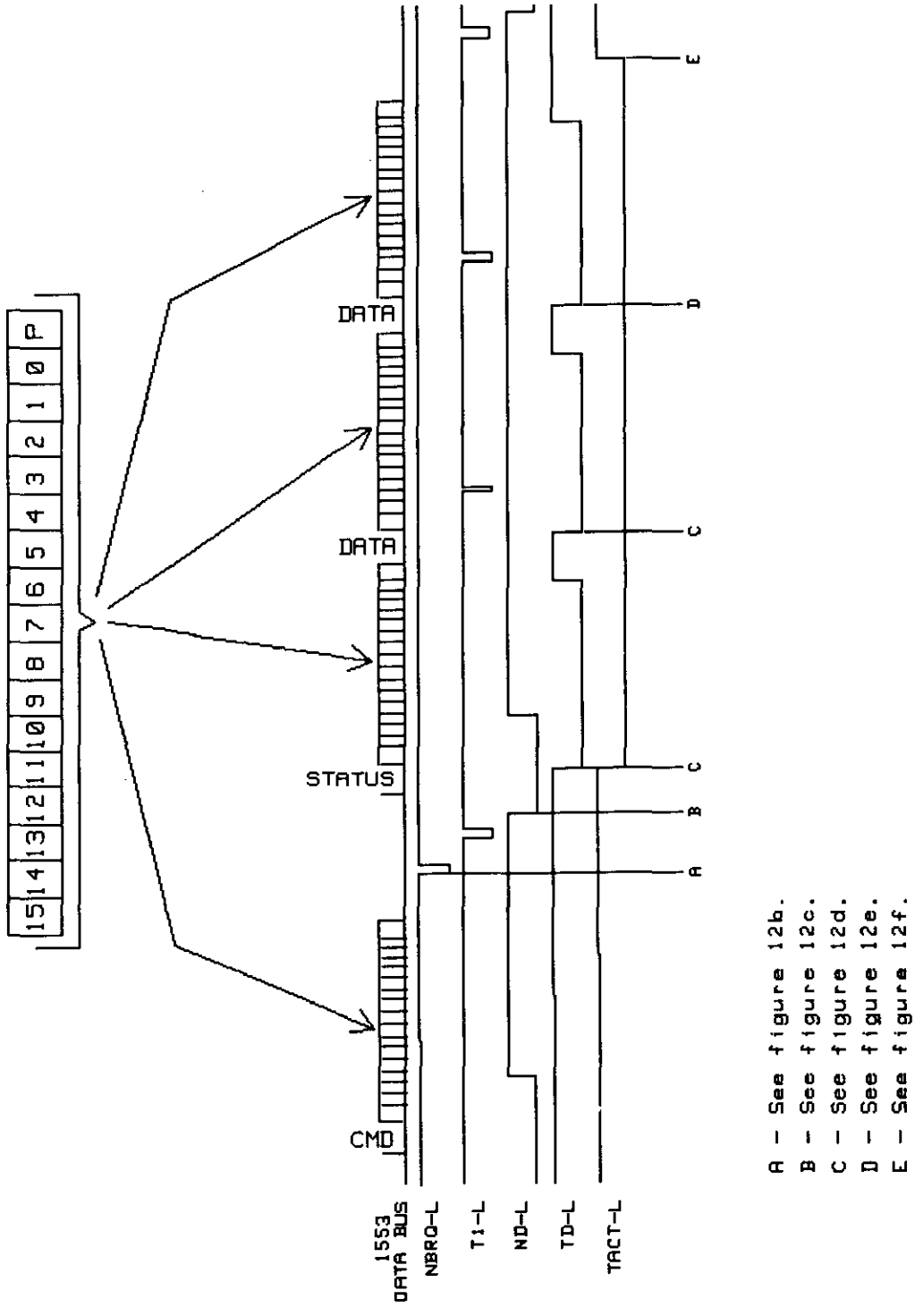


FIGURE 12a. Transmit two words (8 bit mode).

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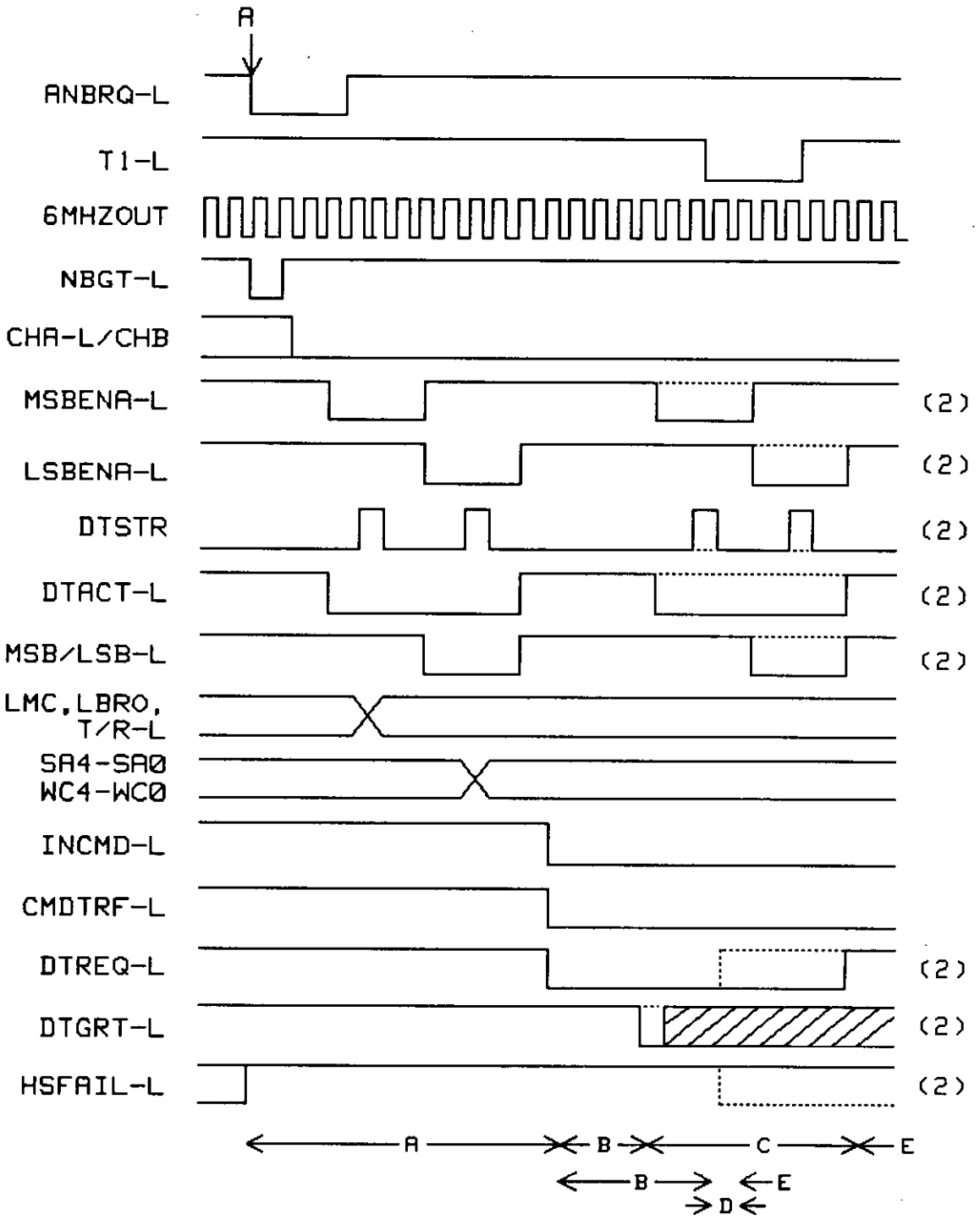


FIGURE 12b. Transmit two words (8 bit mode).

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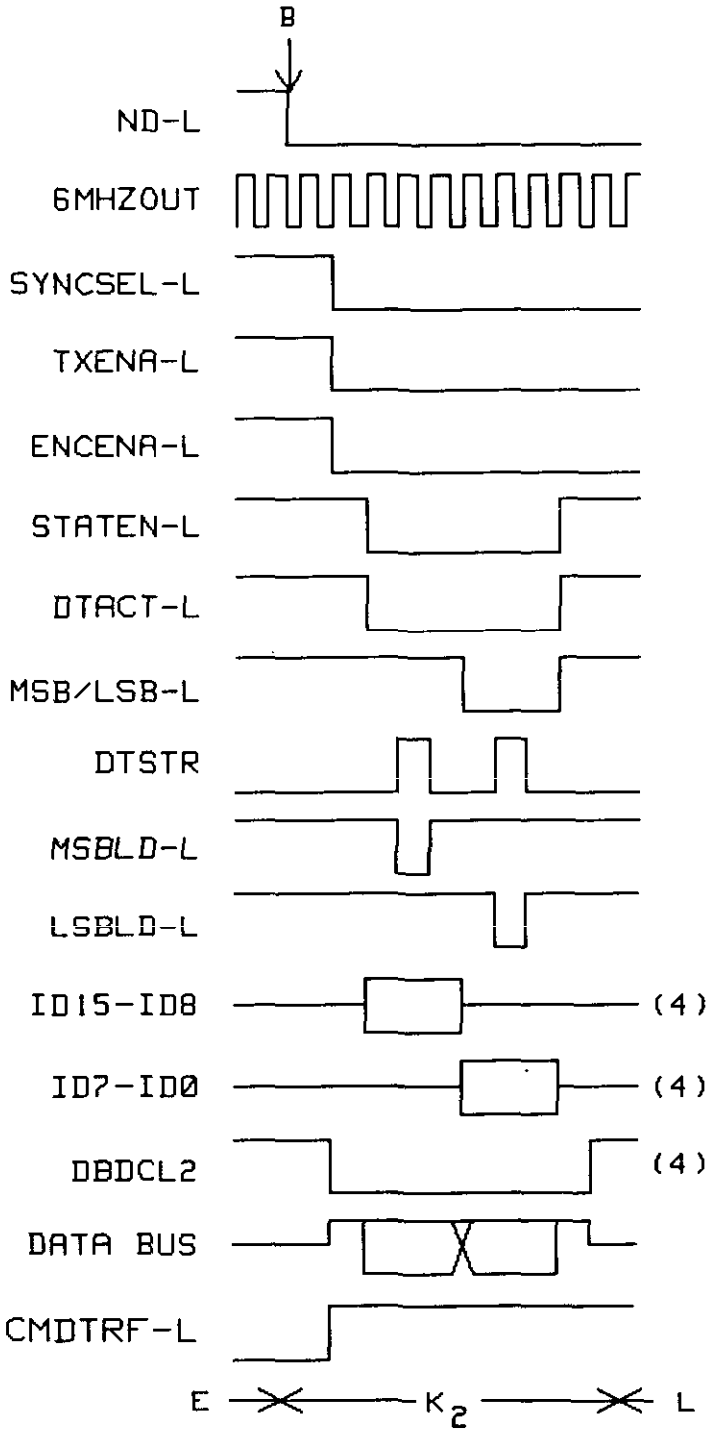


FIGURE 12c. Transmit two words (8 bit mode).

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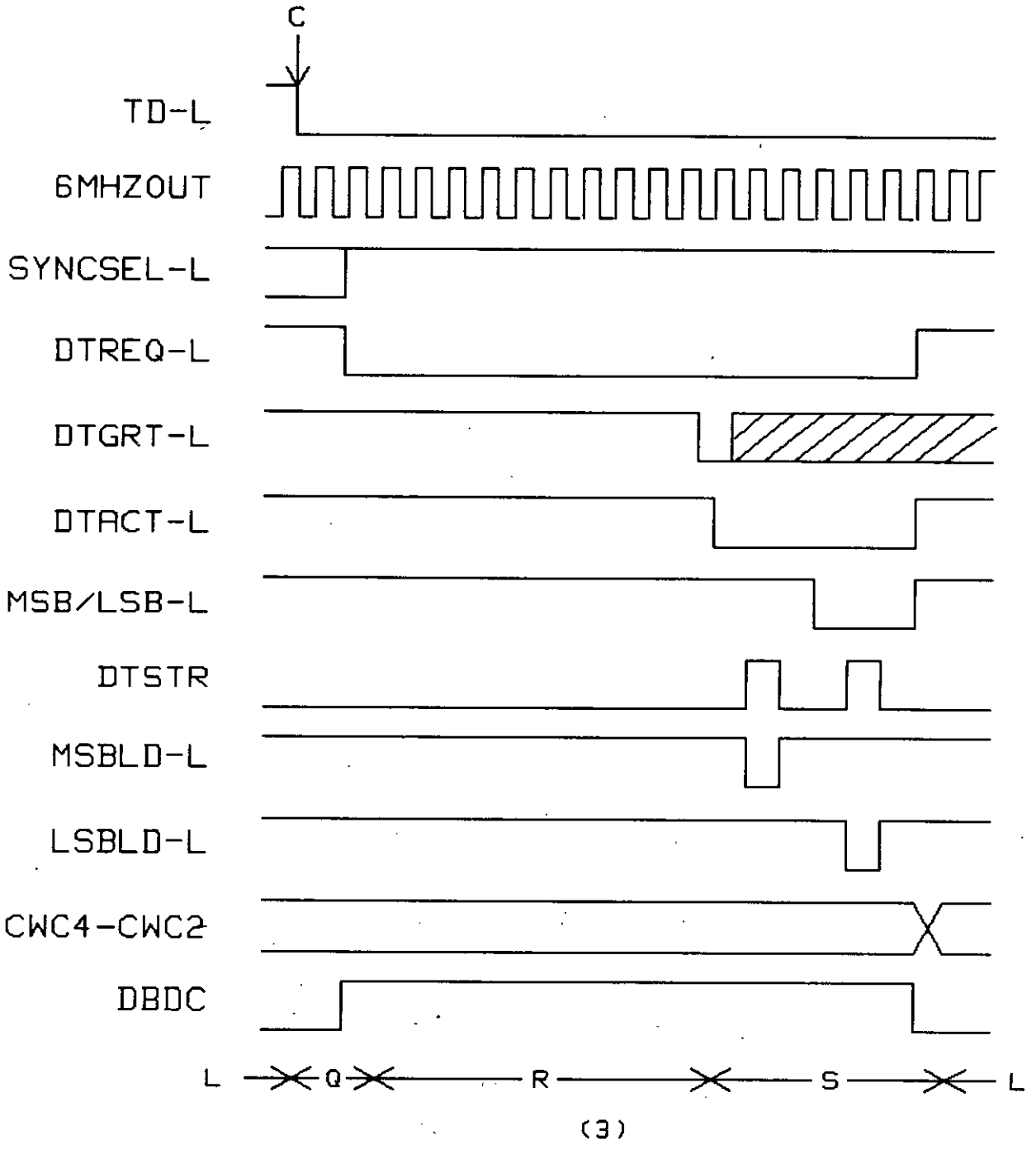


FIGURE 12d. Transmit two words (8 bit mode).

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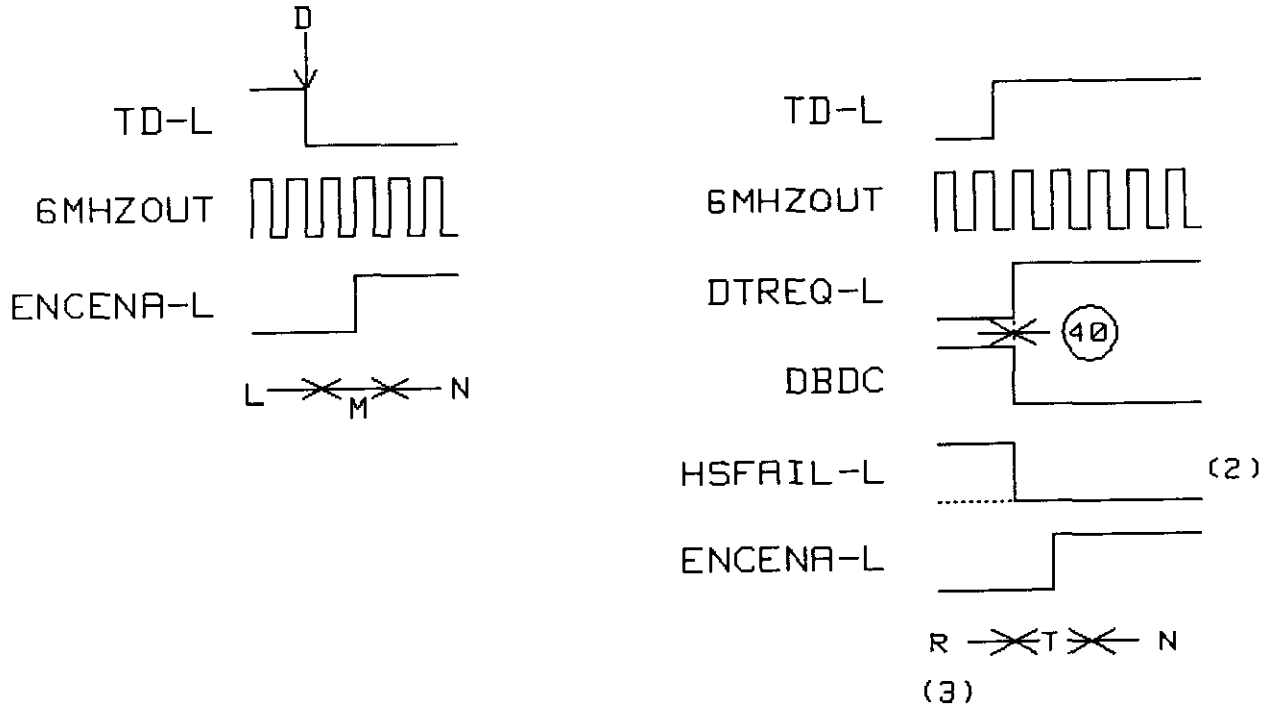


FIGURE 12e. Transmit two words (8 bit mode).

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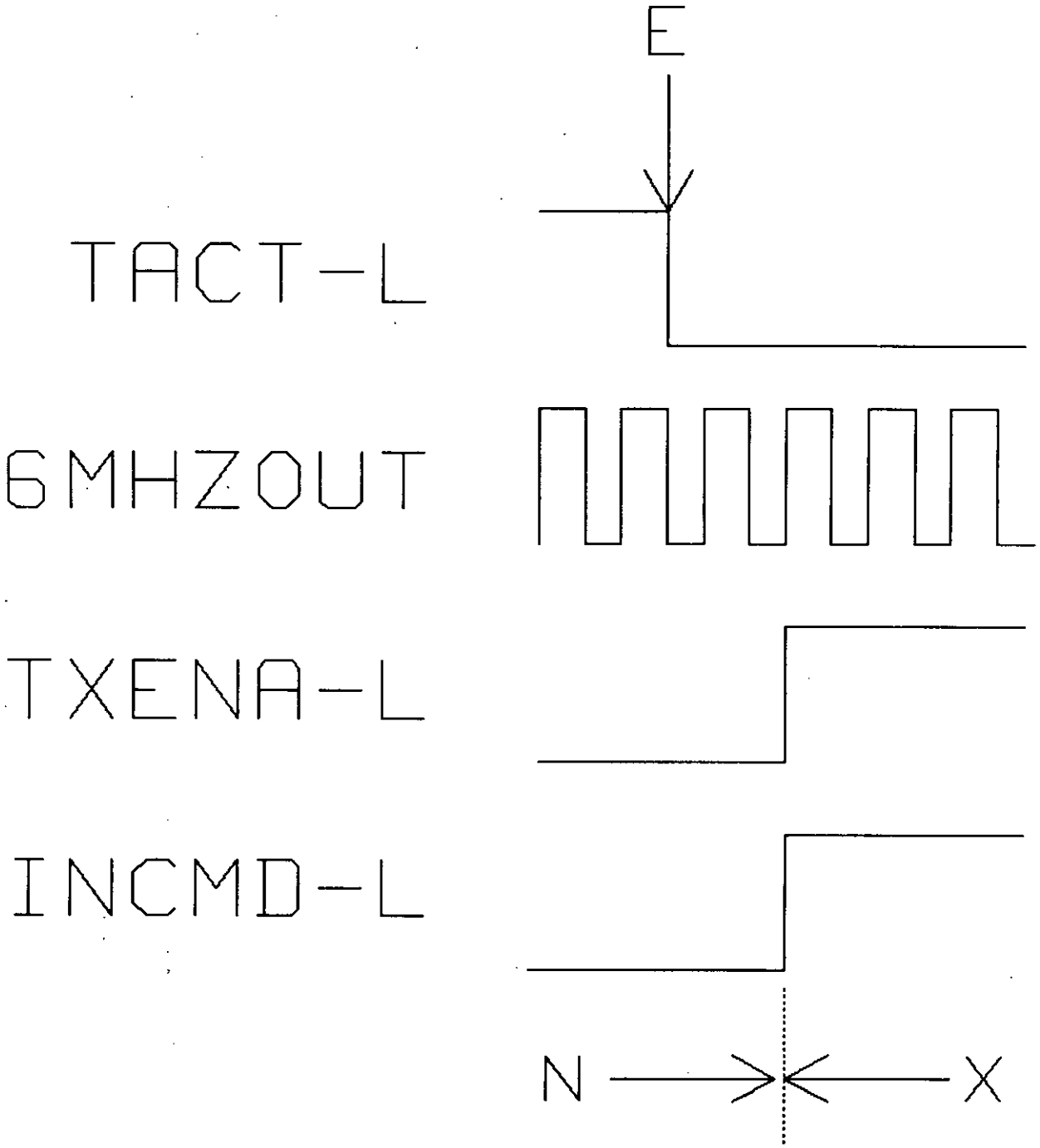


FIGURE 12f. Transmit two words (8 bit mode).

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NOTES:

1. These timing diagrams represent the sequence of events that occur on a typical transmit two word command. The number of data words could vary from 1 to 32 and all that will be repeated is sequence L-Q-R-S-L. If an error condition exists, the message servicing is aborted upon receipt of ND-L (or NW-L if an extra word is sent with the command word). Once the status word has started to be output on the 1553 data bus, and provided the command word is not defined as illegal (through use of the ILLCMD-L input signal) or the busy bit set in the status word, then the unit requests a data word from the subsystem (refer to sequence L-Q-R-S-L). If the command word is defined as illegal or the busy bit is set, sequence L-G-R-S-L is omitted and no data words are sent. If the command word is a mode command to "transmit vector word", the sequence of events is the same as to transmit one word. These timing diagrams reflect an 8 bit data bus structure.

2. Dashed lines (referenced by (2)) represent a handshake failure on the command word transfer to the subsystem.

3. If DTREQ-L is asserted (to request a data word transfer from the subsystem) and TD-L goes high prior to or at the same time that DTGRT-L goes low, then a handshake failure occurs (refer to sequence R-T-N) and all further data word transmissions are aborted. The maximum recommended response time of DTGRT-L to DTREQ-L for a transfer from the subsystem is $\leq$ 15.3 uS (R).

4. Internal signals are shown for reference only.

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Check the source to verify that this is the current version before use.

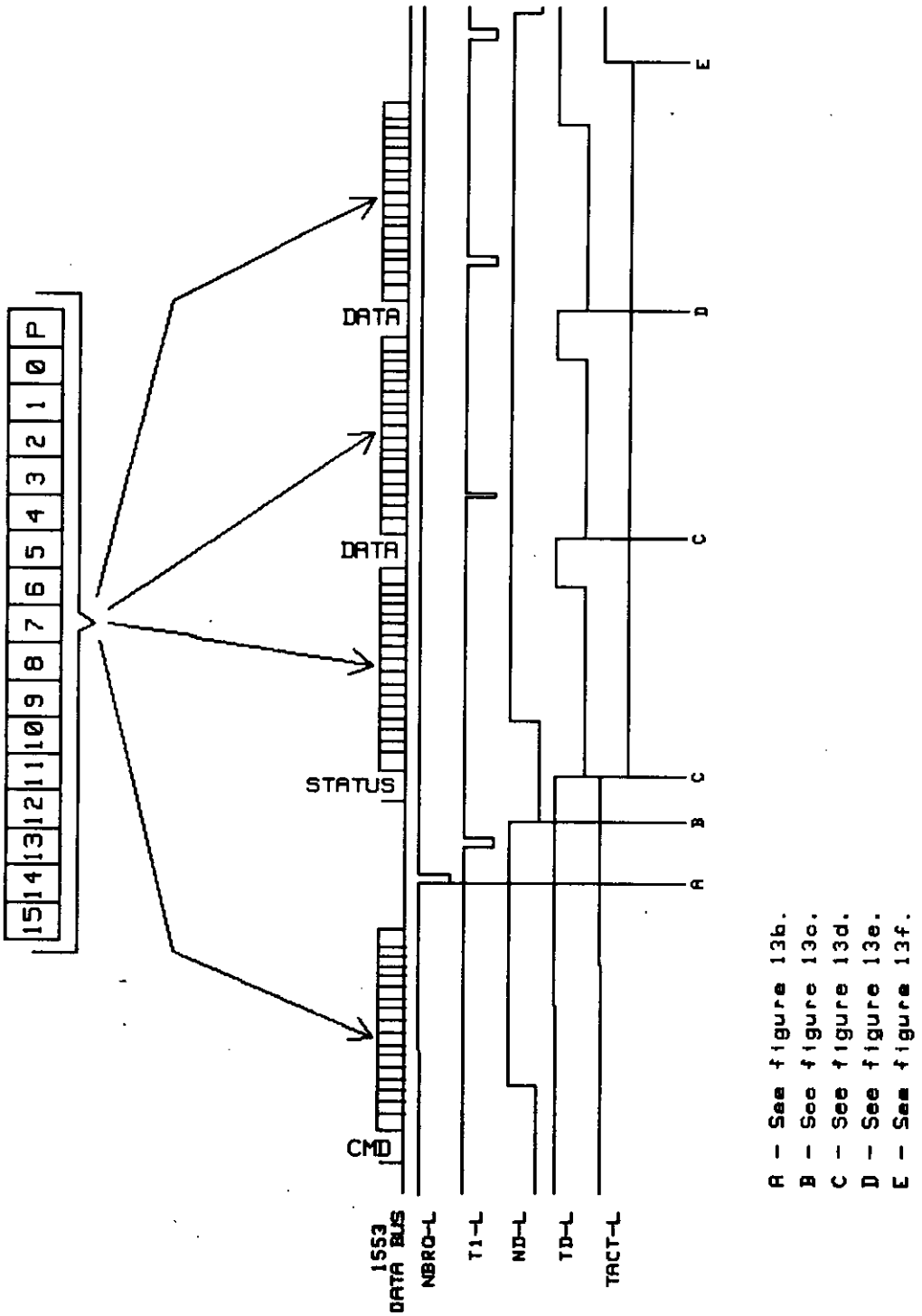


FIGURE 13a. Transmit two words (16 bit mode).

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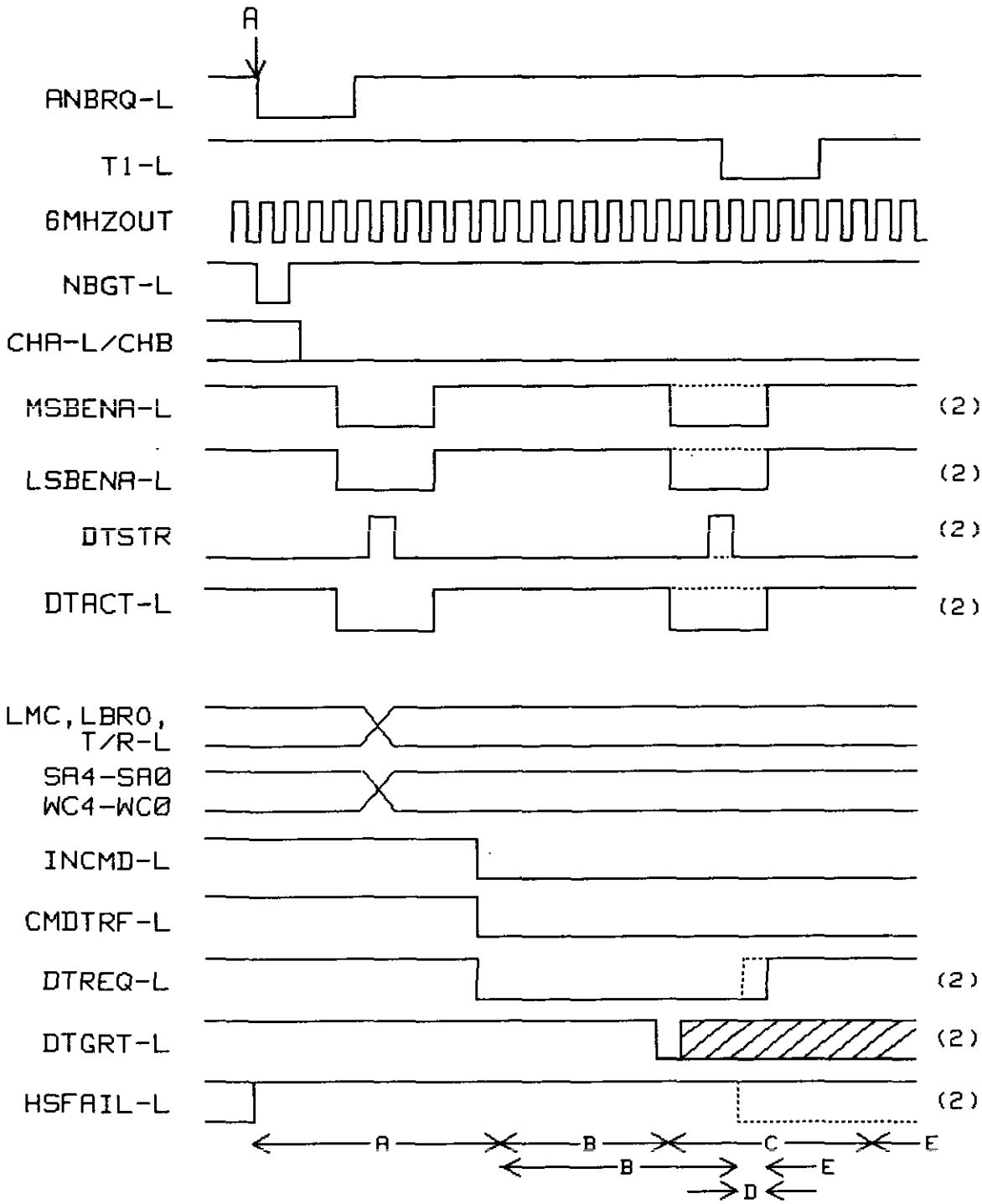


FIGURE 13b. Transmit two words (16 bit mode).

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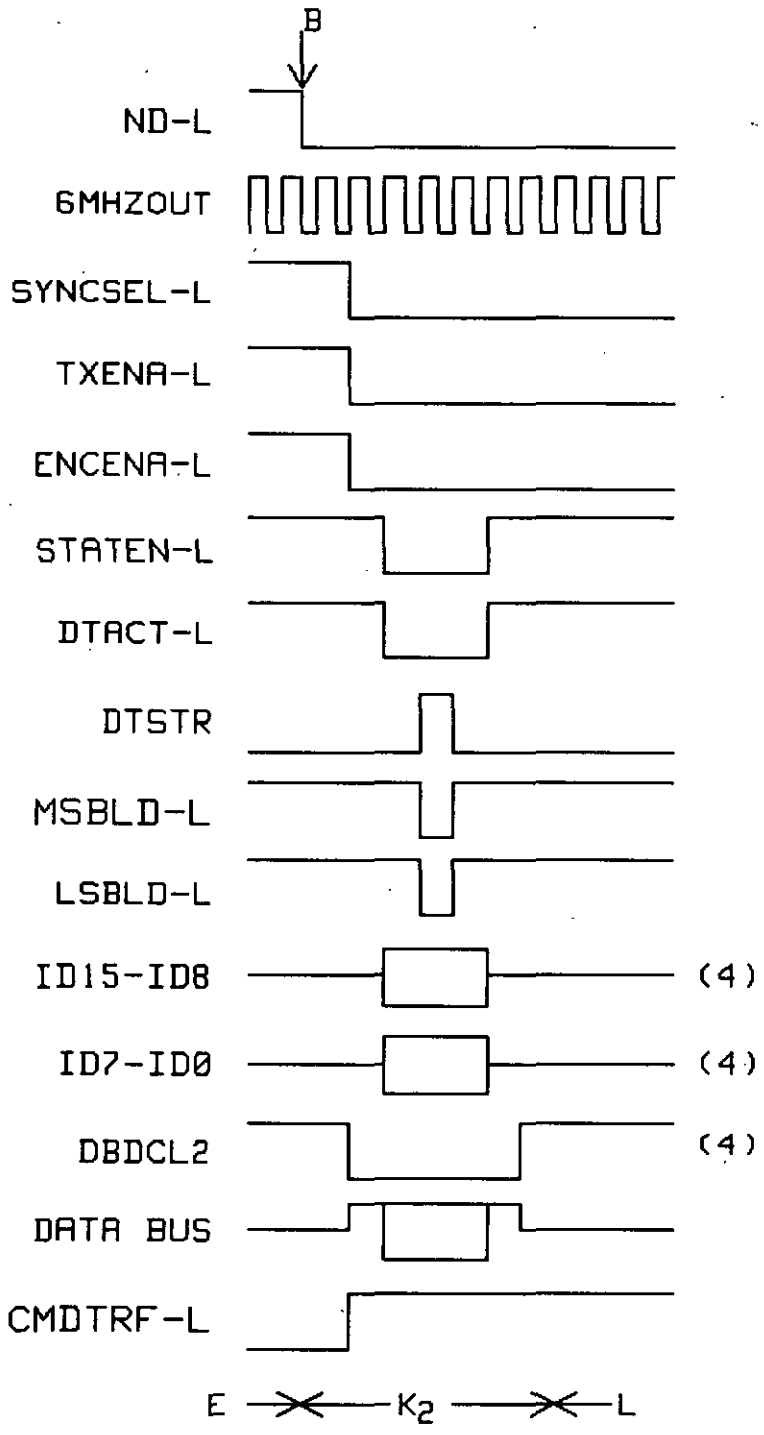


FIGURE 13c. Transmit two words (16 bit mode).

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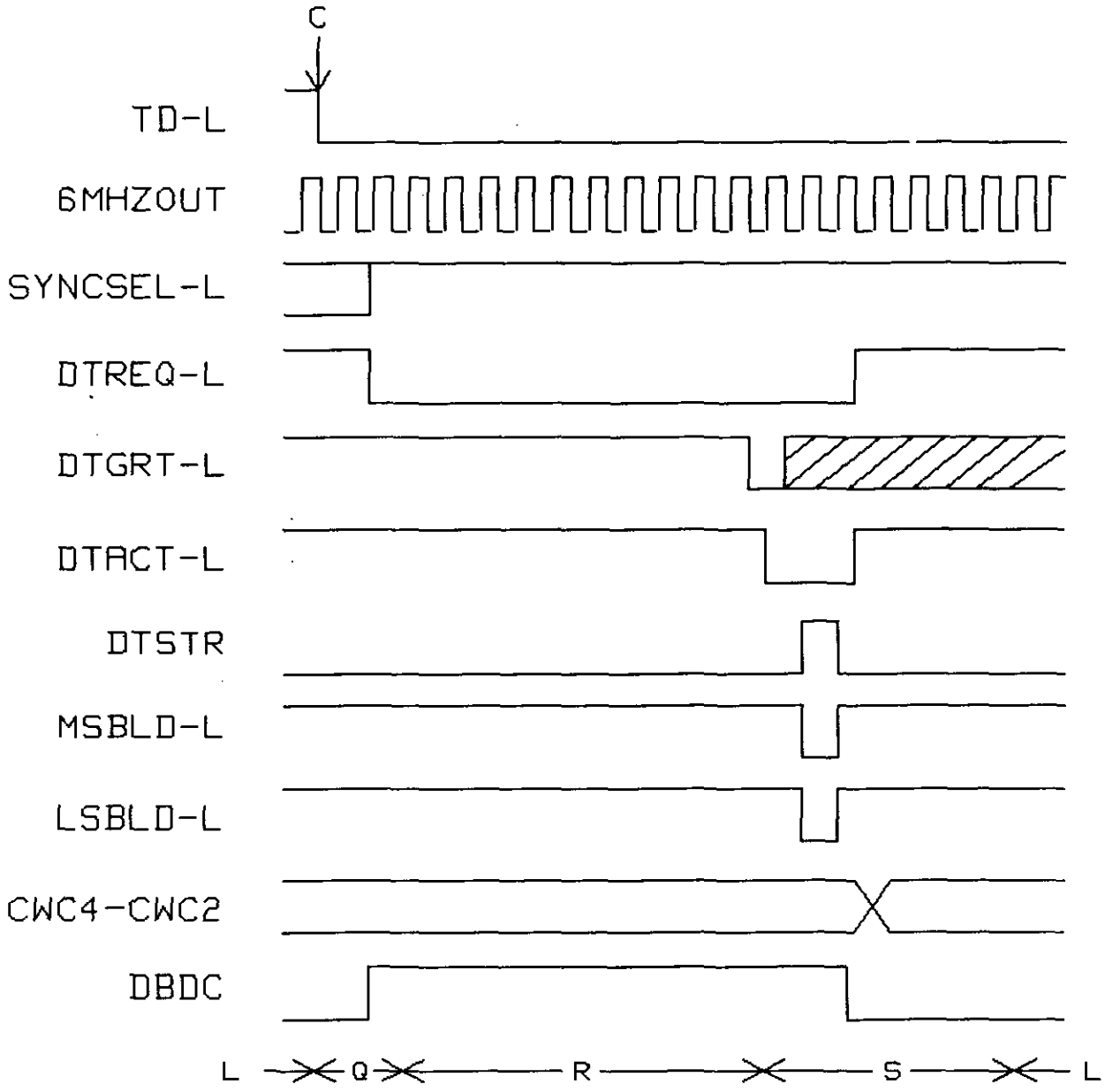


FIGURE 13d. Transmit two words (16 bit mode).

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Check the source to verify that this is the current version before use.

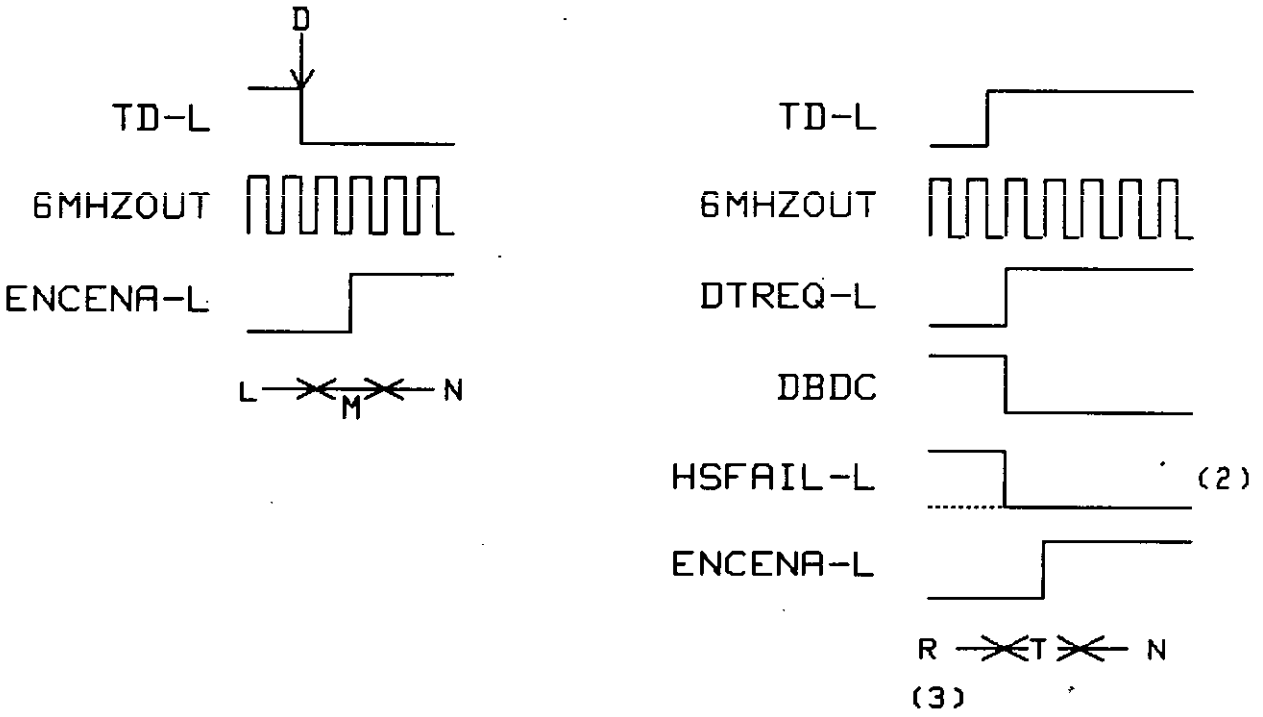


FIGURE 13e. Transmit two words (16 bit mode).

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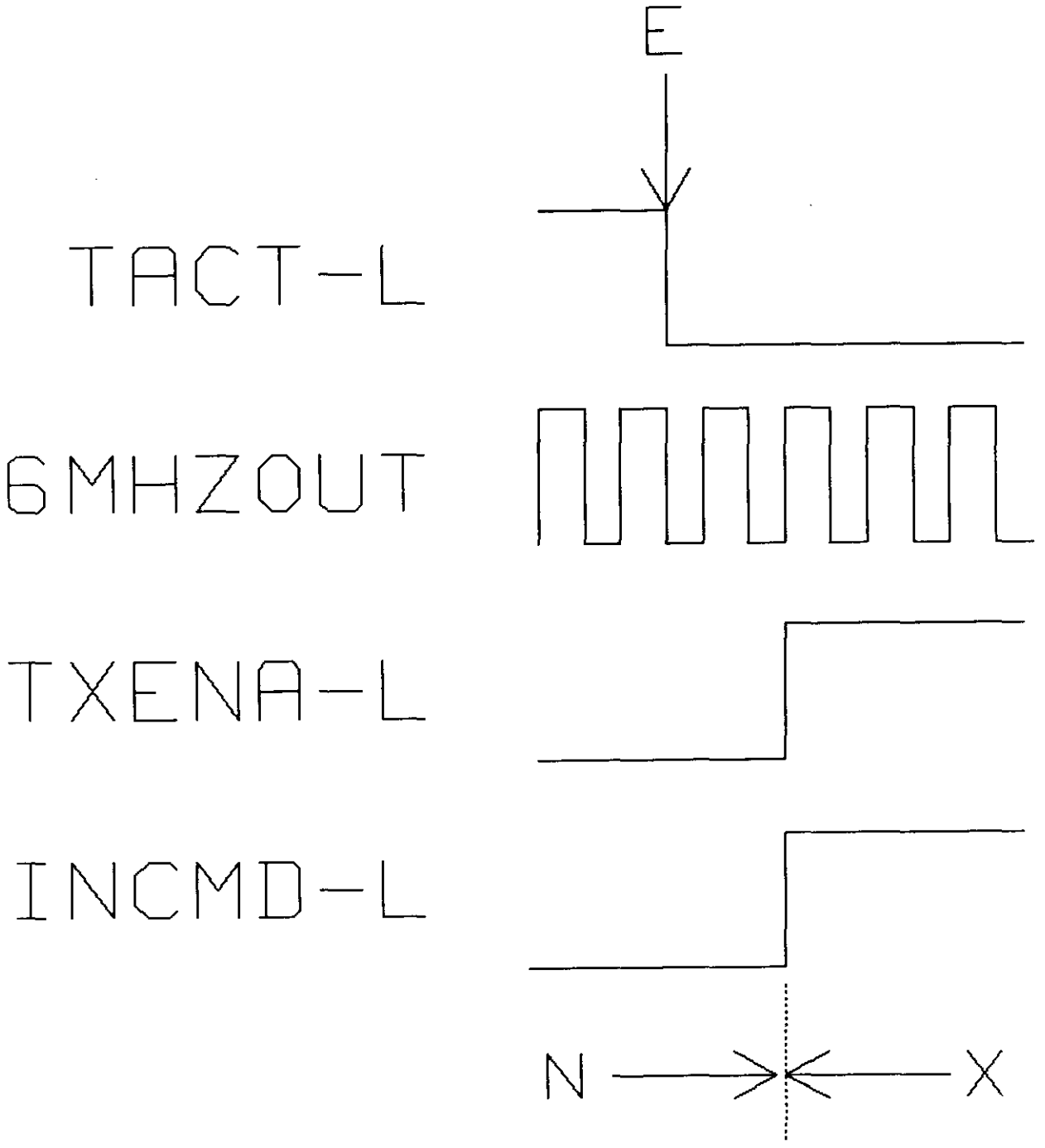


FIGURE 13f. Transmit two words (16 bit mode).

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NOTES:

1. These timing diagrams represent the sequence of events that occur on a typical transmit two word command. The number of data words could vary from 1 to 32 and all that shall be repeated is sequence L-Q-R-S-L. If an error condition exists, the message servicing is aborted upon receipt of ND-L (or NW-L if an extra word is sent with the command word). Once the status word is output on the 1553 data bus, and provided the command word is not defined as illegal (through use of the ILLCMD-L input signal) or the busy bit set in the status word, then the unit requests a data word from the subsystem (refer to sequence L-Q-R-S-L). If the command word is defined as illegal or the busy bit is set, sequence L-G-R-S-L is omitted and no data words are sent. If the command word is a mode command to "transmit vector word", the sequence of events is the same as to transmit one word. These timing diagrams reflect a 16 bit data bus structure.

2. Dashed lines (referenced by (2)) represent a handshake failure on the command word transfer to the subsystem.

3. If DTREQ-L is asserted (to request a data word transfer from the subsystem) and TD-L goes high prior to or at the same time that DTGRT-L goes low, then a handshake failure occurs (refer to sequence R-T-N) and all further data word transmissions are aborted. The maximum recommended response time of DTGRT-L to DTREQ-L for a transfer from the subsystem is $\leq$ 15.3 uS (R).

4. Internal signals are shown for reference only.

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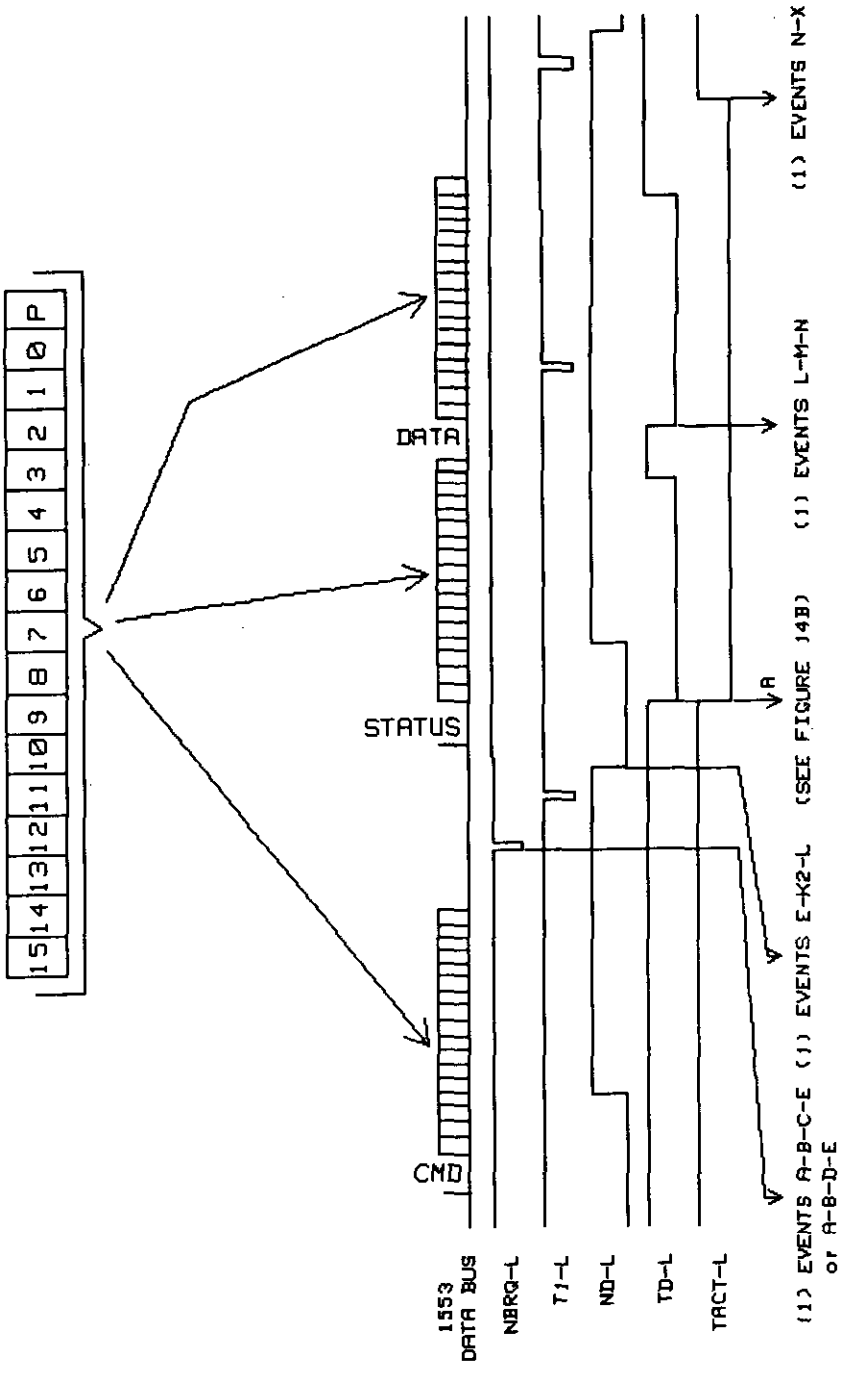


FIGURE 14a. Mode command to "transmit built-in-test word".

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Check the source to verify that this is the current version before use.

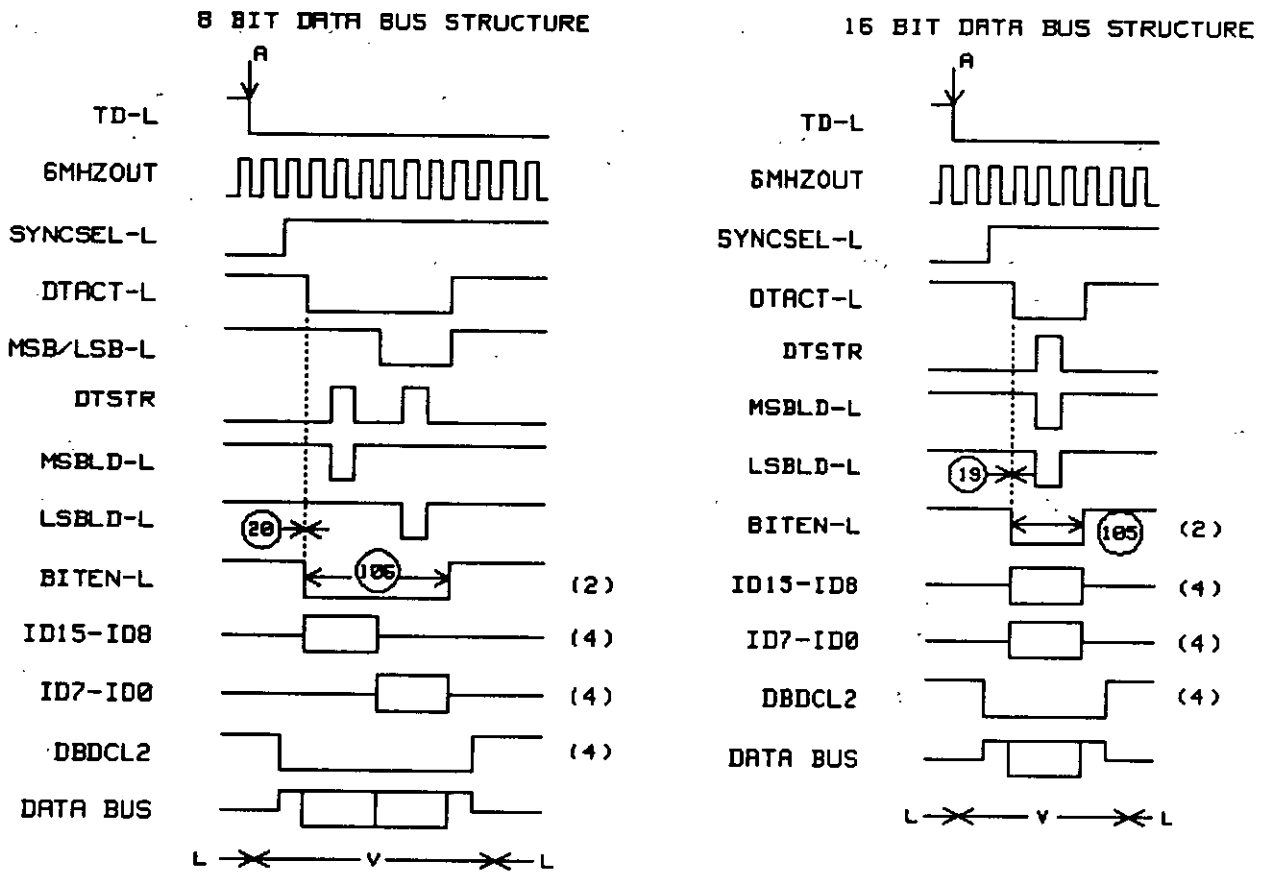


FIGURE 14b. Mode command to "transmit built-in-test word".

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NOTES:

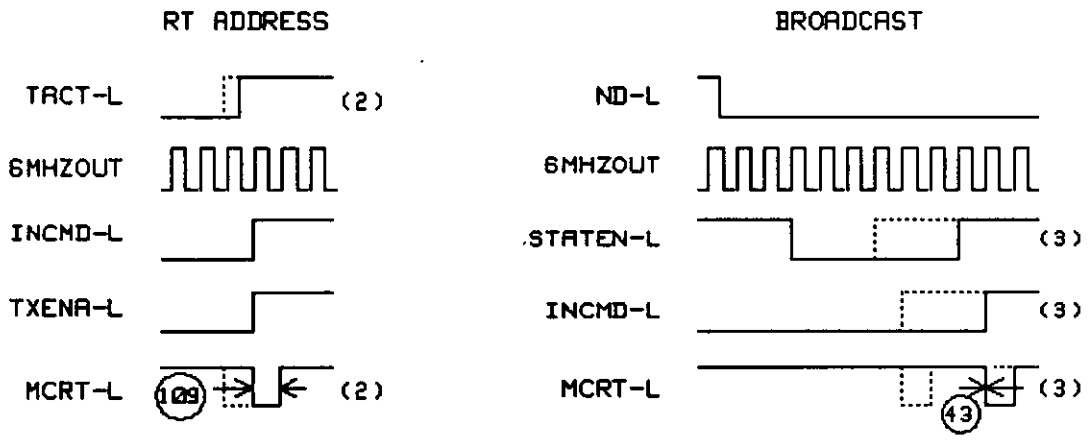
1. All referenced events refer to identical sequences on the "transmit two words" timing diagrams (figures 12 and 13).

2. If command word is a mode command to "transmit last command", the sequence of events is identical except that the output signal BITEN-L is not active (it remains high).

3. The data word is always transmitted on the 1553 data bus regardless of the state of the busy bit in the status register. For a normal transmit command or a mode command to "transmit vectored word", if the busy bit is set, the data word is not transmitted on the 1553 data bus.

4. Internal signals are shown for reference only.

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NOTES:

1. Mode commands without data follow the typical sequence of events as shown on the "receive two words" timing diagrams. The only exception is that sequence E-F-G-H-J is omitted since no data words are expected. The T/R-L bit is set to logic "1". If the command is a mode command to "synchronize" without data, and there is no format error, and the command was not defined as illegal, then a GSR-L pulse is generated in the place of GBR-L, as with the case of the mode command to "synchronize with data".

2. Dashed lines (referenced by (2)) represent a condition that might occur on a mode reset command in the event that TACT-L comes in time to meet the setup time of one internal flip-flop and not the other. All that happens is that MCRT-L pulse would be 330 nS instead of 166 nS.

3. Shown is a broadcast mode reset command for an 8 bit data bus structure. Dashed lines represent a 16 bit data bus structure. All other signals shown on the "receive two words" timing diagram sequence J-K1-X still apply although they are not shown here.

FIGURE 15. Mode command to "reset remote terminal".

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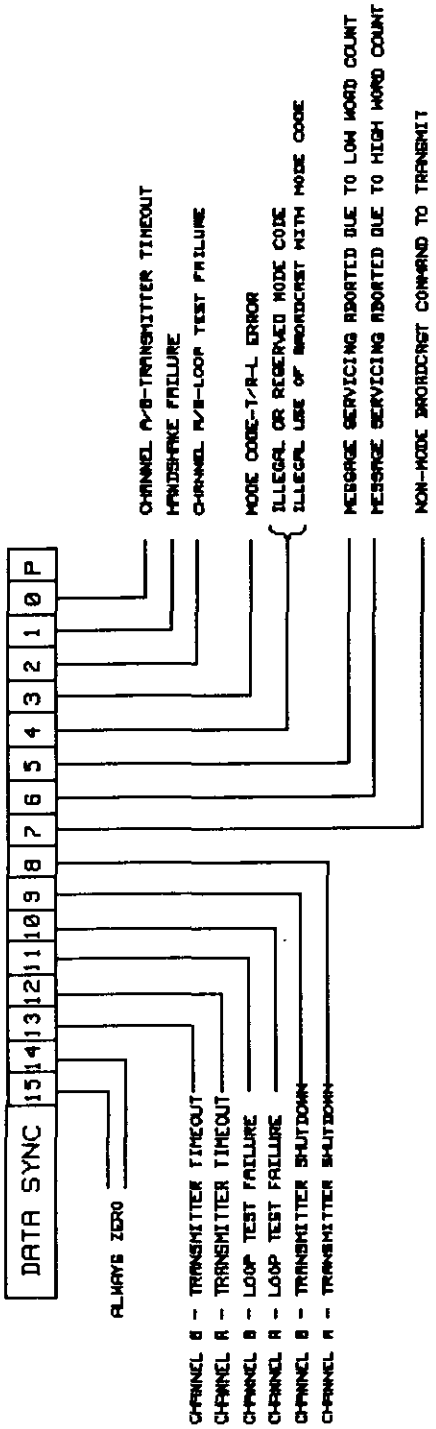


FIGURE 16. Built-in-test word.

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NOTES:

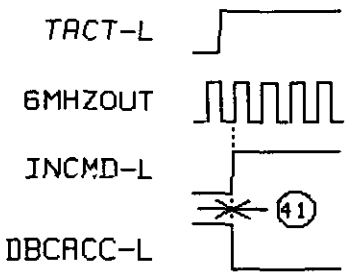
1. Built-in-test word format:

(a) Bits 3-7 are cleared in the beginning of each new message and updated at the end of the message. They only reflect the present command word.

(b) Bits 0-2, 10-13 are latched and are only cleared by a mode reset command or a master reset (RST2-L).

(c) Bits 8 and 9 are set only by the mode command for "transmitter shutdown" and are cleared by the mode command for "override transmitter shutdown" or "reset remote terminal". Bits 8 and 9 are also cleared by RST2-L.

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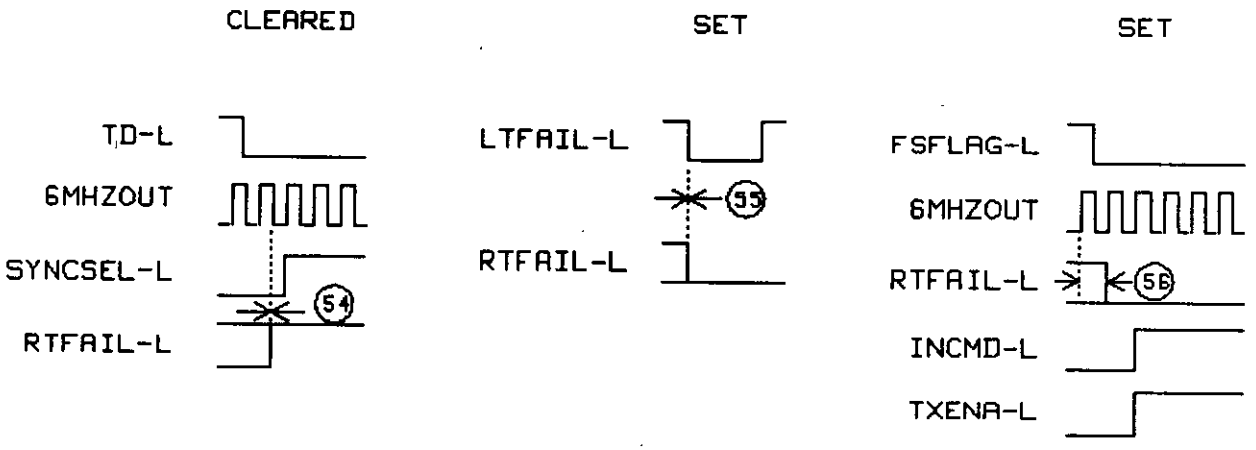
NOTES:

1. Mode commands without data follow the typical sequence of events as shown on the "receive two words" timing diagrams. The only exception is that sequence E-F-G-H-J is omitted since no data words are expected. The T/R-L bit is set to logic "1". If the command is a mode command to "synchronize" without data, and there is no format error, and the command was not defined as illegal, then a GSR-L pulse is generated in the place of GBR-L, as with the case of the mode command to "synchronize with data".

2. If the mode command for "dynamic bus control" comes in without a message error and is not defined as illegal (through the ILLCMD-L input signal) and ADBC-L is low as the status word is updated, the DBCACC-L goes low as shown and remains low until RST1-L or a NBRQ-L comes in.

FIGURE 17. Mode command for "dynamic bus control".

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NOTES:

1. Mode commands without data follow the typical sequence of events as shown on the "receive two words" timing diagrams. The only exception is that sequence E-F-G-H-J is omitted since no data words are expected. The T/R-L bit is set to logic "1". If the command is a mode command to "synchronize" without data, and there is no format error, and the command is not defined as illegal, then a GSR-L pulse is generated in the place of GBR-L, as with the case of the mode command to "synchronize with data".

2. RTFAIL-L is a latched signal that reflects the condition of the built-in-test performed on every transmission. The latch is cleared at the start of the status word transmission and is set whenever LTFAIL-L comes in and the transmitter is still enabled or if the transmitter timeout occurs while the transmitter is enabled. If transmitter timeout occurs, message servicing is aborted.

FIGURE 18. RTFAIL-L signal.

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6.4.2 Functional description of I/O pins.

Signal name	Pin number	I/O	Description
D0	43	I/O	Bi-directional parallel data bus bit 0 (least significant bit).
D1	44	I/O	Bi-directional parallel data bus bit 1.
D2	45	I/O	Bi-directional parallel data bus bit 2.
D3	46	I/O	Bi-directional parallel data bus bit 3.
D4	47	I/O	Bi-directional parallel data bus bit 4.
D5	48	I/O	Bi-directional parallel data bus bit 5.
D6	49	I/O	Bi-directional parallel data bus bit 6.
D7	50	I/O	Bi-directional parallel data bus bit 7.
D8	100	I/O	Bi-directional parallel data bus bit 8.
D9	99	I/O	Bi-directional parallel data bus bit 9.
D10	98	I/O	Bi-directional parallel data bus bit 10.
D11	97	I/O	Bi-directional parallel data bus bit 11.
D12	96	I/O	Bi-directional parallel data bus bit 12.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
D13	95	I/O	Bi-directional parallel data bus bit 13.
D14	94	I/O	Bi-directional parallel data bus bit 14.
D15	93	I/O	Bi-directional parallel data bus bit 15 (most significant bit).
HSFAIL-L	2	O	Handshake fail - output signal that goes low and stays low whenever the subsystem fails to supply DTGRT-L in time to do a successful transfer. Cleared by the next NBGT-L.
MCRT-L	3	O	Mode code reset - low level output pulse (166 nS) to the subsystem used to indicate the reception of a valid, legal mode command to reset.
ME-L	7	O	Message error - output signal that goes low and stays low whenever there is a format or word error with the received message over the 1553 Data Bus. Cleared by the next NBGT-L.
GBR-L	8	O	Good block received - low level output pulse (0.5 uS) used to flag the subsystem that a valid, legal, non-mode receive command with the correct number of data words has been received without a message error and successfully transferred to the subsystem.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
BITEN-L	9	O	Built-in-test word enable - low level output signal present when the built-in-test word is enabled on the parallel data bus.
STATEN-L	12	O	Status word enable - low level active output signal present when the status word is enabled on the parallel data bus.
CMDTRF-L	13	O	Command transfer - low level active output signal present when the command word is being transferred to the subsystem.
INCMD-L	14	O	In command - low level output signal used to inform the subsystem that valid information is present on LMC, LBRO, T/R-L, SA4-SA0 and WC4-WC0 outputs. Stays low until the protocol sequence is completed.
LSBENA-L	18	O	LSB enable - low level output pulse (0.5 uS) to the front end module (BUS 67001) used to enable the least significant byte of the selected BUS 67001 receive register onto the parallel data bus for transfer.
LSBLD-L	19	O	LSB load - low level output pulse (166 nS) to the front end module in the middle of a data transfer (LSB) used to load the transmission register.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MSBENA-L	20	O	MSB enable - low level output pulse (0.5 uS) to the front end module (BUS 67001) used to enable the most significant byte of the selected BUS 67001 receive register onto the parallel data bus for transfer.
MSBLD-L	21	O	MSB load - low level output pulse (166 nS) to the front end module in the middle of a data transfer (MSB) used to load the transmission register.
ENCENA-L	22	O	Encoder enable - active low output signal to BUS 67001 used to initiate a word transmission on the selected channel over the 1553 bus.
TXENA-L	23	O	Transmitter enable - active low output signal to the front end module used to enable the transmitter (TXDATAOUT, TXDATAOUT-L, and transceiver) of the selected channel.
SYNCSEL-L	24	O	Sync select - output signal to the front end module to control the sync type to be output on transmission. A low level selects the command sync.
WC0	26	O	Word count 0 - latched output signal of the least significant bit in the command word, word count field. Valid during INCMD-L.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
WC1	27	O	Word count 1 - latched output signal of the second least significant bit in the command word, word count field. Valid during INCMD-L.
WC2	28	O	Word count 2 - latched output signal of the third most significant bit in the command word, word count field. Valid during INCMD-L.
WC3	29	O	Word count 3 - latched output signal of the second most significant bit in the command word, word count field. Valid during INCMD-L.
WC4	30	O	Word count 4 - latched output signal of the most significant bit in the command word, word count field. Valid during INCMD-L.
GSR-L	55	O	Good sync received - a low level output pulse (0.5 uS) used to flag the subsystem that a valid mode synchronize command (with or without data - 10001 or 00001) has been received without a message error and successfully transferred to the subsystem.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DBDC	56	O	Data bus direction control - normally a low level output signal that goes high when a data word transfer (from subsystem to front end module) is requested. Returns low after transfer is completed or timed out. Can be used as a read/write line or to control data bus isolation buffer direction.
DBCACC-L	57	O	Dynamic bus control accepted - output signal to subsystem that goes low at completion of the present command servicing if it was a valid, legal mode command for dynamic bus control with no message error and the dynamic bus control acceptance bit is set in the status register. Stays low until reset or the next NBGT-L.
DTSTR	58	O	Data strobe - high level output pulse (or pulses if 8 bit data bus structure) present in the middle of every data word (or byte) transfer over the parallel data bus. Used to latch or strobe data. Recommend using the falling edge to clock the data in.
DTACT-L	59	O	Data transfer active - output signal that goes low whenever data is being transferred over the parallel data bus.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DTREQ-L	61	O	Data transfer request - active low output signal to subsystem indicating that the protocol module has data for or needs data from the subsystem and requests a data transfer over the parallel data bus. Will stay low until transfer is completed or a transfer timeout has occurred.
NBGT-L	66	O	New bus grant - low level output pulse (166 nS) used to indicate the start of a new protocol sequence in response to the command word just received.
CWC0	71	O	Current word count 0 - output signal of the least significant bit in the current word counter. Cleared by NBGT-L and incremented after each data word transfer or timeout.
CWC1	72	O	Current word count 1 - output signal of the second least significant bit in the current word counter. Cleared by NBGT-L and incremented after each data word transfer or timeout.
CWC2	73	O	Current word count 2 - output signal of the third most significant bit in the current word counter. Cleared by NBGT-L and incremented after each data word transfer or timeout.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CWC3	74	O	Current word count 3 - output signal of the second most significant bit in the current word counter. Cleared by NBGT-L and incremented after each data word transfer or timeout.
CWC4	75	O	Current word count 4 - output signal of the most significant bit in the current word counter. Cleared by NBGT-L and incremented after each data word transfer or timeout.
CHA-L/CHB	76	O	Channel A/channel B - output signal to the front end module used to select the active channel. Low level indicates channel A.
SA0	82	O	Subaddress 0 - latched output signal of the least significant bit in the command word subaddress field. Valid during INCMD-L.
SA1	83	O	Subaddress 1 - latched output signal of the second least significant bit in the command word subaddress field. Valid during INCMD-L.
SA2	84	O	Subaddress 2 - latched output signal of the third most significant bit in the command word subaddress field. Valid during INCMD-L.
SA3	85	O	Subaddress 3 - latched output signal of the second most significant bit in the command word subaddress field. Valid during INCMD-L.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SA4	86	O	Subaddress 4 - latched output signal of the most significant bit in the command word subaddress field. Valid during INCMD-L.
T/R-L	87	O	Transmit/receive - latched output signal of the T/R-L bit in the command word. Valid during INCMD-L. Low level indicates receive.
LMC	88	O	Latched mode code - inverted latched output signal of MC-L input from BUS 67001. If high, indicates the subaddress field of the command word was all zeros or all ones. Valid during INCMD-L.
LBRO	89	O	Latched broadcast - inverted latched output signal of BRO-L input from BUS 67001. If high, indicates the command word was a broadcast and not an explicit remote terminal address. Valid during INCMD-L.
RTFAIL-L	65	O	Remote terminal failure - latched, active low output signal to subsystem to flag detection of a remote terminal continuous self-test failure. Cleared by the start of the next message transmission (status word) and set if problem is again detected.
6MHZOUT	6	O	6 MHz clock out - output of the internal system clock.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MSB/LSB-L	70	O	Output signal used to flag which byte of the 16 bit word is being transferred. Should be ignored if 16 bit data bus structure is implemented.
RST2-L	92	O	Reset 2 - output signal to front end modules used to reset. Goes low when +5 volt power is first applied or whenever RST1-L is low.
12MHZCLKOUT	54	O	12 MHz clock output - output of the 12 MHz oscillator circuit. Output drive limited to Dual Redundant Bus Terminal configuration plus 40 pF maximum system backpanel capacitance.
Vcc(CLK)	51	-	+5 V power connection for 12 MHz oscillator only.
Vcc	1	-	+5 V power connection
Vcc	31	-	+5 V power connection
GND	5	-	Ground connection
GND	10	-	Ground connection
GND	15	-	Ground connection
GND	25	-	Ground connection
GND	40	-	Ground connection
GND	60	-	Ground connection
GND	11	-	Frame ground
SSFLAG-L	16	I	Subsystem flag - input from the subsystem used to control the subsystem flag bit in the status register. If low when the status word is updated, the subsystem flag will be set; if high, it will be cleared.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SSBUSY-L	17	I	Subsystem busy - input from the subsystem used to control the busy bit in the status register. If low when the status word is updated, the busy bit will be set; if high, it will be cleared. If the busy bit is set in the status register, no data will be requested from the subsystem in response to a transmit command. On receive commands, data will be transferred to subsystem.
TACT-L	32	I	Transmitter active - input signal from front end module (BUS 67001), low while module is transmitting on the 1553 Data Bus. A high level on this input after the last word has been transmitted (or during transmission) will flag the protocol hybrid that command servicing is completed.
T1-L	33	I	Time 1 - active low input pulse (0.5 uS) from front end module (BUS 67001) that occurs 3 uS after NW-L, ANBRQ-L or BNBRQ-L. It is used by the protocol hybrid to abort a command or data word transfer request on the parallel data bus if the transfer has not already started. It is also used as a timing indicator as to when to update the status register after ANBRQ-L or BNBRQ-L.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
TD-L	34	I	Transmitting data - input signal from front end module that goes low for 16 uS during each data word transmission. The falling edge causes the protocol module to request the next data word (if any) from the subsystem. The rising edge causes the protocol module to abort the DTREQ-L if the data transfer has not begun.
MC-L	35	I	Mode code - input signal from front end module that describes the command word received. Low if subaddress field equals all zeros or all ones.
BRO-L	36	I	Broadcast - input signal from the front end module (BUS 67001) that describes the address field of the command word. Goes low if the address field is all ones, informing the protocol module that no status word response is required.
ND-L	37	I	No data - input signal from the front end module. A low level applied to this pin will inform the protocol module that the start of a new word (valid sync and 2 data bits) has not been detected and, therefore, could not be continuous with the previous word.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
VW-L	38	I	Valid word - input signal from the front end module. A low level applied to this input pin during NW-L pulse informs the protocol module that the word just received did not have any Manchester coding errors or a parity error.
CMD-L	39	I	Command - input signal from the front end module. A low level applied to this pin during NW-L pulse informs the protocol module that the word just received had a command sync.
NW-L	41	I	New word - input signal from front end module. A low level pulse (200 nS min) applied to this input pin will inform the protocol module that a new word has just been received.
GAP-L	42	I	Input signal from front end module. Informs protocol module (when low) that the command word was preceded on the 1553 Data Bus by a 2 uS GAP minimum. Used in command word decoding and in RT-RT transfers.
DTGRT-L	62	I	Data transfer grant - active low input signal from the subsystem that informs the protocol module when DTREQ-L is asserted to start the transfer. Once transfer is started, DTGRT-L can be removed.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ADBC-L	64	I	Accept dynamic bus control - active low input signal from subsystem used to set the "dynamic bus control acceptance" bit in the status register if the command word was a valid, legal mode command for dynamic bus control.
RTDIS-L	67	I	Remote terminal disable - active low input signal from the subsystem which places the protocol module in reset and tri-states all outputs except the 12MHZCLKOUT and RST1-L. Used to select between remote terminal mode or bus control mode when BUS 67003 is also sharing the front end modules (BUS 67002).
SRQ-L	68	I	Subsystem service request - input from the subsystem used to control the service request bit in the status register. If low when the status word is updated, the service request bit will be set; if high, it will be cleared.
ILLCMD-L	69	I	Illegal command - active low input signal from the subsystem strobed in on the falling edge of INCMD-L. Used to define the command word as illegal and to set the message error bit in the status register.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
LTFAIL-L	77	I	Loop test fail - input signal from the front end module. A low level on this pin while the transmitter is enabled will cause the RTFAIL-L output to go low and set the corresponding bits in the built-in-test register.
FSFLAG-L	78	I	Fail safe flag - input signal from the front end module indicating a transmitter timeout (768 uS) has occurred. A low level on this pin while the transmitter is enabled will cause the RTFAIL-L output to go low, set the corresponding bits in the built-in-test register and abort all further servicing of the present command word.
BNBRQ-L	79	I	B-New bus request - input signal from the channel B front end module. A low level pulse (200 nS min) applied to this input pin will cause the protocol module to abort whatever it was doing and start a new command word initialization cycle on channel B, to service the command.
ANBRQ-L	80	I	A-New bus request - input signal from the channel A front end module. A low level pulse (200 nS min) applied to this input pin will cause the protocol module to abort whatever it was doing and start a new command word initialization cycle on channel A, to service the command.

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6.4.2 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
VA-L	81	I	Valid address - input signal from the front end module. A low level applied to this input pin during NW-L pulse informs the protocol module that the command word just received had this remote terminal's exclusive address. Used for error detection in RT-RT transfers.
12MHZCLK	4	I	12MHz clock input - input for the master clock used to run the system.
RST1-L	91	I	Reset 1 - input signal from subsystem. A low level applied to this pin will place the protocol hybrid in the reset state as well as causing RST2-L to go low.
T2-L	90	I	Time 2 - input signal from selected front end module. Goes low 12 uS after ND-L goes low, provided a valid sync and two data bits have not been detected on the 1553 Data Bus. Used for RT-RT no response timeout. When low, enables internal 4 uS counter which will trigger no response timeout.
SEL1	63	I	Select 1 - hardwired input signal used to select the parallel data bus structure. Ground for 16 bit operation. For 8 bit operation, connect to Vcc and connect the upper byte of the data bus (D15-D8) to the lower byte (D7-D0).

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6.4.3 UKU input signal information. The input signals on the UKU module can be broken up into three types. The first type of signals are the basic module set up signals that configure the module operation. They are as follows:

SEL1 (Pin 63) - hard-wired high or low.
12MHZCLK (Pin 4) - continuously applied.
RTDIS-L (Pin 67) - should be held high while module is active, if low, module remains reset.
RST1-L (Pin 91) - normally high.

The second type of input signals are key signals that will trigger events if certain sequences have occurred. Their exact timing referenced to the system clock (or set-up and hold time) is not important as long as the module sees them in the proper state. They are as follows:

ANBRQ-L	(Pin 80)	T1-L	(Pin 33)
BNBRQ-L	(Pin 79)	T2-L	(Pin 90)
NW-L	(Pin 41)	LTFAIL-L	(Pin 77)
ND-L	(Pin 37)	FSFLAG-L	(Pin 78)
TD-L	(Pin 34)	DTGRT-L	(Pin 62)
TACT-L	(Pin 32)		

Of the above signals, LTFAIL-L and FSFLAG-L are normally high and go low only if a problem is detected. They have no effect on the module unless the TXENA-L output (Pin 23) is low. The module only has to see them low to set the RTFAIL-L output. T2-L input is only examined during RT-RT transfers while waiting for the transmitting RT to respond. DTGRT-L must just be sensed low while DTREQ-L is low. A minimum pulse of 200 nS is sufficient to guarantee this will happen. The third type of input signals are only used to describe key signals, control protocol, or update status register. They are as follows:

BRO-L	(Pin 36)	ADBC-L	(Pin 64)
MC-L	(Pin 35)	SSFLAG-L	(Pin 16)
GAP-L	(Pin 42)	SSBUSY-L	(Pin 17)
CMD-L	(Pin 39)	SRQ-L	(Pin 68)
VW-L	(Pin 38)	ILLCMD-L	(Pin 69)
VA-L	(Pin 81)		

They will be valid referenced to the key signals as shown on figures 19a-19d. Data bus input parameters shall be valid referenced to the key signals as shown on figure 20.

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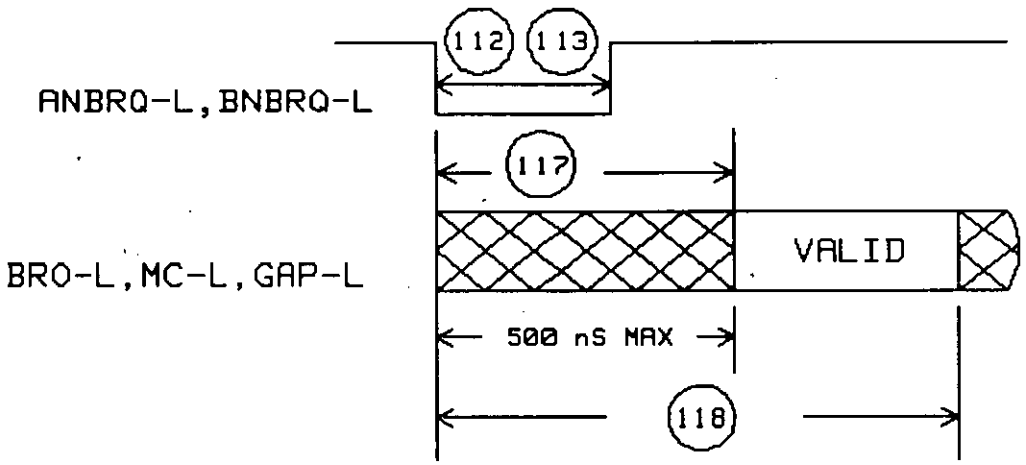
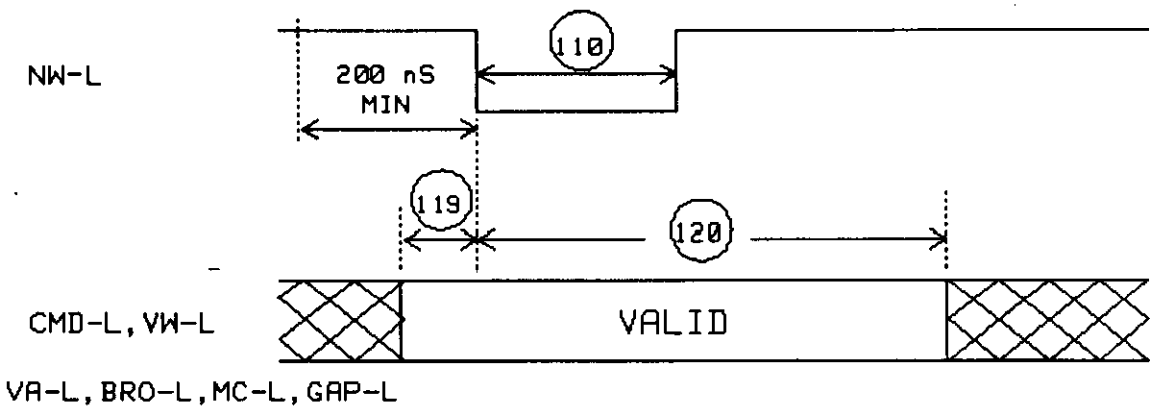


FIGURE 19a. UKU input timing.



NOTE:

1. VA-L, BRO-L, and MC-L are only required on the first NW-L pulse following ANBRQ-L or BNBRQ-L, when doing an RT-RT transfer where the module set is the receiving RT. GAP-L is only required on the 2nd NW-L pulse during an RT-RT transfer, when the transmitting RT responds with its status word.

FIGURE 19b. UKU input timing.

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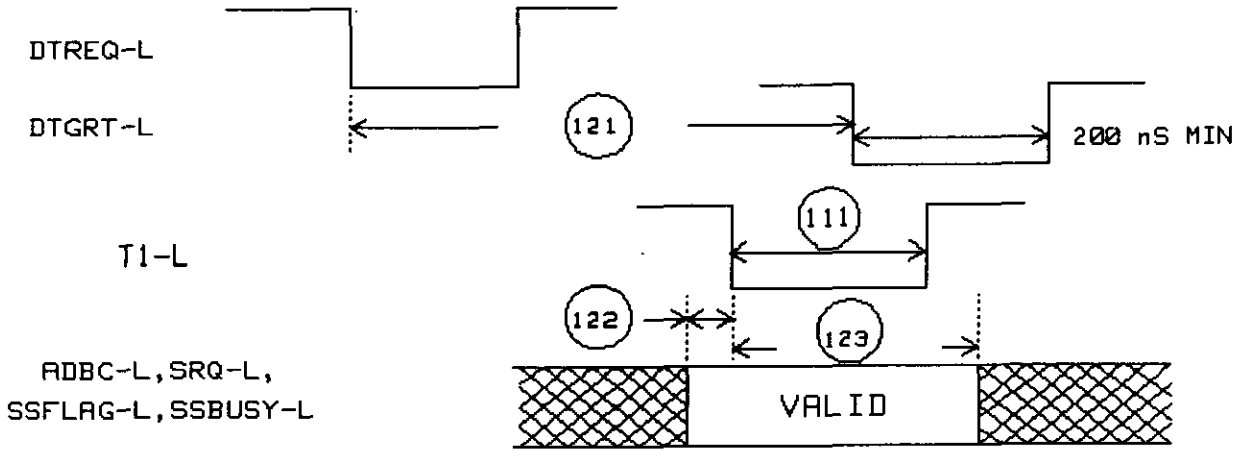


FIGURE 19c. UKU input timing.

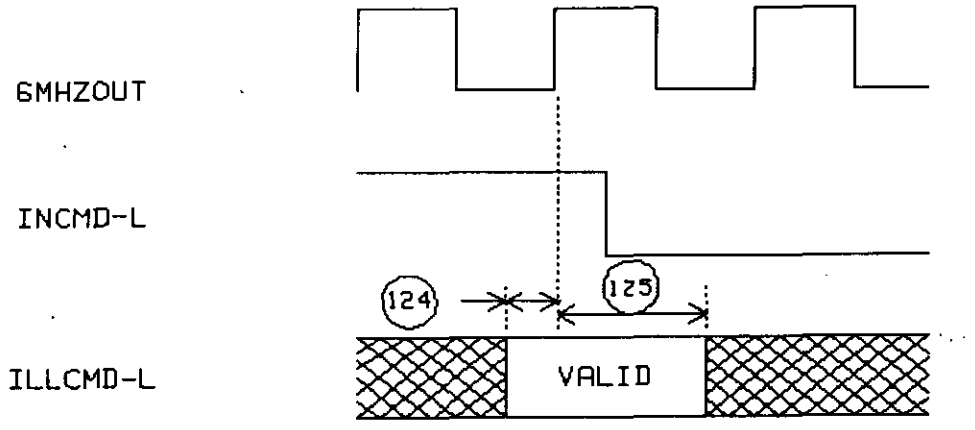


FIGURE 19d. UKU input timing.

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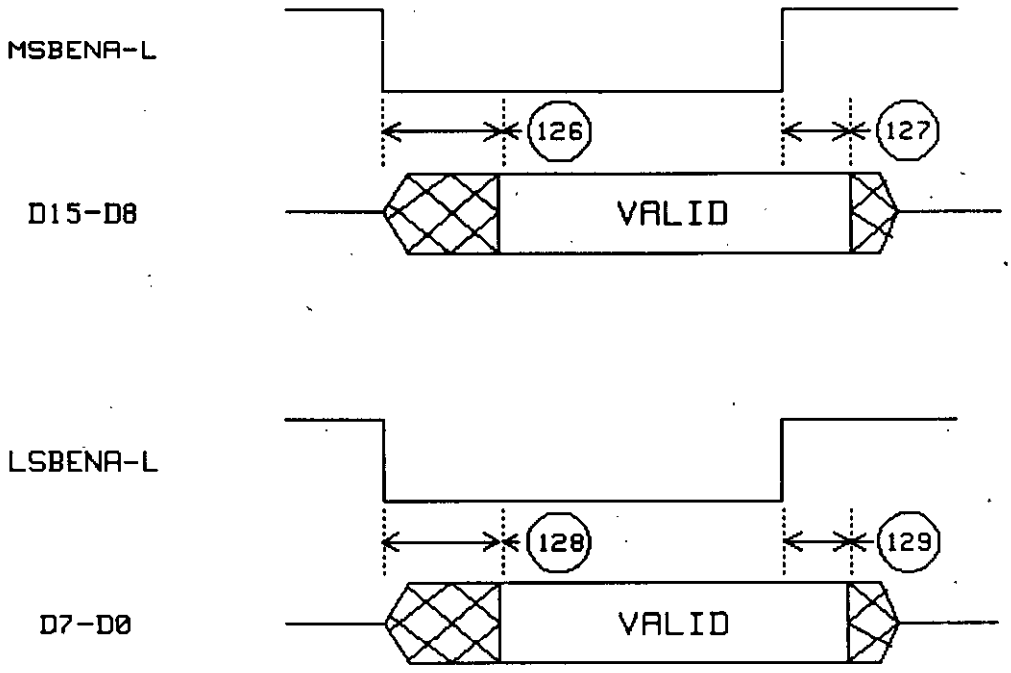


FIGURE 20. UKU data bus input requirements.

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6.5 Module handling procedures. The UKU module contains CMOS devices and is susceptible to damage when exposed to static electrical charge. When handling, shipping, or storing the module, pins should be shorted together with a conductive material or the module should be placed into an electrically conductive bag. The UKU module cannot be plugged into or removed from a hot socket. Failure to comply with the handling and hot socket guidelines could cause irreversible module damage.

6.5.1 Bus contention. Care should be exercised when using the UKT, UKU, and UKS modules to prevent bus contention during normal operation or Vcc stress-up conditions. During software debugging where bus contentions are possible, current limiting of Vcc stress supplies is a suggested method to prevent damage to modules.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N725)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused	52,53	C,11	10	INF	MOhm	1-8
A3	Isolation, unused to unused	52	53	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1
Megohm bridge voltage	30 V	± 10 percent	A2,A3
Isolation measurement	NA	± 500 KOhm	A1-A3
Isolation measurement (See note 7)	NA	± 50 KOhm	A1-A3

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins, except pin 11, shorted together.

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NOTES: - Continued.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The minimum end-of-life limit is 1 MOhm.

8. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0	1	Ohm	1-5
B2	Continuity, GND	5	10				
B3	Continuity, GND	10	15				
B4	Continuity, GND	15	25				
B5	Continuity, GND	25	40				
B6	Continuity, GND	40	60				
B7	Continuity, Frame	11	Frame	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. When performing test condition B7, penetrate the anodized finish on module frame with a sharply pointed probe.

5. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits	Notes
C	Vcc stress	1,31,51	5,10,15, 25,40,60, 11	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(CLOCK)	7.0 V	+0 mV,-70 mV	C
Power supply current limit, (Vcc)	582 mA	NA	C
(CLOCK)	46 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. In the set-up conditions in table IV, 'A' shall correspond to the positive terminal of the power supply, and 'B' shall correspond to the negative terminal.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	SSFLAG-L	16	5.5	-20	20	uA	1-3
D2	SSBUSY-L	17					
D3	TACT-L	32					
D4	T1-L	33					
D5	TD-L	34					
D6	MC-L	35					
D7	BRO-L	36					
D8	ND-L	37					
D9	VW-L	38					
D10	CMD-L	39					
D11	NW-L	41					
D12	GAP-L	42					
D13	DTGRT-L	62					
D14	ADBC-L	64					
D15	RTDIS-L	67					
D16	SRQ-L	68					
D17	ILLCMD-L	69					
D18	LTFAIL-L	77					
D19	FSFLAG-L	78					
D20	BNBRQ-L	79					
D21	ANBRQ-L	80	5.5	-20	20	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	VA-L	81	5.5	-20	20	uA	1-3
D23	12MHZCLK	4	5.5	-20	20	uA	1-3
D24	RST1-L	91	5.5	-11	111	uA	1-3
D25	T2-L	90	5.5	0	200	uA	1-3
D26	SEL1	63	5.5	-11	111	uA	1-3
D27	D0	43	5.5	0	100	uA	1-3
D28	D1	44					
D29	D2	45					
D30	D3	46					
D31	D4	47					
D32	D5	48					
D33	D6	49					
D34	D7	50					
D35	D8	100					
D36	D9	99					
D37	D10	98					
D38	D11	97					
D39	D12	96					
D40	D13	95					
D41	D14	94					
D42	D15	93	5.5	0	100	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:58Z
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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D42
(CLOCK)	5.5 V	± 55 mV	D1-D42
Power supply current limit, (Vcc)	485 mA	NA	D1-D42
(CLOCK)	38 mA	NA	D1-D42
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D42
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D42
low, (VL)	0.0 V	± 100 mV	D1-D42
Input current measurement	NA	± 2.0 uA	D1-D23
	NA	± 6.1 uA	D24
	NA	± 10.0 uA	D25
	NA	± 6.1 uA	D26
	NA	± 5.0 uA	D27-D42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	RST1-L	91	-18.0	-1.5	-0.3	V	1-3
E2	T2-L	90					
E3	D0	43					
E4	D1	44					
E5	D2	45					
E6	D3	46					
E7	D4	47					
E8	D5	48					
E9	D6	49					
E10	D7	50					
E11	D8	100					
E12	D9	99					
E13	D10	98					
E14	D11	97					
E15	D12	96					
E16	D13	95					
E17	D14	94					
E18	D15	93	-18.0	-1.5	-0.3	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E18
(CLOCK)	4.5 V	± 45 mV	E1-E18
Power supply current limit, (Vcc)	485 mA	NA	E1-E18
(CLOCK)	38 mA	NA	E1-E18
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E18
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E18
low, (VL)	0.0 V	± 100 mV	E1-E18
Input voltage measurement	NA	± 60 mV	E1-E18

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	D0	43	0.0	-225	-40	mA	1-4
F2	D1	44					
F3	D2	45					
F4	D3	46					
F5	D4	47					
F6	D5	48					
F7	D6	49					
F8	D7	50					
F9	D8	100					
F10	D9	99					
F11	D10	98					
F12	D11	97					
F13	D12	96					
F14	D13	95					
F15	D14	94					
F16	D15	93	0.0	-225	-40	mA	1-4
F17	RST2-L	92	0.0	-100	-20	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK)	5.5 V 5.5 V	± 55 mV ± 55 mV	F1-F17 F1-F17
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	F1-F17 F1-F17
Forcing voltage, (Vout)	0.0 V	± 50 mV	F1-F17
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F17 F1-F17
Output current measurement	NA NA	± 9.25 mA ± 4.0 mA	F1-F16 F17

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	SSFLAG-L	16	2.7	-20	20	uA	1-3
G2	SSBUSY-L	17					
G3	TACT-L	32					
G4	T1-L	33					
G5	TD-L	34					
G6	MC-L	35					
G7	BRO-L	36					
G8	ND-L	37					
G9	VW-L	38					
G10	CMD-L	39					
G11	NW-L	41					
G12	GAP-L	42					
G13	DTGRT-L	62					
G14	ADBC-L	64					
G15	RTDIS-L	67					
G16	SRQ-L	68					
G17	ILLCMD-L	69					
G18	LTFAIL-L	77					
G19	FSFLAG-L	78					
G20	BNBRQ-L	79					
G21	ANBRQ-L	80	2.7	-20	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	VA-L	81	2.7	-20	20	uA	1-3
G23	12MHZCLK	4	2.7	-20	20	uA	1-3
G24	RST1-L	91	2.7	-260	-70	uA	1-3
G25	T2-L	90	2.7	0	40	uA	1-3
G26	SEL1	63	2.7	-20	50	uA	1-3
G27	D0	43	2.7	0	20	uA	1-3
G28	D1	44					
G29	D2	45					
G30	D3	46					
G31	D4	47					
G32	D5	48					
G33	D6	49					
G34	D7	50					
G35	D8	100					
G36	D9	99					
G37	D10	98					
G38	D11	97					
G39	D12	96					
G40	D13	95					
G41	D14	94					
G42	D15	93	2.7	0	20	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G42
(CLOCK)	5.5 V	± 55 mV	G1-G42
Power supply current limit, (Vcc)	485 mA	NA	G1-G42
(CLOCK)	38 mA	NA	G1-G42
Forcing voltage, (Vih)	2.7 V	± 50 mV	G1-G42
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G42
low, (VL)	0.0 V	± 100 mV	G1-G42
Input current measurement	NA	± 2.0 uA	G1-G23
	NA	± 9.5 uA	G24
	NA	± 2.0 uA	G25
	NA	± 3.5 uA	G26
	NA	± 1.0 uA	G27-G42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	SSFLAG-L	16	0.4	-20	20	uA	1-3
H2	SSBUSY-L	17					
H3	TACT-L	32					
H4	T1-L	33					
H5	TD-L	34					
H6	MC-L	35					
H7	BRO-L	36					
H8	ND-L	37					
H9	VW-L	38					
H10	CMD-L	39					
H11	NW-L	41					
H12	GAP-L	42					
H13	DTGRT-L	62					
H14	ADBC-L	64					
H15	RTDIS-L	67					
H16	SRQ-L	68					
H17	ILLCMD-L	69					
H18	LTFAIL-L	77					
H19	FSFLAG-L	78					
H20	BNBRQ-L	79					
H21	ANBRQ-L	80	0.4	-20	20	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	VA-L	81	0.4	-20	20	uA	1-3
H23	12MHZCLK	4	0.4	-20	20	uA	1-3
H24	RST1-L	91	0.4	-910	-200	uA	1-3
H25	T2-L	90	0.4	-800	0	uA	1-3
H26	SEL1	63	0.4	-810	0	uA	1-3
H27	D0	43	0.4	-400	0	uA	1-3
H28	D1	44					
H29	D2	45					
H30	D3	46					
H31	D4	47					
H32	D5	48					
H33	D6	49					
H34	D7	50					
H35	D8	100					
H36	D9	99					
H37	D10	98					
H38	D11	97					
H39	D12	96					
H40	D13	95					
H41	D14	94					
H42	D15	93	0.4	-400	0	uA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK)	5.5 V 5.5 V	± 55 mV ± 55 mV	H1-H42 H1-H42
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	H1-H42 H1-H42
Forcing voltage, (Vil)	0.4 V	± 50 mV	H1-H42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H42 H1-H42
Input current measurement	NA NA NA NA	± 2.0 uA ± 35.5 uA ± 40.0 uA ± 20.0 uA	H1-H23 H24 H25,H26 H27-H42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	D0	43	-3.0	2.4	4.5	V	1-5
J2	D1	44					
J3	D2	45					
J4	D3	46					
J5	D4	47					
J6	D5	48					
J7	D6	49					
J8	D7	50					
J9	D8	100					
J10	D9	99					
J11	D10	98					
J12	D11	97					
J13	D12	96					
J14	D13	95					
J15	D14	94					
J16	D15	93	-3.0	2.4	4.5	V	1-5
J17	HSFAIL-L	2	-0.4	2.4	4.5	V	1-5
J18	MCRT-L	3					
J19	ME-L	7					
J20	GBR-L	8					
J21	BITEN-L	9	-0.4	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	STATEN-L	12	-0.4	2.4	4.5	V	1-5
J23	CMDTRF-L	13					
J24	INCMD-L	14					
J25	LSBENA-L	18					
J26	LSBLD-L	19					
J27	MSBENA-L	20					
J28	MSBLD-L	21					
J29	ENCENA-L	22					
J30	TXENA-L	23					
J31	SYNCSEL-L	24					
J32	WC0	26					
J33	WC1	27					
J34	WC2	28					
J35	WC3	29					
J36	WC4	30					
J37	GSR-L	55					
J38	DBDC	56					
J39	DBCACC-L	57					
J40	DTSTR	58					
J41	DTACT-L	59					
J42	DTREQ-L	61	-0.4	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	NBGT-L	66	-0.4	2.4	4.5	V	1-5
J44	CWC0	71					
J45	CWC1	72					
J46	CWC2	73					
J47	CWC3	74					
J48	CWC4	75					
J49	CHA-L/CHB	76					
J50	SA0	82					
J51	SA1	83					
J52	SA2	84					
J53	SA3	85					
J54	SA4	86					
J55	T/R-L	87					
J56	LMC	88					
J57	LBRO-	89	-0.4	2.4	4.5	V	1-5
J58	RTFAIL-L	65	-0.39	2.4	4.5	V	1-5
J59	6MHZOUT	6	-0.38	2.4	4.5	V	1-5
J60	MSB/LSB-L	70	-0.38	2.4	4.5	V	1-5
J61	RST2-L	92	-0.39	2.5	4.5	V	1-5

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J61
(CLOCK)	4.5 V	± 45 mV	J1-J61
Power supply current limit, (Vcc)	485 mA	NA	J1-J61
(CLOCK)	38 mA	NA	J1-J61
Forcing current, (Ioh)	-3.0 mA	± 30.0 uA	J1-J16
	-0.4 mA	± 4.0 uA	J17-J57
	-0.39 mA	± 3.9 uA	J58
	-0.38 mA	± 3.8 uA	J59,J60
	-0.39 mA	± 3.9 uA	J61
Pattern voltage	(See note 4)		
high, (VH)			J1-J61
low, (VL)			J1-J61
Output voltage measurement	NA	± 105 mV	J1-J60
	NA	± 100 mV	J61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

4. The high and low level threshold voltage for each input is as specified in 6.3.1.

5. Abbreviations used are defined in 6.1.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	D0	43	12.0	0.0	0.4	V	1-5
K2	D1	44					
K3	D2	45					
K4	D3	46					
K5	D4	47					
K6	D5	48					
K7	D6	49					
K8	D7	50					
K9	D8	100					
K10	D9	99					
K11	D10	98					
K12	D11	97					
K13	D12	96					
K14	D13	95					
K15	D14	94					
K16	D15	93	12.0	0.0	0.4	V	1-5
K17	HSFAIL-L	2	1.6	0.0	0.4	V	1-5
K18	MCRT-L	3					
K19	ME-L	7					
K20	GBR-L	8					
K21	BITEN-L	9	1.6	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K22	STATEN-L	12	1.6	0.0	0.4	V	1-5
K23	CMDTRF-L	13					
K24	INCMD-L	14					
K25	LSBENA-L	18					
K26	LSBLD-L	19					
K27	MSBENA-L	20					
K28	MSBLD-L	21					
K29	ENCENA-L	22					
K30	TXENA-L	23					
K31	SYNCSEL-L	24					
K32	WC0	26					
K33	WC1	27					
K34	WC2	28					
K35	WC3	29					
K36	WC4	30					
K37	GSR-L	55					
K38	DBDC	56					
K39	DBCACC-L	57					
K40	DTSTR	58					
K41	DTACT-L	59					
K42	DTREQ-L	61	1.6	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K43	NBGT-L	66	1.6	0.0	0.4	V	1-5
K44	CWC0	71					
K45	CWC1	72					
K46	CWC2	73					
K47	CWC3	74					
K48	CWC4	75					
K49	CHA-L/CHB	76					
K50	SA0	82					
K51	SA1	83					
K52	SA2	84					
K53	SA3	85					
K54	SA4	86					
K55	T/R-L	87					
K56	LMC	88					
K57	LBRO	89					
K58	RTFAIL-L	65	1.6	0.0	0.4	V	1-5
K59	6MHZOUT	6	1.2	0.0	0.4	V	1-5
K60	MSB/LSB-L	70	1.2	0.0	0.4	V	1-5
K61	RST2-L	92	4.0	0.0	0.4	V	1-5

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K61
(CLOCK)	4.5 V	± 45 mV	K1-K61
Power supply current limit, (Vcc)	485 mA	NA	K1-K61
(CLOCK)	38 mA	NA	K1-K61
Forcing current, (Iol)	12.0 mA	± 120 uA	K1-K16
	1.6 mA	± 16 uA	K17-K58
	1.2 mA	± 12 uA	K59,K60
	4.0 mA	± 40 uA	K61
Pattern voltage	(See note 4)		
high, (VH)			K1-K61
low, (VL)			K1-K61
Output voltage measurement	NA	± 20 mV	K1-K61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input must be verified.

4. The high and low level threshold voltage for each input is as specified in 6.3.1.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	D0	43	2.7	-20	20	uA	1-3
L2	D1	44					
L3	D2	45					
L4	D3	46					
L5	D4	47					
L6	D5	48					
L7	D6	49					
L8	D7	50					
L9	D8	100					
L10	D9	99					
L11	D10	98					
L12	D11	97					
L13	D12	96					
L14	D13	95					
L15	D14	94					
L16	D15	93					
L17	HSFAIL-L	2					
L18	MCRT-L	3					
L19	ME-L	7					
L20	GBR-L	8					
L21	BITEN-L	9	2.7	-20	20	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	STATEN-L	12	2.7	-20	20	uA	1-3
L23	CMDTRF-L	13					
L24	INCMD-L	14					
L25	LSBENA-L	18					
L26	LSBLD-L	19					
L27	MSBENA-L	20					
L28	MSBLD-L	21					
L29	ENCENA-L	22					
L30	TXENA-L	23					
L31	SYNCSEL-L	24					
L32	WC0	26					
L33	WC1	27					
L34	WC2	28					
L35	WC3	29					
L36	WC4	30					
L37	GSR-L	55					
L38	DBDC	56					
L39	DBCACC-L	57					
L40	DTSTR	58					
L41	DTACT-L	59					
L42	DTREQ-L	61	2.7	-20	20	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L43	NBGT-L	66	2.7	-20	20	uA	1-3
L44	CWC0	71					
L45	CWC1	72					
L46	CWC2	73					
L47	CWC3	74					
L48	CWC4	75					
L49	CHA-L/CHB	76					
L50	SA0	82					
L51	SA1	83					
L52	SA2	84					
L53	SA3	85					
L54	SA4	86					
L55	T/R-L	87					
L56	LMC	88					
L57	LBRO	89	2.7	-20	20	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L57
(CLOCK)	5.5 V	± 55 mV	L1-L57
Power supply current limit, (Vcc)	485 mA	NA	L1-L57
(CLOCK)	38 mA	NA	L1-L57
Forcing voltage, (Voh)	2.7 V	± 50 mV	L1-L57
Pattern voltage high, (VH)	3.0 V	± 100 mV	L1-L57
low, (VL)	0.0 V	± 100 mV	L1-L57
Output current measurement	NA	± 2 uA	L1-L57

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	D0	43	0.4	-400	20	uA	1-3
M2	D1	44					
M3	D2	45					
M4	D3	46					
M5	D4	47					
M6	D5	48					
M7	D6	49					
M8	D7	50					
M9	D8	100					
M10	D9	99					
M11	D10	98					
M12	D11	97					
M13	D12	96					
M14	D13	95					
M15	D14	94					
M16	D15	93	0.4	-400	20	uA	1-3
M17	HSFAIL-L	2	0.4	-20	20	uA	1-3
M18	MCRT-L	3					
M19	ME-L	7					
M20	GBR-L	8					
M21	BITEN-L	9	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M22	STATEN-L	12	0.4	-20	20	uA	1-3
M23	CMDTRF-L	13					
M24	INCMD-L	14					
M25	LSBENA-L	18					
M26	LSBLD-L	19					
M27	MSBENA-L	20					
M28	MSBLD-L	21					
M29	ENCENA-L	22					
M30	TXENA-L	23					
M31	SYNCSEL-L	24					
M32	WC0	26					
M33	WC1	27					
M34	WC2	28					
M35	WC3	29					
M36	WC4	30					
M37	GSR-L	55					
M38	DBDC	56					
M39	DBCACC-L	57					
M40	DTSTR	58					
M41	DTACT-L	59	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M42	DTREQ-L	61	0.4	-20	20	uA	1-3
M43	NBGT-L	66					
M44	CWC0	71					
M45	CWC1	72					
M46	CWC2	73					
M47	CWC3	74					
M48	CWC4	75					
M49	CHA-L/CHB	76					
M50	SA0	82					
M51	SA1	83					
M52	SA2	84					
M53	SA3	85					
M54	SA4	86					
M55	T/R-L	87					
M56	LMC	88					
M57	LBRO	89	0.4	-20	20	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK)	5.5 V 5.5 V	± 55 mV ± 55 mV	M1-M57 M1-M57
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	M1-M57 M1-M57
Forcing voltage, (Vol)	0.4 V	± 50 mV	M1-M57
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M57 M1-M57
Output current measurement	NA NA	± 21 uA ± 2 uA	M1-M16 M17-M57

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Oscillator test (12MHZCLKOUT).

Test cond	Parameter	Connect A to pin	25 ± 5°C Initial limits Min	Max	Units	Figure
N1	PWH	54	28.0	55.33	nS	21,22
N2	PWL	54	28.0	55.33	nS	21,22
N3	PRR	54	83.25	83.42	nS	21,22
N4	Voh	54	5.0	--	V	21,22
N5	Vol	54	--	0.4	V	21,22

NOTES:

1. Oscillator test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	NA	NA	N1-N5
(CLOCK)	5.5 V	± 55 mV	N1-N5
Power supply current limit, (Vcc)	NA	NA	N1-N5
(CLOCK)	38.0 mA	NA	N1-N5
Oscillator test measurements	NA	± 1.3 nS	N1,N2
	NA	± 8.5 pS	N3
	NA	± 250.0 mV	N4
	NA	± 20.0 mV	N5

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified on figure 21. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The output waveform for pin 54 shall be as indicated on figure 22.

4. The load circuit for pin 54 shall be as specified on figure 23.

5. Abbreviations used are defined in 6.1.

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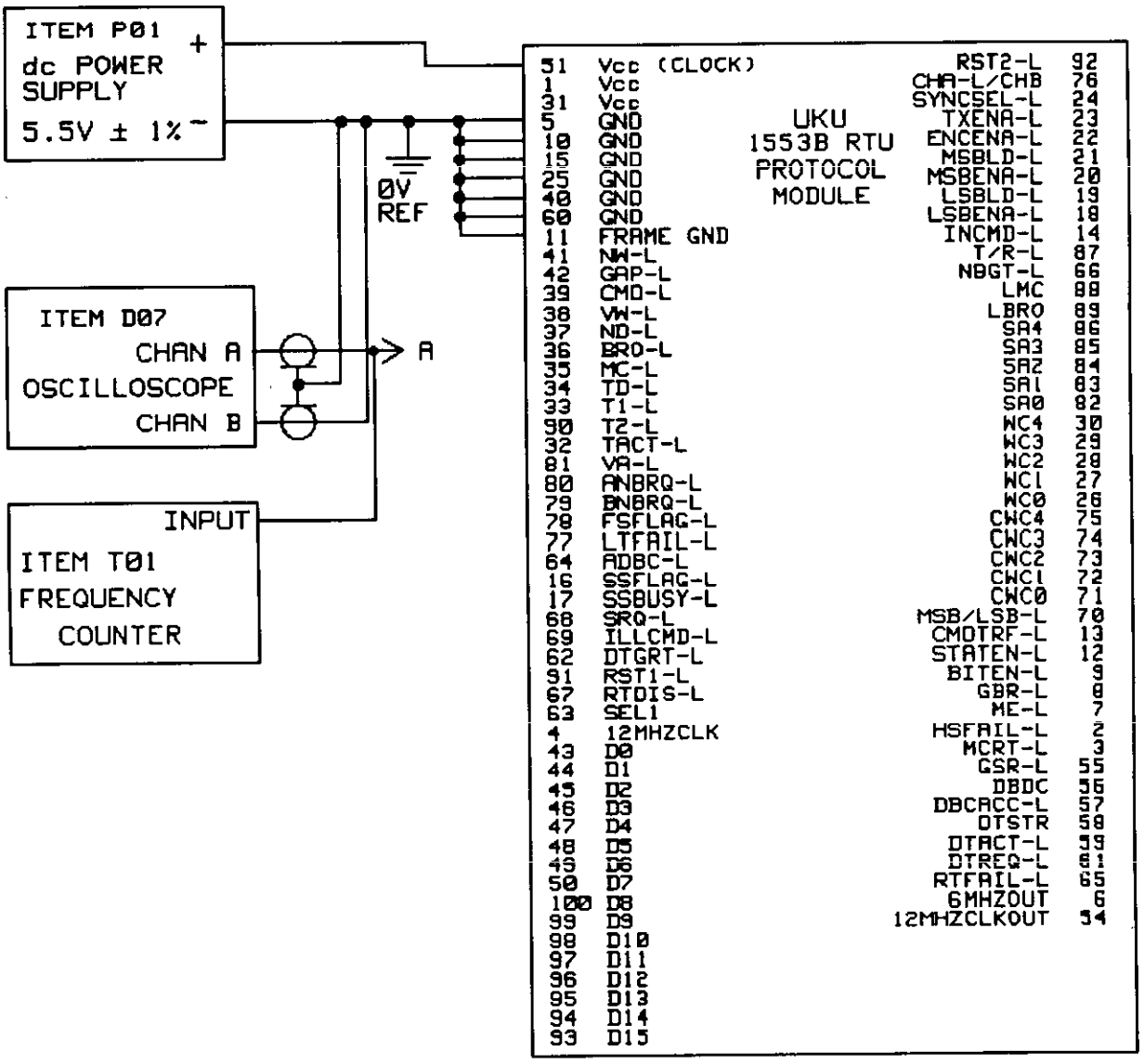


FIGURE 21. Oscillator test circuit.

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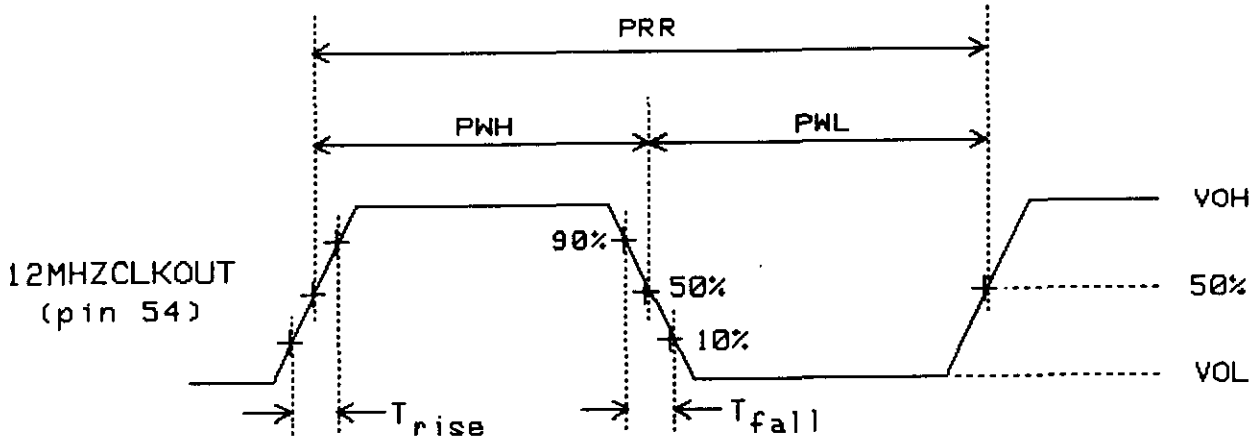
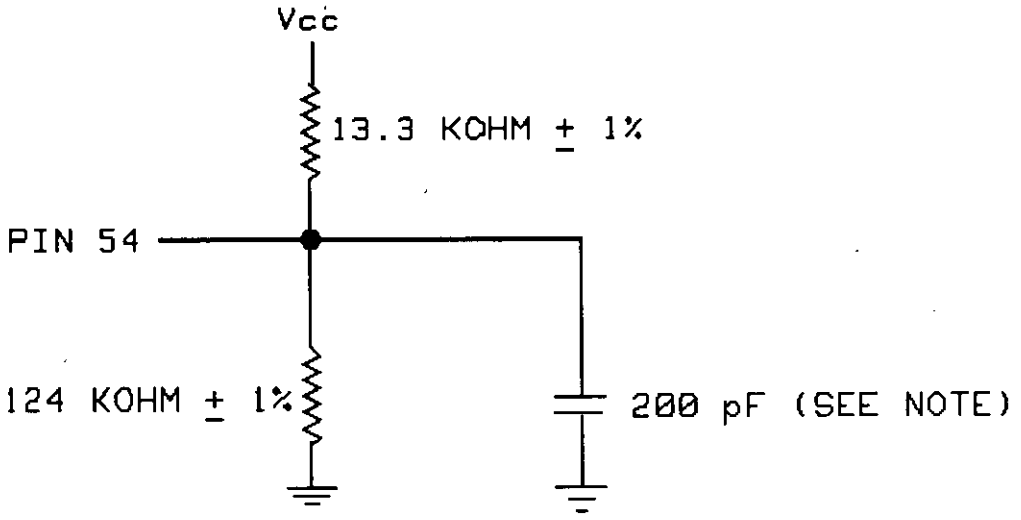


FIGURE 22. Oscillator test waveform.

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NOTE:

1. Includes probe capacitance.

FIGURE 23. Load circuit for pin 54.

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TABLE XV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
P1	Power supply current, (Vcc)	1,31	30	323	mA	1-3
P2	Power supply current, (CLOCK)	51	2	25	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK)	5.5 V 5.5 V	± 55 mV ± 55 mV	P1,P2 P1,P2
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	P1,P2 P1,P2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	P1,P2 P1,P2
Power supply current measurement	NA NA	± 14.65 mA ± 1.15 mA	P1 P2

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with maximum power supply current as indicated in table XV.

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Functional test.

Test cond	Test Description	Notes	Figs
Q1	Functional test minimum power supply voltage maximum clock frequency (12 MHz)	1-3	24
Q2	Functional test maximum power supply voltage maximum clock frequency (12 MHz)	1-3	24

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK) Power supply, (Vcc) (CLOCK)	4.5 V 4.5 V 5.5 V 5.5 V	± 45 mV ± 45 mV ± 55 mV ± 55 mV	Q1 Q1 Q2 Q2
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	Q1,Q2 Q1,Q2
Pattern voltage high, (VH) low, (VL)	2.4 V 0.4 V	± 50 mV ± 50 mV	Q1,Q2 Q1,Q2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	NA NA NA	± 105 mV ± 155 mV ± 20 mV	Q1 Q2 Q1,Q2

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

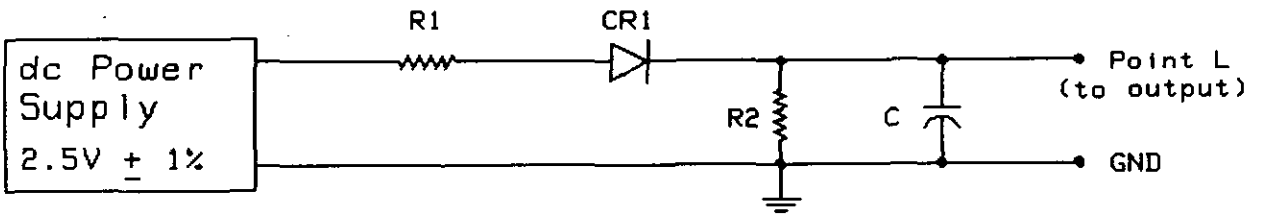
(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XVI using only a capacitive load of 150 pF (see figure 24). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$.
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 pF ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

FIGURE 24. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (Ohms)	Iol (mA)	Vol (V)
43,44,45,46,47,48, 49,50,100,99,98,97, 96,95,94,93	806	12.0	0.4
2,3,7,8,9,12,13,14, 18,19,20,21,22,23, 24,26,27,28,29,30, 55,56,57,58,59,61, 66,71,72,73,74,75, 76,82,83,84,85,86 87,88,89	6040	1.6	0.4
65	6190	1.6	0.4
6,70	6340	1.2	0.4
92	6490	4.0	0.4
54	412	10.0	0.4

1/ All resistors are metal film, ± 1 percent, 0.25 W.

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TABLE XVII. Dynamic test.

Test cond	Test description	Notes
R1	Dynamic test, sample maximum clock frequency (12 MHz) minimum power supply	1-6
R2	Dynamic test, sample maximum clock frequency (12 MHz) maximum power supply	1-6
R3	Dynamic test, qualification maximum clock frequency (12 MHz) minimum power supply	1-6
R4	Dynamic test, qualification maximum clock frequency (12 MHz) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	R1,R3
(CLOCK)	4.5 V	± 45 mV	R1,R3
Power supply, (Vcc)	5.5 V	± 55 mV	R2,R4
(CLOCK)	5.5 V	± 55 mV	R2,R4
Power supply current limit, (Vcc)	485 mA	NA	R1-R4
(CLOCK)	38 mA	NA	R1-R4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 50 mV	R1-R4
low, (VL)	0.4 V	± 50 mV	R1-R4
Output voltage verification			
high, (Voh)	NA	± 100 mV	R1,R3
high, (Voh)	NA	± 100 mV	R2,R4
low, (Vol)	NA	± 100 mV	R1-R4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

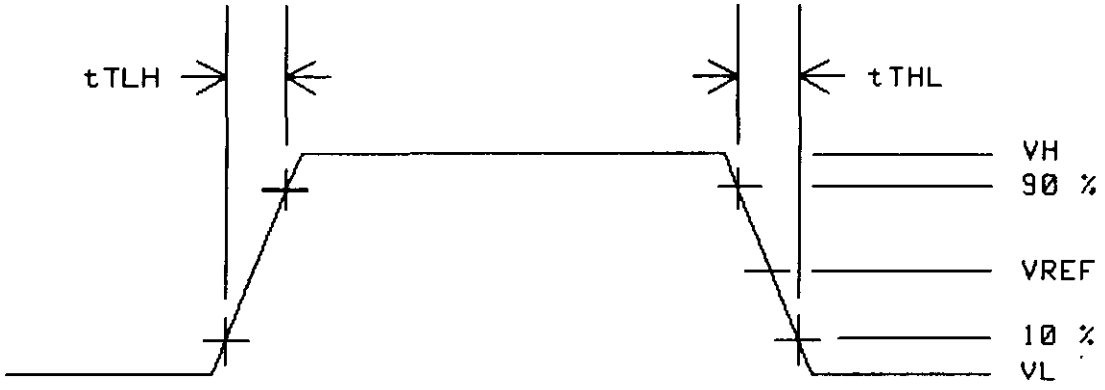
High level: $2.4\ V \leq Voh \leq Vcc$.
Low level: $0.0\ V \leq Vol \leq 0.4\ V$.

(c) The dynamic parameters are found in 6.3.3.

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NOTES: - Continued.

3. Input/Output (I/O) waveforms are as specified on figures 5 and 6. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

4. The load circuit for this test is specified on figure 24.

5. This test shall be performed as indicated in table XVII.

6. Abbreviations used are defined in 6.1.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSFC	1,31	4.8	8.8	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	S
Capacitance bridge voltage	0.3 Vp	± 30 mVp	S
Capacitance measurement	NA	± 200 nF	S

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1,31	0	400	mV	1-3	25
T2	PSTA	51	0	400	mV	1-3	25

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (CLOCK)	5.0 V 5.0 V	± 250 mV ± 250 mV	T1,T2 T1,T2
Power supply current limit, (Vcc) (CLOCK)	485 mA 38 mA	NA NA	T1,T2 T1,T2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together and the scope point, Y01, shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, set at a maximum strobe rate of one-tenth of the cycle time per division, and visually inspect each pattern row for the most negative spike as specified on figure 25.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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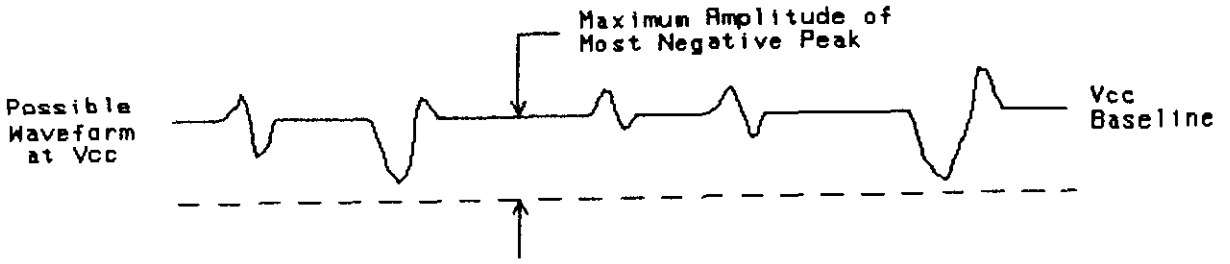


FIGURE 25. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U1	D0	43	0	400	mV	1-5	26,27
U2	D1	44					
U3	D2	45					
U4	D3	46					
U5	D4	47					
U6	D5	48					
U7	D6	49					
U8	D7	50					
U9	D8	100					
U10	D9	99					
U11	D10	98					
U12	D11	97					
U13	D12	96					
U14	D13	95					
U15	D14	94					
U16	D15	93					
U17	HSFAIL-L	2					
U18	MCRT-L	3					
U19	ME-L	7					
U20	GBR-L	8					
U21	BITEN-L	9	0	400	mV	1-5	26,27

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U22	STATEN-L	12	0	400	mV	1-5	26,27
U23	CMDTRF-L	13					
U24	INCMD-L	14					
U25	LSBENA-L	18					
U26	LSBLD-L	19					
U27	MSBENA-L	20					
U28	MSBLD-L	21					
U29	ENCENA-L	22					
U30	TXENA-L	23					
U31	SYNCSEL-L	24					
U32	WC0	26					
U33	WC1	27					
U34	WC2	28					
U35	WC3	29					
U36	WC4	30					
U37	GSR-L	55					
U38	DBDC	56					
U39	DBCACC-L	57					
U40	DTSTR	58					
U41	DTACT-L	59					
U42	DTREQ-L	61	0	400	mV	1-5	26,27

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U43	NBGT-L	66	0	400	mV	1-5	26,27
U44	CWC0	71					
U45	CWC1	72					
U46	CWC2	73					
U47	CWC3	74					
U48	CWC4	75					
U49	CHA-L/CHB	76					
U50	SA0	82					
U51	SA1	83					
U52	SA2	84					
U53	SA3	85					
U54	SA4	86					
U55	T/R-L	87					
U56	LMC	88					
U57	LBRO	89					
U58	RTFAIL-L	65					
U59	6MHZOUT	6					
U60	MSB/LSB-L	70	0	400	mV	1-5	26,27

See notes following table XXI.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
V1	D0	43	0	400	mV	1-5	26,27
V2	D1	44					
V3	D2	45					
V4	D3	46					
V5	D4	47					
V6	D5	48					
V7	D6	49					
V8	D7	50					
V9	D8	100					
V10	D9	99					
V11	D10	98					
V12	D11	97					
V13	D12	96					
V14	D13	95					
V15	D14	94					
V16	D15	93					
V17	HSFAIL-L	2					
V18	MCRT-L	3					
V19	ME-L	7					
V20	GBR-L	8					
V21	BITEN-L	9	0	400	mV	1-5	26,27

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
V22	STATEN-L	12	0	400	mV	1-5	26,27
V23	CMDTRF-L	13					
V24	INCMD-L	14					
V25	LSBENA-L	18					
V26	LSBLD-L	19					
V27	MSBENA-L	20					
V28	MSBLD-L	21					
V29	ENCENA-L	22					
V30	TXENA-L	23					
V31	SYNCSEL-L	24					
V32	WC0	26					
V33	WC1	27					
V34	WC2	28					
V35	WC3	29					
V36	WC4	30					
V37	GSR-L	55					
V38	DBDC	56					
V39	DBCACC-L	57					
V40	DTSTR	58					
V41	DTACT-L	59					
V42	DTREQ-L	61	0	400	mV	1-5	26,27

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
V43	NBGT-L	66	0	400	mV	1-5	26,27
V44	CWC0	71					
V45	CWC1	72					
V46	CWC2	73					
V47	CWC3	74					
V48	CWC4	75					
V49	CHA-L/CHB	76					
V50	SA0	82					
V51	SA1	83					
V52	SA2	84					
V53	SA3	85					
V54	SA4	86					
V55	T/R-L	87					
V56	LMC	88					
V57	LBRO	89					
V58	RTFAIL-L	65					
V59	6MHZOUT	6					
V60	MSB/LSB-L	70	0	400	mV	1-5	26,27

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NOTES:

1. Output crosstalk, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	U1-U60,V1-V60
(CLOCK)	5.0 V	± 250 mV	U1-U60,V1-V60
Power supply current limit, (Vcc)	485 mA	NA	U1-U60,V1-V60
(CLOCK)	38 mA	NA	U1-U60,V1-V60
Pattern voltage high, (VH)	3.0 V	± 100 mV	U1-U60,V1-V60
low, (VL)	0.0 V	± 100 mV	U1-U60,V1-V60
Output crosstalk measurement	NA	± 20 mV	U1-U60,V1-V60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in tables XX and XXI.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in tables XX and XXI, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 26.

(e) The maximum probe capacitance is 10 pF.

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NOTES: - Continued.

3. All outputs shall be loaded as specified on figure 24 with C = 50 pF. Output signals D0-D15 are bidirectional and will be loaded when programmed to be outputs.

4. The test circuit shall be as indicated on figure 27.

5. Abbreviations used are defined in 6.1.

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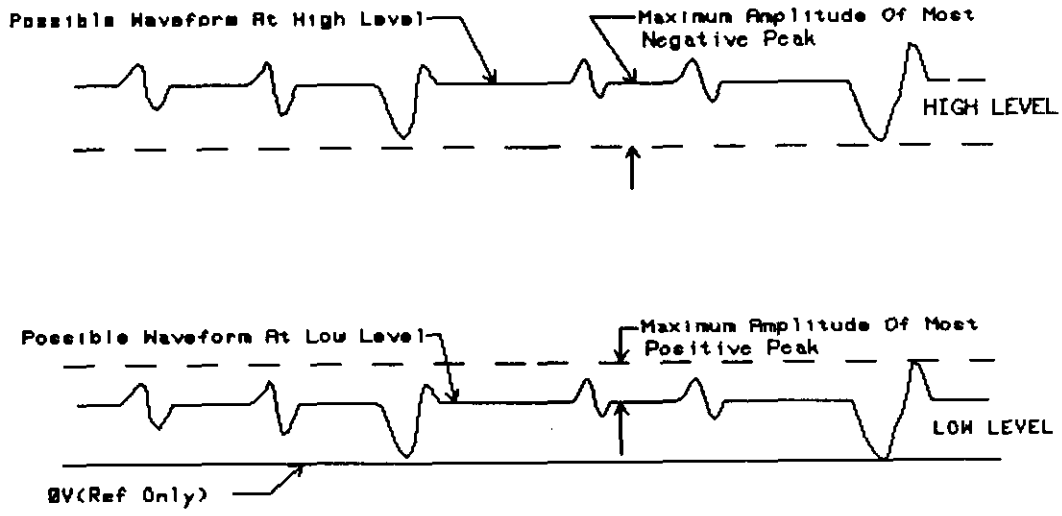


FIGURE 26. Output crosstalk test circuit waveforms, high and low level, for tables XX and XXI.

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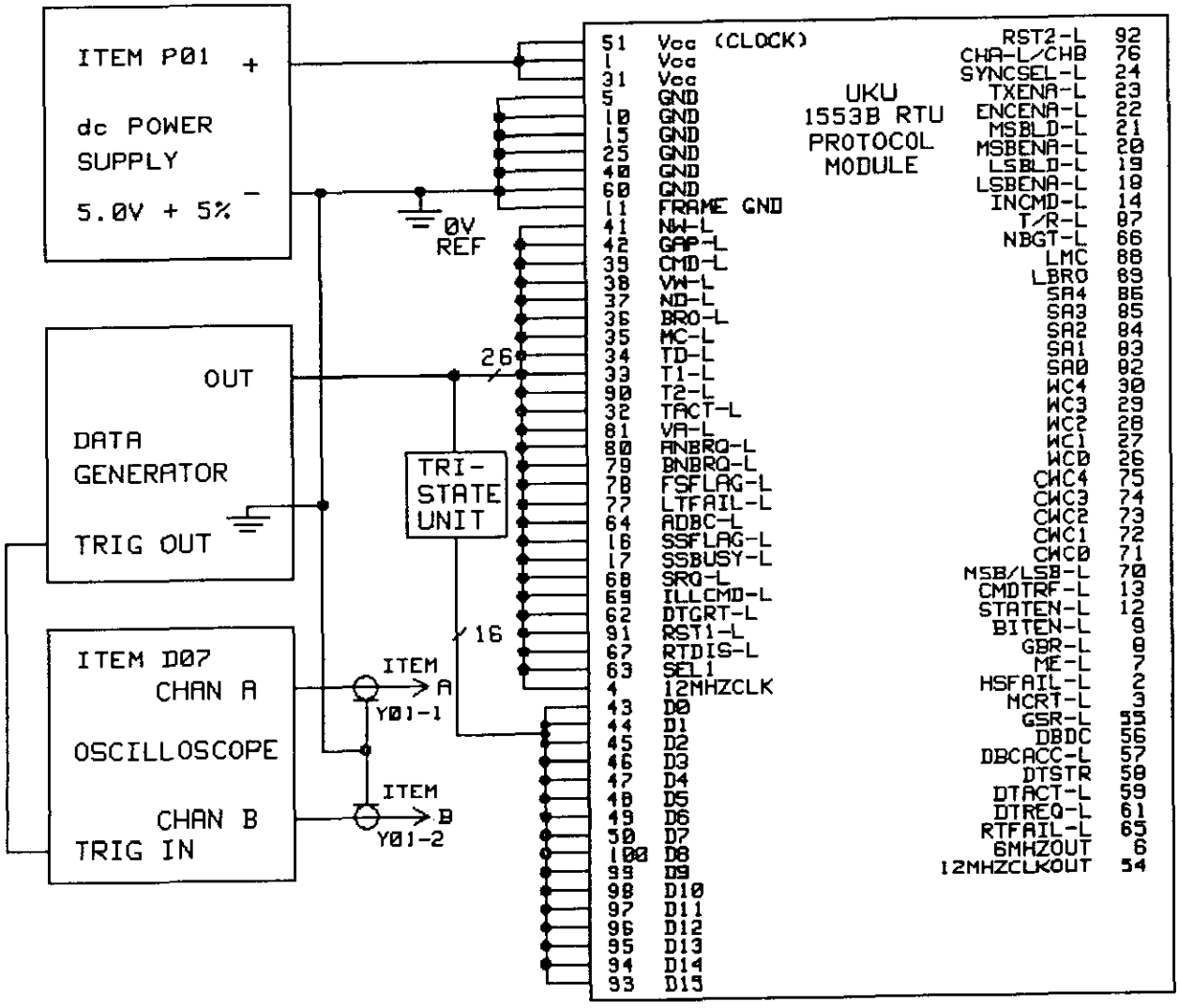


FIGURE 27. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	SSFLAG-L	16	2	40	pF	1-3
W2	SSBUSY-L	17				
W3	TACT-L	32				
W4	T1-L	33				
W5	TD-L	34				
W6	MC-L	35				
W7	BRO-L	36				
W8	ND-L	37				
W9	VW-L	38				
W10	CMD-L	39				
W11	NW-L	41				
W12	GAP-L	42				
W13	DTGRT-L	62				
W14	ADBC-L	64				
W15	RTDIS-L	67				
W16	SRQ-L	68				
W17	ILLCMD-L	69				
W18	LTFAIL-L	77				
W19	FSFLAG-L	78				
W20	BNBRQ-L	79				
W21	ANBRQ-L	80	2	40	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W22	VA-L	81	2	40	pF	1-3
W23	12MHZCLK	4				
W24	RST1-L	91				
W25	T2-L	90				
W26	SEL1	63	2	40	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W26
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W26
Capacitance measurement	NA	± 1.90 pF	W1-W26

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	D0	43	2	40	pF	1-3
X2	D1	44				
X3	D2	45				
X4	D3	46				
X5	D4	47				
X6	D5	48				
X7	D6	49				
X8	D7	50				
X9	D8	100				
X10	D9	99				
X11	D10	98				
X12	D11	97				
X13	D12	96				
X14	D13	95				
X15	D14	94				
X16	D15	93				
X17	HSFAIL-L	2				
X18	MCRT-L	3				
X19	ME-L	7				
X20	GBR-L	8				
X21	BITEN-L	9	2	40	pF	1-3

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TABLE XXIII. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X22	STATEN-L	12	2	40	pF	1-3
X23	CMDTRF-L	13				
X24	INCMD-L	14				
X25	LSBENA-L	18				
X26	LSBLD-L	19				
X27	MSBENA-L	20				
X28	MSBLD-L	21				
X29	ENCENA-L	22				
X30	TXENA-L	23				
X31	SYNCSEL-L	24				
X32	WC0	26				
X33	WC1	27				
X34	WC2	28				
X35	WC3	29				
X36	WC4	30				
X37	GSR-L	55				
X38	DBDC	56				
X39	DBCACC-L	57				
X40	DTSTR	58				
X41	DTACT-L	59				
X42	DTREQ-L	61	2	40	pF	1-3

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TABLE XXIII. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X43	NBGT-L	66	2	40	pF	1-3
X44	CWC0	71				
X45	CWC1	72				
X46	CWC2	73				
X47	CWC3	74				
X48	CWC4	75				
X49	CHA-L/CHB	76				
X50	SA0	82				
X51	SA1	83				
X52	SA2	84				
X53	SA3	85				
X54	SA4	86				
X55	T/R-L	87				
X56	LMC	88				
X57	LBRO	89				
X58	RTFAIL-L	65				
X59	6MHZOUT	6				
X60	MSB/LSB-L	70				
X61	RST2-L	92				
X62	12MHZCLKOUT	54	2	40	pF	1-3

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NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X62
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X62
Capacitance measurement	NA	± 1.90 pF	X1-X62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Power-on reset test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Figure
		A to pin	B to pin	Min	Max		
Y	Power-on reset	1,31	92	240	1790	uS	28,29

NOTES:

1. Power-on reset test tolerance table.

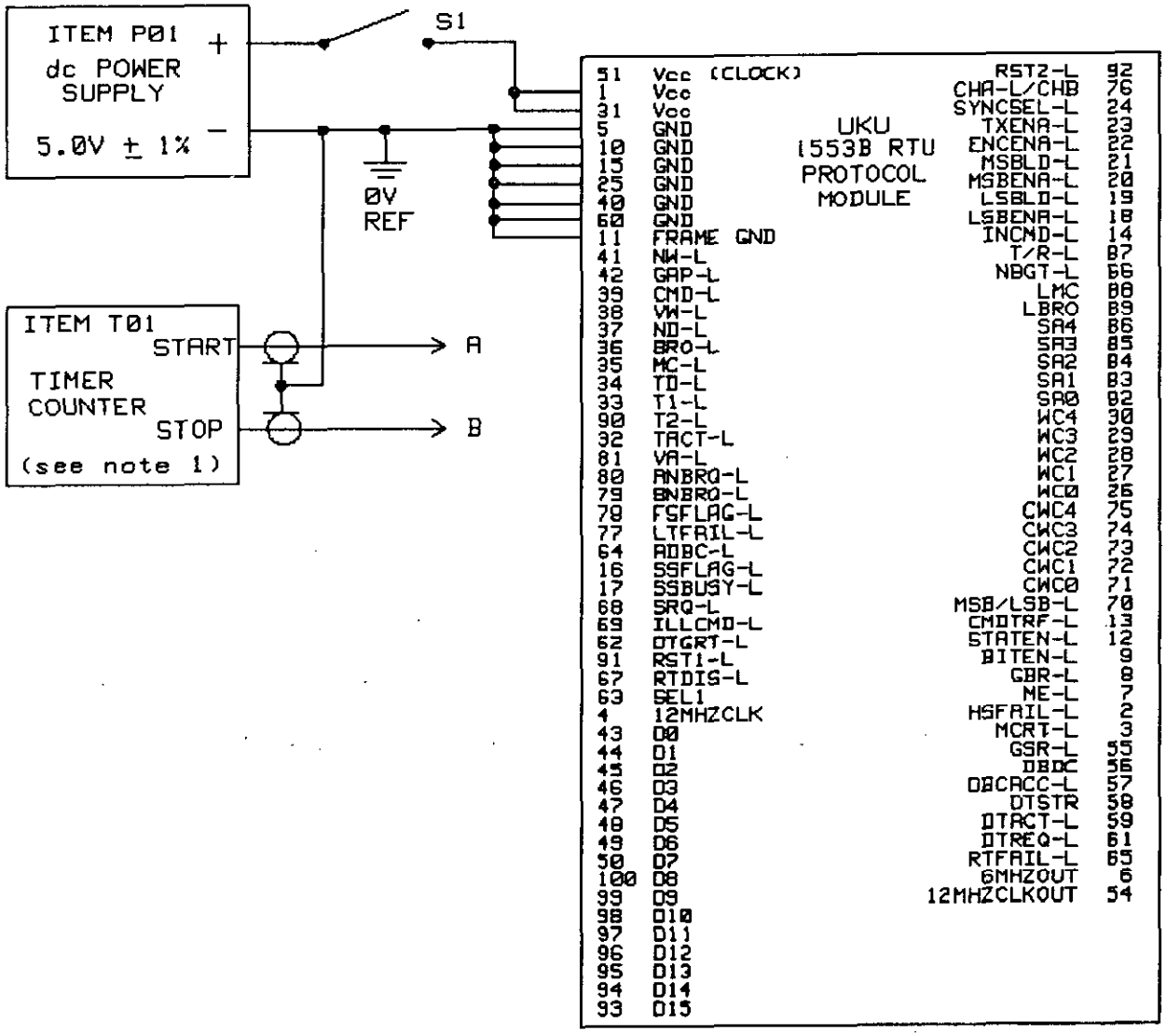
Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 55 mV	Y
Power supply current limit, (Vcc)	485 mA	NA	Y
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	Y Y
Power-on reset time measurement	NA	± 77.5 uS	Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified on figure 28. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XXIV. Make the measurements as indicated on figure 29.

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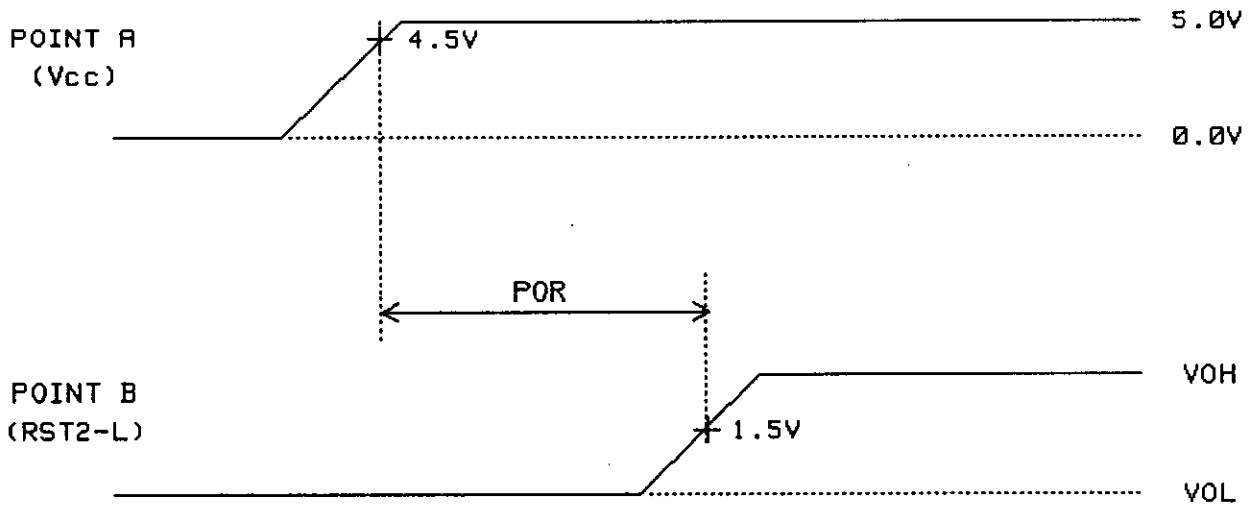


NOTES:

1. Make all connections with S1 open. Close S1 and make the required measurements.

FIGURE 28. Power-on reset test circuit for table XXIV.

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NOTES:

1. Have point A (start) triggered at Vcc = 4.5 V.
2. POR is the power-on reset parameter indicated in table XXIV.
3. Rise time for Vcc (0 to 4.5 V) shall be less than 10 mS.

FIGURE 29. Power-on reset test circuit waveform.

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TABLE XXV. Schmitt trigger threshold voltage test.

Test cond	Parameters	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
Z1	RST1-L positive-going threshold,Vt(+)	91	-	2.0	V	1-3	30
Z2	RST1-L negative-going threshold,Vt(-)	91	0.5	-	V	1-3	30
Z3	Hysteresis (Vt(+)-Vt(-))	91	0.4	-	V	1-3	30

NOTES:

1. Schmitt trigger threshold voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	Z1-Z3
Power supply, (CLOCK)	NA	NA	Z1-Z3
Forcing voltage	See note 2	± 5 percent	Z1-Z3
Power supply current limit, (Vcc)	485 mA	NA	Z1-Z3
(CLOCK)	NA	NA	Z1-Z3
Output voltage measurement	NA	± 25 mV	Z1,Z2
	NA	± 50 mV	Z3

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NOTES: - Continued.

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

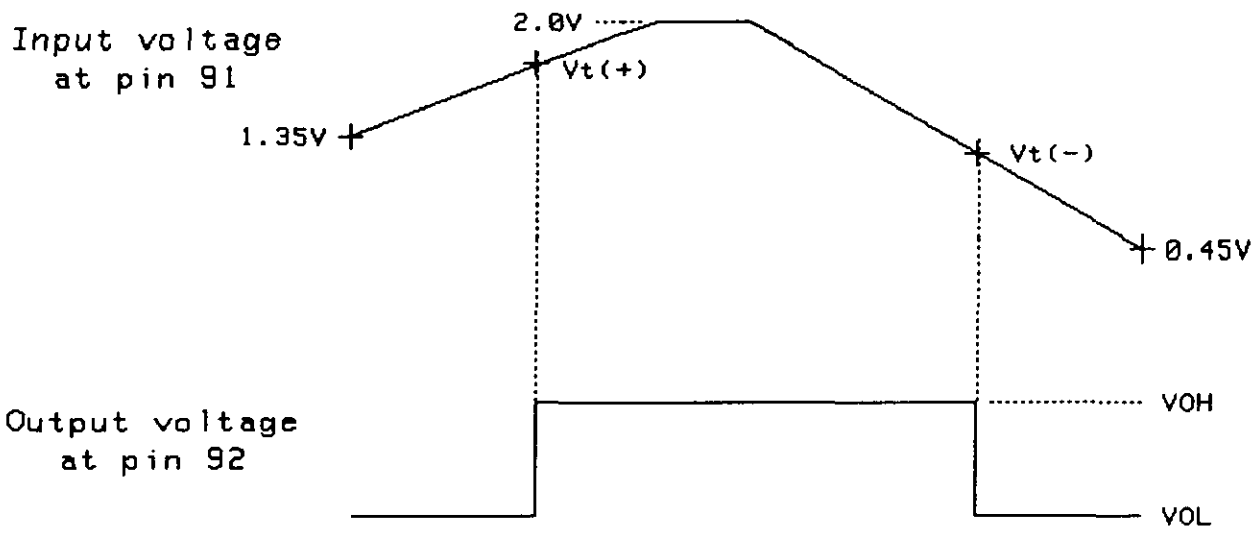
(b) For test condition Z1 (Vt(+)), increase the input voltage on pin 91 in 50 mV increments starting at 1.35 V and ending at 2.0 V. Observe pin 92 (RST2-L) for a low-to-high transition at least 50 nS after each 50 mV increase on pin 91 as indicated in table XXV (see figure 30).

(c) For test condition Z2 (Vt(-)), pin 92 (RST2-L) should initially be at a high level. Decrease the input voltage in 50 mV intervals starting at 2.0 V and ending at 0.45 V. Observe pin 92 (RST2-L) for a high-to-low transition at least 50 nS after each 50 mV decrease on pin 91 as indicated in table XXV (see figure 30).

(d) Test condition Z3 is the difference between the voltage found in Z1 and the voltage found in Z2.

3. Abbreviations used are defined in 6.1.

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NOTE:

1. 2.5 V $\leq$ VOH $\leq$ 5.0 V
   0.0 V $\leq$ VOL $\leq$ 0.4 V

FIGURE 30. Test waveform for table XXV.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/349	Modules, Standard Electronic, MIL-STD-1553 RTU Protocol Module, Key Code UKU

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. *Paragraph Number and Wording:*

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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